

Unit 8

DATABASES

59 items · 810 marks

Syllabus sections in this unit

8.1 Database Concepts

8.2 Database Management Systems (DBMS)

8.3 Data Definition Language (DDL) and Data Manipulation Language (DML)

Unit 8: Databases

1. M/J 26/P11/Q4/a,b,c

- 4 A shop allows customers to pay to borrow toys. The shop uses a database to record information about the customers and the toys they have borrowed.

CUSTOMER(CustomerID, ContactFirstName, Phone)

TOY(ToyID, ToyName, CostPerDay)

TOY_INSTANCE(UniqueToyID, ToyID, Quality)

BORROW(BorrowID, CustomerID, UniqueToyID, StartDate, NumberDays)

- (a) Complete the table by describing the relationship between each pair of tables. Include the primary key(s) and foreign key(s) and how these create each relationship.

pair of tables	description of relationship
CUSTOMER and BORROW	
TOY_INSTANCE and TOY	

[4]

- (b) Write a Structured Query Language (SQL) script to return the total number of days of all toys borrowed by each customer. The customer ID and first name for each customer also need to be returned.

The result needs to include a suitable field name for the total number of days.

[4]

- (c) The table BORROW needs amending to include an additional field.

- (i) The field will store whether the toy has been returned and will store either Yes or No.

Write an SQL script to include the additional field in the table BORROW.

[2]

- (ii) The toy that was borrowed with the borrow ID A2226 has been returned.

Write an SQL script to amend the record and store that the toy has been returned.

[3]

2. M/J 26/P12/Q3/a,b,c

- 3** A computer game developer stores data about a computer game in a database. The database has three entities:

- the users, who are called players
- the characters that users can create
- the groups that users can create and allow characters to join.

(a) Describe the relationships between the **three** entities.

[2]

(b) The database is created with **four** tables:

PLAYER(PlayerID, PlayerName, DateJoined)

CHARACTER(CharacterID, CharacterName, DateCreated, Level, NumberOfPoints, PlayerID)

GROUP(GroupID, GroupName, Category)

CHARACTER_GROUP(CharacterGroupID, CharacterID, GroupID, DateJoinedGroup)

Sample data for the table CHARACTER is given:

CharacterID	CharacterName	DateCreated	Level	NumberOfPoints	PlayerID
00123	Crystal	05/05/2025	5	35687	85AD
00124	Diamond	06/06/2025	4	21055	69CD
00125	Ruby	07/10/2025	1	1002	85AD

- (i) The given Structured Query Language (SQL) script is written to define the table CHARACTER.

```

01  CREATE TABLE CHARACTER(
02      CharacterID INT,
03      CharacterName VARCHAR,
04      DateCreated DATE,
05      TheLevel INT,
06      NumberOfPoints INT,
07      PlayerID VARCHAR,
08      PRIMARY KEY CharacterID,
09      FOREIGN KEY (PlayerID)
10  );

```

The SQL script contains **four** errors.

Give the line number of each error and the corrected line.

error line 1

corrected line

error line 2

corrected line

error line 3

corrected line

error line 4

corrected line

[4]

- (ii) Write an SQL script to return each player's name and the number of characters each player has created.

The result needs to include a suitable field name for the number of characters.

Order the results in descending order of player ID.

[5]

- (c) Complete the description of database key terms by writing the missing words.

The stores data about the data in the database. The table names, field names and relationships will be stored. Two other examples of data that will be stored are and

When designing a database, the schema is the design of the database tables, fields and relationships independent of any Database Management System (DBMS).

A DBMS provides security features to protect the data.

are used to give user accounts permission to read data, some to read and write data and others may be unable to read the data.

When creating a database, a provides tools to create visual components for example forms and reports.

[5]

3. M/J 26/P13/Q4/a,b,c

- 4 A database needs designing for a school library.

The library stores data about books including the author, title, category and number of pages.

The library will also store data about the students and the books they have borrowed. The data needs to include the student's name and student's class.

Books are borrowed for 14 days. A book is late if it has not been returned within the 14 days. The database needs to store whether the book is late, or not late.

- (a) Create a 3-table design for the library database normalised to Third Normal Form (3NF).

Give your table design in the format:

TableName (Field1, Field2, Field3, ...)

A primary key is underlined with a solid line.

A foreign key is underlined with a broken line.

table 1

table 2

table 3

[6]

(b) Explain the reasons why the library database should enforce referential integrity.

[2]

(c) The school is also creating a database to store data about the departments, the staff and classes in the school. The database is called `SCHOOLDATA`. The database has four tables:

`STAFF(StaffID, FirstName, FamilyName, DepartmentID)`

`DEPARTMENT(DepartmentID, DepartmentName, DepartmentLeader)`

`CLASS(ClassID, YearGroup, SubjectCode)`

`SUBJECT(SubjectCode, SubjectName, DepartmentID)`

(i) The table shows some sample data for `SUBJECT`.

SubjectCode	SubjectName	DepartmentID
CS1	Computer Science IGCSE	ITCS
CS2	Computer Science IAS-level	ITCS
MA3	Mathematics IGCSE	MTS
ME4	Media IGCSE	ITCS

The Structured Query Language (SQL) scripts to create the database and the table `SUBJECT` are shown.

```

01  CREATE DATA BASE SCHOOLDATA;
02  CREATE TABLE SUBJECT(
03      SubjectCode VARCHAR NONULL,
04      SubjectName VARCHAR
05      DepartmentID VARCHAR,
06      PRIMARY KEY (SubjectCode),
07      FOREIGN KEY (DepartmentID) REFERENCES (DepartmentID)
08  );

```

The SQL scripts contain **four** errors.

Give the line number of each error and the corrected line.

error line 1

corrected line

error line 2

corrected line

error line 3

corrected line

error line 4

corrected line

[4]

- (ii) Write an SQL script to return each department's name and the number of subject codes for that department.

The result needs to include a suitable field name for the number of subject codes.

Order the results in ascending order of department name.

[5]

4. O/N 25/P11/Q2/a,b,c,d

- 2 A relational database, `WORKEXP`, stores data about students and the companies where they complete work experience.

Students can complete multiple work experience placements but can only complete one placement at a time.

Students can complete more than one placement at the same company.

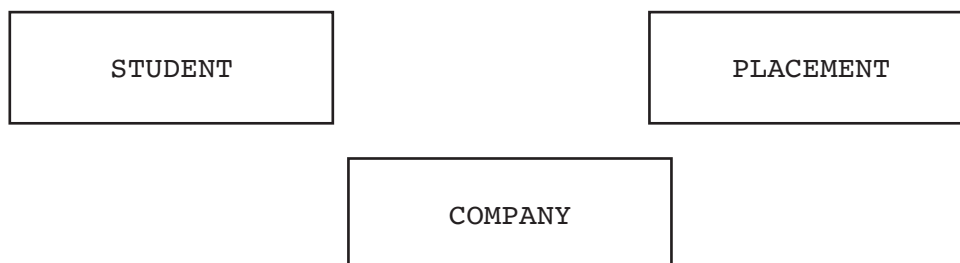
Part of the database is shown:

`STUDENT` (`StudentID`, `FirstName`, `TelephoneNumber`, `UniversityName`)

`PLACEMENT` (`PlacementID`, `StudentID`, `CompanyID`, `StartDate`, `EndDate`, `Complete`)

`COMPANY` (`CompanyID`, `CompanyName`, `MaxStudentsPerPlacement`)

- (a) Complete the entity-relationship (E-R) diagram for the database `WORKEXP`.



[2]

- (b) Explain why the database `WORKEXP` is in Third Normal Form (3NF).

[2]

- (c) Some example data from the `PLACEMENT` table is shown:

<code>PlacementID</code>	<code>StudentID</code>	<code>CompanyID</code>	<code>StartDate</code>	<code>EndDate</code>	<code>Complete</code>
PC001	LDEA01	MSCM	01/01/2023	12/07/2023	TRUE
PC002	LDEA01	MSCM	01/04/2024	08/05/2024	FALSE
PC003	ALAU02	NEAM	09/03/2020	10/03/2021	FALSE
PC004	LOLI75	GOUZ	07/06/2018	11/09/2018	TRUE

- (i) Write a Structured Query Language (SQL) script to delete all placements that have been completed.

[2]

- (ii) Write an SQL script to return the total number of placements completed by the student with ID LDEA01 at the company with ID NEAM. The total should be given an appropriate name.

[4]

- (d) The database tables are repeated here:

STUDENT(StudentID, FirstName, TelephoneNumber, UniversityName)

PLACEMENT(PlacementID, StudentID, CompanyID, StartDate, EndDate, Complete)

COMPANY(CompanyID, CompanyName, MaxStudentsPerPlacement)

Explain what is meant by referential integrity, and how it applies to this database.

[3]

5. O/N 25/P12/Q4/a,b,c,d

- 4 A relational database, SHIPPING, stores data about the ships in a company and the containers that are carried on the ships.

The database has the following tables:

CONTAINER(ContainerID, Type, Weight, OwnerName, ShipID)

SHIP(ShipID, Type, Capacity, ShipName)

- (a) Describe the relationship between the two tables. Refer to the primary and foreign keys in your answer.

[2]

- (b) The table CONTAINER needs an additional field to store the data for the last inspection date.

Write a Structured Query Language (SQL) script to add **one** field to the table CONTAINER to store the date of last inspection of the container, for example 08/07/2019.

[2]

- (c) Write an SQL script to return the number of containers stored in the database for the ship with the name Caledonia.

[4]

- (d) Describe the purpose of a developer interface in a Database Management System (DBMS).

[2]

6. O/N 25/P13/Q5/a,b,c,d

5 A company creates a relational database to store data about its customers.

The database, `REVIEWS`, stores data about the customers, products, complaints and staff.

The database has four tables:

`CUSTOMER`(`CustomerID`, `CustomerFirstName`, `CustomerLastName`, `CustomerEmail`)

`PRODUCT`(`ProductID`, `ProductName`, `ProductDetail`, `Price`, `Rating`)

`COMPLAINT`(`ComplaintID`, `ProductID`, `CustomerID`, `ComplaintDetails`, `StaffID`)

`STAFF`(`StaffID`, `StaffFirstName`, `StaffLastName`, `Department`, `RemoteWorker`)

(a) Complete the entity-relationship (E-R) diagram for the database `REVIEWS`.



[3]

(b) Some example data from the `STAFF` table is shown.

<code>StaffID</code>	<code>StaffFirstName</code>	<code>StaffLastName</code>	<code>Department</code>	<code>RemoteWorker</code>
1	Ralph	Jura	E	Yes
2	Luca	Emcee	A	Yes
3	Darwin	Acula	F	No

Write a Structured Query Language (SQL) script to define the table `STAFF`.

[4]

(c) Products are given a rating between 1 and 10 inclusive.

Write an SQL script to return only the `ProductID`, `ProductName` and `ComplaintDetails` for all products with a rating of 5 or less. The results need to be displayed in descending order of rating.

[5]

(d) A Database Management System (DBMS) is used to maintain and manage the database.

Describe **two** ways in which the DBMS can be used to ensure the security of the customer data.

1

2

[4]

7. M/J 25/P11/Q5/a,b,c,d,e

- 5** A company builds and sells furniture to customers. The company stores data about customers, their payment cards and their furniture orders in a database.

The database, FURNITURE, has the following tables:

CUSTOMER(CustomerID, Name, Phone)

CUSTOMER_CARD_DATA(CardID, CustomerID, CardType, CardNumber, EndDate)

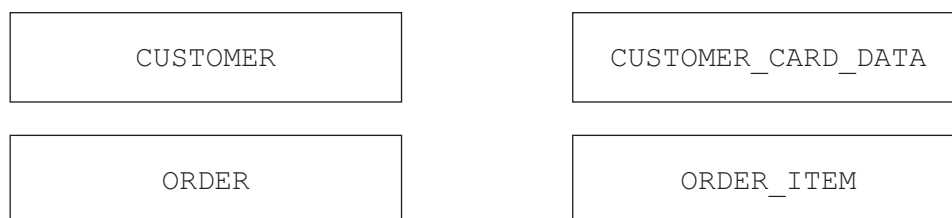
ORDER(OrderID, CustomerID, TotalCost, Paid, OrderDate, Complete)

ORDER_ITEM(OrderItemID, OrderID, Type, Height, Width, Depth, Details)

The primary keys are underlined in each table.

The attribute `Complete` in the table `ORDER` stores the Boolean value `TRUE` if the order has been built and `FALSE` if the order has not been built.

- (a) Complete the entity-relationship (E-R) diagram for the database.



[3]

- (b) Identify **one** attribute in the table `CUSTOMER_CARD_DATA` that could be a candidate key.

[1]

- (c) Identify **two** tables in the database that contain one or more foreign keys.
Give **one** attribute that is a foreign key in each table.

	Table	Foreign key
1		
2		

[2]

- (d) Explain the reasons why the data in the table `ORDER_ITEM` cannot be stored in the table `ORDER`

[3]

- (e) Write an Structured Query Language (SQL) script to output the customer ID, the customer's name and the total cost of the customer's orders that have **not** been paid.

The output of the total cost must have an appropriate title.

[4]

8. M/J 25/P12/Q5/a,b,c,d

- 5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

USER (Username, Password, DateOfBirth)

CHARACTER (CharacterName, CharacterID, Username, Level, Money)

ITEM (ItemName, MinimumLevel, Cost)

CHARACTER_ITEM (CharacterID, ItemName)

- (a) Explain the purpose of the table CHARACTER_ITEM in the database.

[2]

- (b) Underline the attribute, or attributes, that form the primary key in each of the tables.

USER (Username, Password, DateOfBirth)

CHARACTER (CharacterName, CharacterID, Username, Level, Money)

ITEM (ItemName, MinimumLevel, Cost)

CHARACTER_ITEM (CharacterID, ItemName)

[2]

- (c) A Database Management System (DBMS) provides data security.

- (i) Identify **two** methods the DBMS can use to protect the data in the table USER from unauthorised access.

Explain how each method protects the data.

Method 1

Explanation

Method 2

Explanation

[4]

- (ii) The DBMS also supports data integrity.

Give **two** ways that a DBMS can support data integrity.

1

2

[2]

- (d) (i) Write a Structured Query Language (SQL) script to count the number of items purchased by the user with the username "KAT123".

[3]

- (ii) The following changes need to be made to the character with the ID "0002":

- level changed to 3
- money changed to 10000.00

Write an SQL script to change the character's data.

[3]

9. M/J 25/P13/Q6/a,b,c,d

- 6** A shop sells pens to customers. Customers place an order with the shop and collect the items the next day. The shop uses a database to store the information about the orders.

The database contains the following tables:

CUSTOMER(CustomerID, CustomerName, Email)

ORDER(OrderID, CustomerID, Date, Collected)

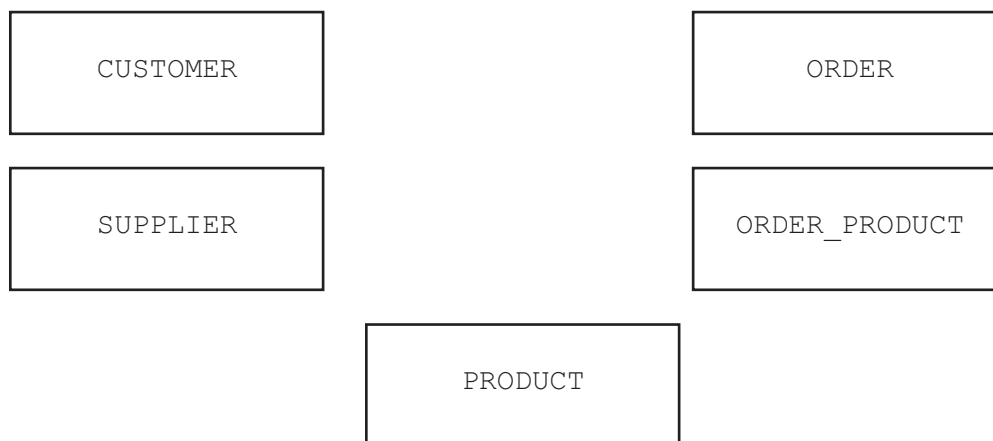
ORDER_PRODUCT(OrderID, ProductID, Quantity)

PRODUCT(ProductID, ProductName, QuantityInBox, Cost, SupplierID)

SUPPLIER(SupplierID, SupplierName, SupplierEmail)

The primary keys are underlined.

- (a)** Complete the entity-relationship (E-R) diagram for the database.



[4]

- (b)** A new product needs to be entered into the database. The product has the ID 002323, the product name 'Blue ball point 2mm', there are 50 in a box, the product costs \$5.00 and the supplier has the ID SFX223.

Write a Structured Query Language (SQL) script to enter the new product into the table PRODUCT.

[4]

- (c)** The attribute `Collected` in the table `ORDER` stores the Boolean value `TRUE` if the order has been collected and `FALSE` if the order has not been collected.

Write an SQL script to return the customer name for each customer that has orders they have not collected. Include the number of orders each customer has not collected with an appropriate title.

An example output might be:

CustomerName	NotCollected
Jack Wright	2
Lin Cho	1
Santaya Yui	1

[4]

(d) The shop uses a Database Management System (DBMS).

(i) One feature provided by a DBMS is a data dictionary.

The data dictionary stores the attribute names, table names, foreign keys and primary keys in the database.

Identify **three** other items stored in a data dictionary.

1

2

3

[3]

(ii) The DBMS provides a developer interface.

Explain how a database designer can make use of the developer interface.

[3]

10. O/N 24/P11/Q2/a,b,c

2 A shop repairs electronic devices, for example mobile phones and tablet computers. The shop owner stores the data about the repairs using a file-based approach.

(a) Give **one** limitation of using a file-based approach to store the data **and** explain how a relational database addresses this limitation.

Limitation

Explanation

[3]

(b) The shop owner creates a relational database called `FIXIT`.

The database stores data about the customers and the devices for repair.

Some devices need new parts that are ordered from suppliers.

The database `FIXIT` is designed to include the following tables:

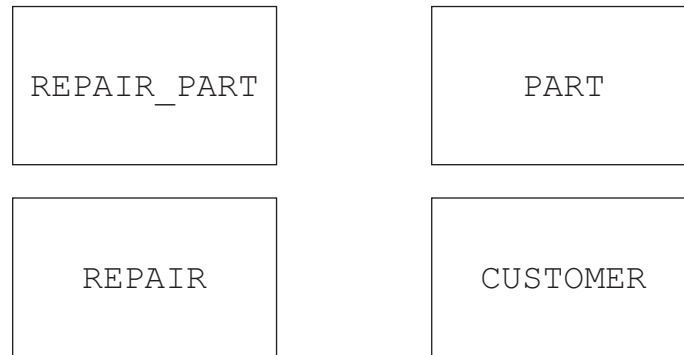
`PART(PartID, Description, Price, SupplierID)`

`CUSTOMER(CustomerID, FirstName, LastName, ContactNumber)`

`REPAIR(RepairNumber, StartDate, EndDate, CustomerID, Device)`

`REPAIR_PART(PartID, RepairNumber, Quantity)`

- (i) Complete the entity-relationship (E-R) diagram for the given tables.



[3]

- (ii) The table shows sample data for the table `REPAIR_PART`.

PartID	RepairNumber	Quantity
ACD128SA	0022	3
PPOR543DWW	0022	1
TR453	0023	1
PPOR543DWW	0023	2
WED5	0024	5

Write a Structured Query Language (SQL) script to define the table `REPAIR_PART`.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

[5]

- (iii) Suppliers send invoices to the company for the parts that are used. A new table, `INVOICE`, stores the data about each invoice and whether it has been paid or not.

The design for the table `INVOICE` is shown:

```
INVOICE(InvoiceID, SupplierID, AmountDue, Paid, DatePaid)
```

The table shows sample data for the table `INVOICE`.

InvoiceID	SupplierID	AmountDue	Paid	DatePaid
000001	JK675	22.50	TRUE	01/01/2024
000002	WR443	358.99	FALSE	
000003	JK675	10.21	FALSE	

Write an SQL script to return the total amount due to the supplier with the ID of JK675 for all the invoices that have **not** currently been paid.

[3]

(c) Complete the table by writing a definition for each of the database terms.

Term	Definition
Referential integrity	
Candidate key	
Tuple	

[3]

11. O/N 24/P12/Q6/a,b

- 6 A company uses a relational database to store data about its customers, employees and the individual repair jobs that customers have booked.

(a) Explain the benefits of using a relational database instead of a file-based approach.

[3]

- (b) The company decides which employees will work on each repair job. An employee can log into the database to access information about their repair jobs.

The database is normalised and includes these tables:

- CUSTOMER stores personal data about each customer
- EMPLOYEE stores personal data about each employee
- LOGIN_DATA stores the username and password for each employee
- JOB stores the data about each repair job
- JOB_EMPLOYEE stores the employees that are working on each repair job.

- (i) Identify each relationship between the database tables **and** explain how each relationship can be implemented in the normalised database.

[6]

- (ii) The database also has the table INVOICE that stores data about each invoice that is sent to a customer.

Example data from the table INVOICE is given.

InvoiceID	DateSent	Amount	Paid	JobID
29262	12/12/2023	105.20	Y	221
26765	11/11/2023	200.00	Y	315
13290	02/01/2024	50.00	Y	315
34090	05/02/2024	25.95	N	569

Write a Structured Query Language (SQL) script to return the total amount of all the invoices sent in the year 2023 that have been paid.

[3]

12. O/N 24/P13/Q4/a,b,c,d,e

- 4 A company makes ice cream and sells it to shops.

The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

The company's owner has designed a relational database, ICECREAM, to store data about their ice cream and customers.

Some of the tables in the database are given. The database is not normalised.

BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

CUSTOMER(CustomerID, CompanyName, EmailAddress, TelephoneNumber)

SALE(SaleID, BatchID, CustomerID, Quantity, Date)

- (a) Identify **two** foreign keys in the table SALE **and** the table that each foreign key references.

Foreign key 1

Table name 1

Foreign key 2

Table name 2

[2]

- (b) Write an SQL script to return the total quantity of ice cream sold to the customer with the ID of 0034E in the year 2023.

[3]

- (c) The table definition for `BATCH` is repeated here:

`BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)`

Sample data for the table `BATCH` is given:

BatchID	Type	Flavour	Size	SellingPrice	EndDate
KIV12	Plain	Vanilla	1	2.20	12/12/2024
RIC14	Plain	Chocolate	0.5	1.80	12/12/2024
TYL1	Non-dairy	Lemon	0.5	2.10	01/02/2025
FYV2	Non-dairy	Vanilla	0.25	1.50	02/02/2025
BIV13	Plain	Vanilla	1	2.20	02/02/2024

- (i) Write an SQL script to define the table `BATCH`.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

[5]

- (ii) The table `BATCH` is not normalised.

Normalise the database table `BATCH`.

Write the table definitions for your new tables.

Identify any primary and foreign keys in your tables.

Do **not** change or include the tables `CUSTOMER` and `SALE`.

[4]

(d) Complete the following table by defining each database term.

Database term	Definition
Entity	
Attribute	

[2]

(e) A Database Management System (DBMS) supports data integrity.

Explain how a DBMS supports data integrity.

[3]

13. M/J 24/P11/Q6/a,b,c

- 6 A company is developing a website that will allow users to create an account and then play a quiz every day. The data about the users and the quizzes are stored in a database.

A user must select a unique username and enter a valid email address to create an account. All users must be over the age of 16. A new quiz is given to the users every day. Each quiz is stored in its own text file.

The database stores the filename of each quiz and the date it can be played. The user gets a score for each quiz they complete, which is stored in the database. The scores are used to give each user a rating, for example Gold.

(a) Create a 3-table design for this database normalised to Third Normal Form (3NF).

Give your table design in the format:

TableName (PrimaryKey, Field1, Field2, ...)

[6]

(b) The company is using a Database Management System (DBMS) to set up the database.

Describe what is meant by the following DBMS features:

Data dictionary

Logical schema

[4]

(c) The company has another database, FARMING, for a different game.

The database `FARMING` has a table named `EVENT` which is shown with some sample data.

PlayerID	EventID	Category	Points
000123	3	Build	100
000124	1	Grow	36
000123	4	Grow	22
000123	7	Create	158
000125	3	Grow	85
000125	4	Build	69

- (i) The database `FARMING` has a second table created named `PLAYER` that has the primary key `PlayerID`.

The field `PlayerID` in `EVENT` needs to be set up as a foreign key to link to `PlayerID` in `PLAYER`.

Write a Structured Query Language (SQL) script to change the table definition for `EVENT` to link the foreign key to `PLAYER`.

[2]

- (ii) Write an SQL script to return the number of events that each player has completed.

[3]

14. M/J 24/P12/Q4/a,b,c,d

- 4** An assessment board wants to store the marks students achieved in exams in a database named RECORDS.

Part of the database design includes these two tables:

EXAM(ExamID, Subject, Level, TotalMarks)

EXAM_QUESTION(ExamQuestionID, ExamID, QuestionNumber, Question, MaxMark)

- (a)** Identify the relationship between EXAM and EXAM_QUESTION.

[1]

- (b)** Sample data for the table EXAM is shown:

ExamID	Subject	Level	TotalMarks
00956124	Computer Science	2	75
00956125	Computer Science	3	120
00956126	Mathematics	2	100
00956127	Mathematics	3	150
00956128	Physics	2	70
00956129	Physics	3	80

Write a Structured Query Language (SQL) script to define the table EXAM.

[3]

- (c)** The table EXAM_QUESTION has been created but the foreign key has not been linked.

Write an SQL script to update EXAM_QUESTION **and** link the foreign key to EXAM.

[2]

- (d)** The database also needs to store data about the students, the exams the students have taken and the marks the students achieved in each question of each exam.

Describe the additional tables that will need to be included in the database **and** explain how all the tables in the database will be linked.

[5]

15. M/J 24/P13/Q4/a,b,c,d

- 4** A theatre wants to use a database to store data about the shows that are scheduled, their customers and the seats that the customers have booked.

In the theatre:

- Each show can take place on multiple dates.
- Some dates can have more than one performance.
- There are multiple rows of seats.
- Each seat can be individually booked by its row letter and seat number, for example row E seat 2.

Part of the database design includes these tables:

SHOW(ShowID, Title, Duration)

SEAT(SeatID, RowLetter, SeatNumber)

PERFORMANCE(PerformanceID, ShowID, ShowDate, StartTime)

- (a)** Identify the relationship between the tables PERFORMANCE and SHOW.

[1]

- (b)** Sample data for the table PERFORMANCE is shown:

PerformanceID	ShowID	ShowDate	StartTime
0001	MK12	5/5/2025	13:00
0002	MK12	5/5/2025	19:30
0003	MK12	6/5/2025	19:00
0004	OP3	7/5/2025	18:30
0005	OP3	8/5/2025	18:30
0006	OP3	9/5/2025	13:00

Write a Structured Query Language (SQL) script to define the table PERFORMANCE.

[4]

- (c)** Write an SQL script to return the number of times each show is scheduled. For example, in the sample data in part **(b)**, the show MK12 is scheduled three times.

The result needs to include the show name and a suitable field name for the number of times it is scheduled.

[4]

- (d)** Customers give their first name, last name and email address when they are making a booking. One booking can include multiple seats.

Describe the additional tables that will need to be included in the database **and** explain how these tables will be linked within the database.

[5]

16. O/N 23/P11/Q3/a,b,c

- 3** A shop manager has designed a relational database to store customer orders.

The database will have the following tables:

CUSTOMER(CustomerID, FirstName, LastName, Town)

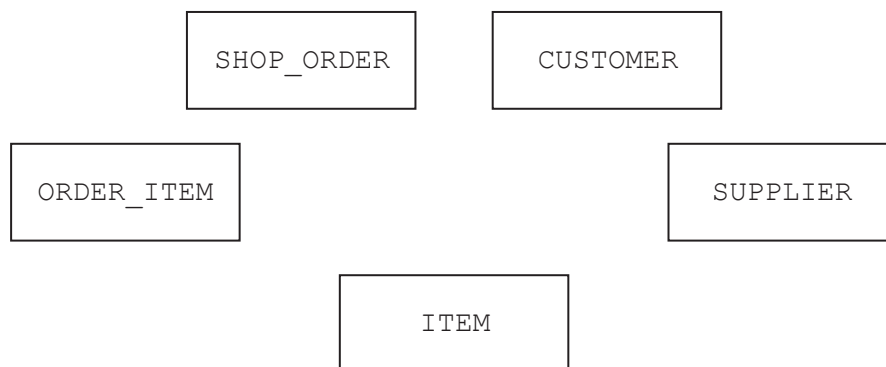
SHOP_ORDER(OrderNo, CustomerID, OrderDate)

SUPPLIER(SupplierID, EmailAddress, TelephoneNumber)

ITEM(ItemNumber, SupplierID, Description, Price)

ORDER_ITEM(ItemNumber, OrderNo, Quantity)

- (a)** Complete the entity-relationship (E-R) diagram for the relational database.



[3]

- (b)** Identify **three** advantages of a relational database compared to a file-based approach.

1

2

3

[3]

- (c) (i)** Write a Structured Query Language (SQL) script to define the database called *SHOP*.

[1]

- (ii)** Write the SQL script to return the total quantity of items that the customer with the ID of HJ231 has ordered.

[4]

17. O/N 23/P12/Q2/a,b,c

- 2 (a)** State what is meant by the following terms in a relational database model.

Entity

Primary key

Referential integrity

[3]

- (b)** Authentication is one method a Database Management System (DBMS) can use to improve the security of a database.

Describe **other** methods that a DBMS can use to improve the security of a database.

[4]

- (c)** The following database table is not normalised.

StudentName	DateOfBirth	TutorGroup	Subject	SubjectCode
Yuwei Chen	01/09/2004	SMH	English, Maths, Computer Science	EN, MA, CS
Claudia Raj	23/02/2005	JMB	Maths, Physics, Art	MA, PY, AR
Aamil Akram	24/01/2005	KMB	Art, Design, English language	AR, DE, EN
Areeba Faraz	21/12/2004	SMH	English language, Chemistry, Design	EN, CH, DE

Explain how to modify the table to put it into First Normal Form (1NF).

[4]

18. O/N 23/P13/Q3/b

- 3** A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

- (b)** The company uses a database, `COURSES`, to store data about the courses and their tutors.

Each course starts at different times of the year and may have a different tutor.

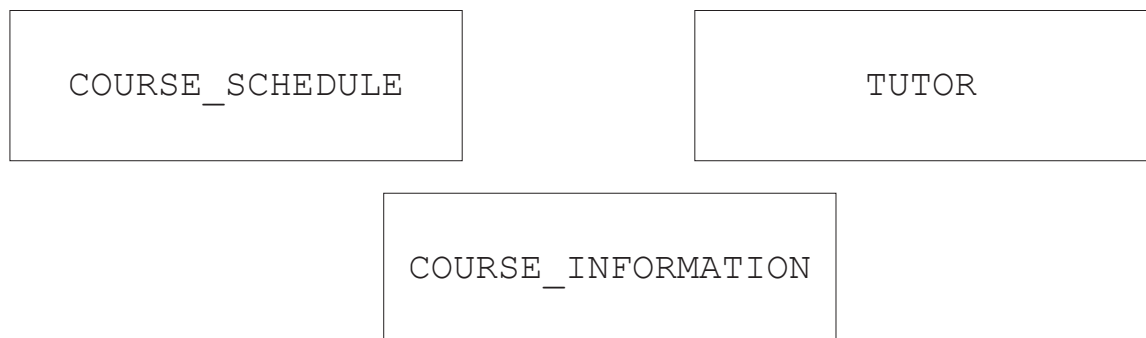
The database has the following structure:

```
COURSE_INFORMATION(CourseID, Description, Cost)
```

```
TUTOR(TutorID, TelephoneNumber, EmailAddress, TutorName)
```

```
COURSE_SCHEDULE(CourseID, DateStarted, TutorID)
```

- (i)** Complete the entity-relationship (E-R) diagram for the database `COURSES`.



[1]

- (ii)** Write the Structured Query Language (SQL) script to return the total number of courses that have started after 9 September 2023.

The value returned must have an appropriate field name.

[4]

19. M/J 23/P11/Q2/a,b

- 2** An organisation uses a database to store data about the types of bird that people have seen.

- (a)** The database is managed using a Database Management System (DBMS).

- (i)** State what is meant by a data dictionary **and** give **one** example of an item typically found in a data dictionary.

Definition

Example

[2]

- (ii)** State what is meant by data integrity **and** give **one** example of how this is implemented in a database.

Definition

Example

[2]

- (b) The database, `Birds`, stores information about the types of bird and the people who have seen them.

Data about each bird seen is stored with its location and data about the person who saw the bird.

Database `Birds` has the following tables:

```
BIRD_TYPE(BirdID, Name, Size)
```

```
BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)
```

```
PERSON(PersonID, FirstName, LastName, EmailAddress)
```

- (i) Complete the table by identifying **two** foreign keys and the database table where each is found.

Foreign key	Database table

[2]

- (ii) The database `Birds` has been normalised.

Draw **one** line from each Normal Form to the most appropriate definition.

Normal Form

Definition

First Normal Form (1NF)

All fields are fully dependent on the primary key.

Second Normal Form (2NF)

There are no repeating groups of attributes.

Third Normal Form (3NF)

There are no partial dependencies.

[1]

(iii) Part of the database table `BIRD_TYPE` is shown:

BirdID	Name	Size
0123	Blackbird	Medium
0035	Jay	Large
0004	Raven	Large
0085	Robin	Small

The database only supports these data types:

- character
- varchar
- Boolean
- integer
- real
- date
- time

Write a Structured Query Language (SQL) script to define the table `Bird_Type`.

[4]

(iv) The database tables are repeated here for reference:

`BIRD_TYPE(BirdID, Name, Size)`

`BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)`

`PERSON(PersonID, FirstName, LastName, EmailAddress)`

Complete the SQL script to return the number of birds of each size seen by the person with the ID of `J_123`.

```
SELECT BIRD_TYPE.Size, ..... (BIRD_TYPE.BirdID)

      AS NumberOfBirds

FROM BIRD_TYPE, .....

WHERE ..... = "J_123"

AND BIRD_TYPE.BirdID = .....

..... BIRD_TYPE.Size;
```

[5]

20. M/J 23/P12/Q2/a,b,c(ii,iii)

2 A horse riding school uses a database, `Lessons`, to store data about lesson bookings.

This database is created and managed using a Database Management System (DBMS).

(a) The table contains names and descriptions of DBMS features and tools.

Complete the table by writing down the missing names and descriptions.

Name	Description
Data dictionary	
Query processor	
	A model of a database that is not specific to one DBMS.
.....	A software tool that allows the user to create items such as tables, forms and reports.

[4]

(b) Explain the reasons why referential integrity is important in a database.

[3]

(c) The database `Lessons` has the following tables:

```
HORSE(HorseID, Name, Height, Age, HorseLevel)
```

```
STUDENT(StudentID, FirstName, LastName, RiderLevel, PreferredHorseID)
```

```
LESSON(LessonID, Date, Time, StudentID, HorseID, LessonContent)
```

Dates in this database are stored in the format `#DD/MM/YYYY#`.

The fields `RiderLevel` and `HorseLevel` can only have the values: `Beginner`, `Intermediate` or `Advanced`.

(ii) Write a Structured Query Language (SQL) script to return the names of all the horses that have the horse level intermediate or beginner.

[4]

- (iii) The following SQL script should return the number of riders that have the rider level beginner and have a lesson booked on 09/09/2023.

```
SELECT SUM(STUDENT.RiderLevel) AS NumberOfRiders  
  
FROM STUDENT, LESSON  
  
WHERE StudentID = StudentID  
  
OR Date = #09/09/2023#  
  
AND STUDENT.RiderLevel = Beginner;
```

There are **four** errors in the script.

Identify **and** correct each error.

- 1
- 2
- 3
- 4

[4]

21. M/J 23/P13/Q4/a,b,c

- 4 A shop rents cars to customers. The shop uses a relational database to store information about the rentals.
- (a) Describe **two** ways in which a relational database addresses the limitations of a file-based approach.

- 1
- 2

[4]

(b) Complete the table by writing the missing term or description for each database feature.

Term	Description
	An object that data is stored about.
Tuple	
Secondary key	
.....	A field in one table that is linked to a primary key in another table.

[4]

(c) The car rental database is not normalised. The current database design is:

```
BOOKING (CarRegistration, StartDate, EndDate,
         CarModel, CarColour, CustomerFirstName)
```

```
CUSTOMER (CustomerFirstName, CustomerLastName, EmailAddress,
          TelephoneNumber)
```

Write a normalised database design for this database.

All tables must be in Third Normal Form (3NF).

Use the field names given **and** underline the primary key fields.

[4]

22. O/N 22/P11/Q4/a,b,c,d

- 4 A photographer creates a relational database to store data about photographs taken at birthday parties.

The database, PHOTOGRAPHS, stores details of the customer, the party, the photographs taken and the cameras used.

The photographer has several cameras that are used for taking the photographs at the parties.

Each camera has a specific lens type (for example, XY32Z) and lighting type (for example, F1672).

Data about each photograph is stored in the database including the party at which it was taken, the time it was taken and the camera used.

The database has these four tables:

CUSTOMER(CustomerID, FirstName, LastName, Telephone)

PARTY(PartyID, CustomerID, PartyDate, StartTime)

PHOTO_DATA(PhotoID, PartyID, TimeTaken, CameraID)

CAMERA_DATA(CameraID, LensType, LightingType)

(a) Complete the entity-relationship (E-R) diagram for the database PHOTOGRAPHS.



[3]

(b) The database is normalised and is in Third Normal Form (3NF).

Describe the characteristics of a database that is in Third Normal Form (3NF).

[3]

(c) The table shows some sample data for the table PHOTO_DATA.

PhotoID	PartyID	TimeTaken	CameraID
ST23-56	BD987	08:34	NIK-02
ST23-57	BD987	08:55	NIK-02
ST23-60	BC08	09:01	CAN-01
ST23-61	BC08	10:23	CAN-12
ST23-62	BC08	10:56	NIK-01

(i) State what is meant by a **tuple**. Give an example of a tuple from PHOTO_DATA.

Tuple

Example

[2]

(ii) Complete the Structured Query Language (SQL) script to display the total number of photographs that have been taken using a camera with a camera ID starting with CAN.

```

SELECT ..... (.....)

FROM .....

WHERE CameraID LIKE ..... ;

```

[4]

- (d) Write an SQL script to include two new fields in CAMERA_DATA to store the number of photographs currently on the camera **and** the date the camera was last used.

[3]

23. O/N 22/P12/Q5/a,b,c,d

- 5 A relational database, GARDEN, has the following tables:

OWNER(OwnerID, FirstName, TelephoneNo, TreeID, TreePosition)

TREE(TreeID, ScientificName, MaxHeight, FastGrowing)

- (a) The database is **not** in Third Normal Form (3NF).

Explain how the database can be normalised to 3NF.

[3]

- (b) Write the Structured Query Language (SQL) script to add a new record in the table TREE to store the following data:

Attribute	Value
TreeID	LOW_1276
ScientificName	Salix_Alba
MaxHeight	30.00
FastGrowing	TRUE

[3]

- (c) State what is meant by a **candidate key** in a relational database.

[1]

- (d) (i) Describe, using an example, what is meant by a **data dictionary**.

[2]

- (ii) Describe what is meant by a **logical schema**.

[2]

24. O/N 22/P13/Q2/a,b,c,d,f

- 2** The relational database `ASTRONOMY` is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

`COMPANY (TelephoneNumber, CompanyID, CompanyName)`

`PHOTOGRAPH (PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)`

`TELESCOPE (TelescopeID, CompanyID, SerialNumber)`

- (a)** Complete the following table by writing the correct answer for each item.

Item	Answer
a suitable field for the primary key in <code>COMPANY</code>	
a candidate key in <code>TELESCOPE</code>	
the degree of relationship between <code>TELESCOPE</code> and <code>PHOTOGRAPH</code>	

[3]

- (b)** A Database Management System (DBMS) has several features.

Identify the feature that describes the relationship between data and its structure.

[1]

- (c)** Complete the SQL script to return the total number of telescopes owned by the company whose ID begins with `HW`.

```
SELECT ..... (.....)
FROM TELESCOPE
WHERE ..... LIKE .....
```

[4]

- (d)** Write the SQL script to add **one** field to the table `PHOTOGRAPH` to store the resolution of the photograph, e.g. `1920 × 1068`.

[2]

- (f)** Describe the purpose of a query processor in a DBMS.

[2]

25. O/N 21/P11/Q5/a,b,c

- 5 Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database `CARS` to store the data about his business.

Part of the database is shown:

`SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)`

`MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)`

`CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)`

- (a) Tick (✓) **one** box in each row to identify whether each field is a primary key or a foreign key.

Table	Field name	Primary key	Foreign key
MANAGER	ManagerID		
SHOP	ManagerID		
CAR	RegistrationNumber		
CAR	ShopID		

[2]

- (b) Describe the ways in which access rights can be used to protect the data in Javier's database from unauthorised access.

[3]

- (c) Javier uses Data Definition Language (DDL) and Data Manipulation Language (DML) statements in his database.

- (i) Complete the following DML statements to return the number of cars for sale in each shop.

`SELECT COUNT (.....)`

`FROM .....`

`..... ShopID`

[3]

- (ii) Complete the DML statement to include the following car in the table `CAR`.

Field	Data
RegistrationNumber	123AA
Make	Tiger
Model	Lioness
NumberOfMiles	10500
ShopID	12BSTREET

..... CAR
 ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")

[2]

26. O/N 21/P12/Q6/a,b,c

- 6 A shop sells plants to customers. The shop manager has a relational database to keep track of the sales.

The database, PLANTSALLES, has the following structure:

PLANT(PlantName, QuantityInStock, Cost)

CUSTOMER(CustomerID, FirstName, LastName, Address, Email)

PURCHASE(PurchaseID, CustomerID)

PURCHASE_ITEM(PurchaseID, PlantName, Quantity)

(a) The database is normalised.

(i) The table lists the following three stages of normalisation:

- The first stage is from a database that is not normalised (0NF) to First Normal Form (1NF).
- The second stage is from 1NF to Second Normal Form (2NF).
- The third stage is from 2NF to Third Normal Form (3NF).

Tick (✓) **one** box in each row to identify the appropriate stage for each task.

Task	Normalisation stage		
	0NF to 1NF	1NF to 2NF	2NF to 3NF
Remove any partial key dependencies			
Remove any repeating groups of attributes			
Remove any non-key dependencies			

[2]

(ii) Draw an entity-relationship (E-R) diagram for the database PLANTSALLES.



[3]

(b) The shop manager uses a Database Management System (DBMS).

Describe the purpose **and** contents of the data dictionary in the DBMS.

[3]

(c) The shop manager uses both Data Definition Language (DDL) and Data Manipulation Language (DML) statements to create and search the database.

(i) Complete the DML statements to return the total number of items purchased with the purchase ID of 3011A.

SELECT SUM (.....)

FROM

WHERE =;

[4]

(ii) Write DDL statements to include a field in the table `PURCHASE` to store the date of the order.

[3]

27. O/N 21/P13/Q5/a,b,c

5 Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database `CARS` to store the data about his business.

Part of the database is shown:

`SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)`

`MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)`

`CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)`

(a) Tick (✓) **one** box in each row to identify whether each field is a primary key or a foreign key.

Table	Field name	Primary key	Foreign key
MANAGER	ManagerID		
SHOP	ManagerID		
CAR	RegistrationNumber		
CAR	ShopID		

[2]

(b) Describe the ways in which access rights can be used to protect the data in Javier's database from unauthorised access.

[3]

(c) Javier uses Data Definition Language (DDL) and Data Manipulation Language (DML) statements in his database.

- (i) Complete the following DML statements to return the number of cars for sale in each shop.

SELECT COUNT (.....)

FROM

..... ShopID

[3]

- (ii) Complete the DML statement to include the following car in the table CAR.

Field	Data
RegistrationNumber	123AA
Make	Tiger
Model	Lioness
NumberOfMiles	10500
ShopID	12BSTREET

..... CAR

..... ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")

[2]

28. 9608 O/N 21/P11/Q9/a,b

- 9 An airline company uses a relational database to store data about passengers and flights.

Part of the database is shown.

PASSENGER(PassengerID, FirstName, LastName)

FLIGHT(FlightID, FlightDate, FlightTime)

PASSENGER_LIST(FlightID, PassengerID, SeatNo)

- (a) Complete the entity-relationship (E-R) diagram to show the relationships between the given tables.



[2]

- (b) The following is example data for the table FLIGHT:

FlightID	FlightDate	FlightTime
MO126	05/05/21	09:00
GK6708	06/09/21	00:30
BA0897	08/12/21	15:30

Write Data Definition Language (DDL) statements to define the table FLIGHT.

[5]

29. M/J 21/P11/Q7/a,b,c

7 Bobby and Kim are discussing databases.

- (a)** Bobby tells Kim that a file-based approach is usually better than a relational database.

Explain why Bobby is incorrect.

[3]

- (b)** Bobby has a shop that sells products to customers. His database will store data about his customers, their payment details, orders and the products he sells. Customers will have login details to access their accounts. The database will update customers' payment and login details without keeping any historical records.

- (i)** Give **one** example of each of the following relationships from Bobby's database.

one-to-one

one-to-many

many-to-many

[3]

- (ii)** Tick (✓) **one** box to identify the relationship that cannot be directly implemented in a normalised relational database.

Relationship	Tick (✓)
one-to-one	
one-to-many	
many-to-many	

[1]

- (iii)** Bobby wants to name his database `SHOPORDERS`.

Write a Data Definition Language (DDL) statement to define a new database with the name `SHOPORDERS`.

[1]

- (c)** A database has a data dictionary.

Give **three** items that are stored in a data dictionary.

1

2

3

[3]

30. M/J 21/P12/Q1/a,b,c

- 1 Raj owns houses that other people rent from him. He has a database that stores details about the people who rent houses, and the houses they rent. The database, HOUSE_RENTALS, has the following structure:

CUSTOMER(CustomerID, FirstName, LastName, DateOfBirth, Email)

HOUSE(HouseID, HouseNumber, Road, Town, Bedrooms, Bathrooms)

RENTAL(RentalID, CustomerID, HouseID, MonthlyCost, DepositPaid)

- (a) Give the definition of the following database terms, using an example from the database HOUSE_RENTALS for each definition.

Term	Definition and example
Field	
Entity	
Foreign key	

[6]

- (b) Tick (✓) **one** box to identify whether the database HOUSE_RENTALS is in Third Normal Form (3NF) or not in 3NF.

Justify your choice using one or more examples from the database HOUSE_RENTALS.

In 3NF	
Not in 3NF	

Justification

[2]

(c) Example data from the table RENTAL are given:

RentalID	CustomerID	HouseID	MonthlyCost	DepositPaid
1	22	15B5L	1000.00	Yes
2	13	3F	687.00	No
3	1	12AB	550.00	Yes
4	3	37	444.50	Yes

(i) Complete the following Data Definition Language (DDL) statement to define the table RENTAL.

```
CREATE ..... (
    RentalID INTEGER NOT NULL,
    CustomerID INTEGER NOT NULL,
    HouseID ..... (5) NOT NULL,
    MonthlyCost ..... NOT NULL,
    DepositPaid BOOLEAN NOT NULL,
    ..... (RentalID)
);
```

[4]

(ii) Write a Data Manipulation Language (DML) script to return the first name and last name of all customers who have **not** paid their deposit.

[4]

31. M/J 21/P13/Q7/a,b,c

7 Bobby and Kim are discussing databases.

- (a)** Bobby tells Kim that a file-based approach is usually better than a relational database.

Explain why Bobby is incorrect.

[3]

- (b)** Bobby has a shop that sells products to customers. His database will store data about his customers, their payment details, orders and the products he sells. Customers will have login details to access their accounts. The database will update customers' payment and login details without keeping any historical records.

- (i)** Give **one** example of each of the following relationships from Bobby's database.

one-to-one

one-to-many

many-to-many

[3]

- (ii)** Tick (✓) **one** box to identify the relationship that cannot be directly implemented in a normalised relational database.

Relationship	Tick (✓)
one-to-one	
one-to-many	
many-to-many	

[1]

- (iii)** Bobby wants to name his database `SHOPORDERS`.

Write a Data Definition Language (DDL) statement to define a new database with the name `SHOPORDERS`.

[1]

- (c)** A database has a data dictionary.

Give **three** items that are stored in a data dictionary.

1

2

3

[3]

32. 9608 M/J 21/P11/Q5/a,b,c,d

- 5** A company runs activity courses. It is creating a relational database to store details of the courses it runs.

The database has five tables:

EMPLOYEE (EmployeeID, FirstName, LastName, Role, Language)

BOOKING (BookingID, CustomerID, CourseID)

CUSTOMER (CustomerID, FirstName, LastName)

COURSE (CourseID, Title, Level, Date)

COURSE_EMPLOYEE (CourseID, EmployeeID)

- (a)** Complete the entity relationship (E-R) diagram for the database.



[4]

- (b)** Describe what is meant by **referential integrity**.

Give an example from the CUSTOMER and BOOKING tables in your answer.

[2]

- (c)** A Data Definition Language (DDL) is used to create the structure of the database. One item that can be created is the database.

Identify **three other** items that can be created in the database using the DDL.

1

2

3

[3]

- (d)** Part of the EMPLOYEE table is shown.

EmployeeID	FirstName	LastName	Role	Language
001	Jasmine	Chen	Leader	French
002	Kenton	Archer	Leader	English
003	Michael	Roux	Cook	French
004	Conrad	Slavorski	Leader	Russian

Write a Data Manipulation Language (DML) statement to return the first name and last name of all employees, who are leaders, and speak either French or English.

[4]

33. 9608 M/J 21/P12/Q4/a,b,c

- 4** K2 Mountain Guiding is a company that runs courses teaching people how to climb mountains.

The company uses a file-based approach to store and retrieve data.

- (a)** Describe **three** drawbacks of a file-based approach compared to a relational database.

Drawback 1

Drawback 2

Drawback 3

[6]

- (b)** Each course has a destination and a trip date, and is run by a guide. Customers can book a place on one or more courses.

The following table shows part of the stored file:

Customer name	Customer date of birth	Guide	Destination	Trip date
Jay Patel	10/10/1976	Artem	Elbrus	06/03/2023
Paul Schloss	10/04/1999	Kenton	K2	01/03/2022
Mina Wang	03/03/2000	Kenton	K2	01/03/2022
Paul Schloss	10/04/1999	Safia	Mont Blanc	07/07/2024
Jay Patel	10/10/1976	Safia	K2	04/04/2023

The company has decided to create a relational database to replace the current file-based approach.

Identify **three** reasons why the data in this table is not in First Normal Form (1NF).

1

2

3

[3]

- (c)** The table, `GUIDE_TABLE`, is created.

Each guide has a unique guide ID.

Part of the table `GUIDE_TABLE` is shown:

GuideID	Guide	DateOfBirth	Gender
1	Artem	07/03/1992	M
2	Kenton	08/04/1984	M
3	Safia	10/10/1999	F

Write a Data Definition Language (DDL) statement to define the table `GUIDE_TABLE`.

[5]

34. 9608 M/J 21/P13/Q8/a,b,c,d

8 A charity has a relational database, `CHARITY_PROJECT`.

The database has three tables to store information about the charity's employees, the projects and the project teams.

```
EMPLOYEE(EmployeeID, FirstName, LastName, Salary, DOB)
```

```
PROJECT(ProjectID, Name, Location, ProjectStartDate)
```

```
PROJECT_TEAM(EmployeeID, ProjectID, JoinedProjectDate)
```

(a) Complete the entity relationship (E-R) diagram to show the relationships between these tables.



[2]

(b) Explain how primary and foreign keys are used to link the tables in `CHARITY_PROJECT`.

[4]

(c) The database needs to store the gender of each employee (either `M` for Male or `F` for Female).

Write a Data Definition Language (DDL) statement to add the attribute `Gender` to the `EMPLOYEE` table.

[2]

(d) Part of the `EMPLOYEE` table is shown:

EmployeeID	FirstName	LastName	Salary	DOB	Gender
001	Jasmine	Chen	25 000	25/12/2000	F
002	Kenton	Archer	20 000	01/04/1993	M
003	Michael	Roux	10 000	10/03/1990	M
004	Conrad	Slavorski	15 000	30/03/1989	M

Each employee is paid a salary in dollars (\$).

Write a Data Manipulation Language (DML) statement to return the first name, last name and salary of all employees that are paid more than \$17 500.

[3]

35. M/J 20/P11/Q7/a,b,c,d,e

- 7 A driving school teaches people how to drive cars. The school has a relational database, DRIVING_SCHOOL, to store information about instructors, students, lessons and the cars used by instructors.

INSTRUCTOR(InstructorID, FirstName, LastName, DateOfBirth, Level)

CAR(Registration, Make, Model, EngineSize)

INSTRUCTOR_CAR(InstructorID, Registration)

STUDENT(StudentID, FirstName, LastName, DateOfBirth, Address1)

LESSON(LessonID, StudentID, InstructorID, LessonDate, LessonTime)

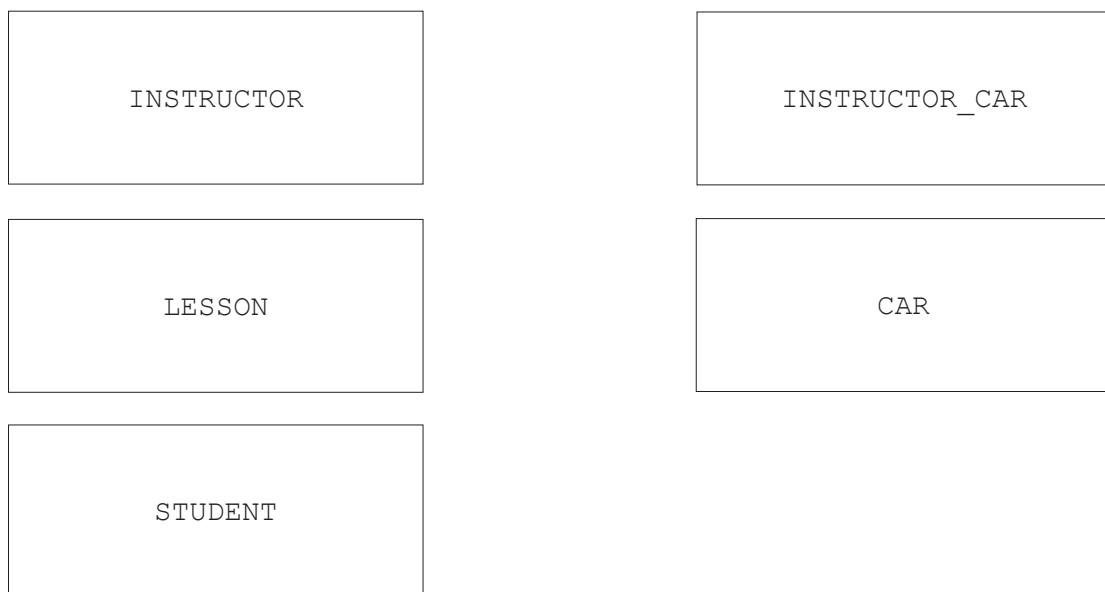
- (a) Give **two** benefits to the **driving school** of using a relational database instead of a flat file.

1

2

[2]

- (b) Complete the entity-relationship diagram for the database DRIVING_SCHOOL.



[4]

- (c) The table shows some sample data for the table INSTRUCTOR.

InstructorID	FirstName	LastName	DateOfBirth	Level
Ins01	Jayden	Han	05/06/1974	1
Ins02	Freda	Choi	06/02/1978	2
Ins03	Kelly	Kim	01/12/1966	1
Ins04	Santana	Thompson	09/09/1985	3

Complete the Data Definition Language (DDL) statement to create the table `INSTRUCTOR`.

```
..... TABLE INSTRUCTOR (

    InstructorID VARCHAR(5),

    FirstName VARCHAR(15),

    LastName VARCHAR(15),

    DateOfBirth DATE,

    Level ..... ,

    ..... (InstructorID)

);
```

[3]

- (d) The table `STUDENT` needs an additional field to store the student's telephone number, for example 012-3456.

Write a Data Definition Language (DDL) statement to add the new field to the table `STUDENT`.

[2]

- (e) Write a Data Manipulation Language (DML) statement to return the date and time of all future lessons booked with the instructor whose `InstructorID` is `Ins01`.

[4]

36. M/J 20/P12/Q6/a,b,c

- 6 A software development company has a relational database, `SOFTWARE_MANAGEMENT`. The database stores details of the customers who have purchased software, as well as the software and licences that customers have purchased.

The `SOFTWARE_MANAGEMENT` database has the following tables:

```
CUSTOMER_DETAILS (CustomerID, CompanyName, Address1, Address2, City)
```

```
SOFTWARE_PURCHASED (SoftwareName, SoftwareDescription, CustomerID,
                    LicenceType, LicenceCost, RenewalDate)
```

- (a) Explain why this database is **not** in Third Normal Form (3NF). Refer to the tables in your answer.

Do **not** attempt to normalise the tables.

[2]

- (b) Give an example from the database `SOFTWARE_MANAGEMENT` for each of the following database terms.

Term	Example
Entity	
Foreign key	
Attribute	

[3]

- (c) The company also develops computer games. They extend the relational database `SOFTWARE_MANAGEMENT` by adding a new table. The new table, `GAME_DEVELOPMENT`, stores details about the games and the software development teams creating them.

The table shows example data in `GAME_DEVELOPMENT`.

GameName	Genre	TeamNumber	DevelopmentStage	ManagerID
Bunny Hop	Platform	4	Analysis	23KP
Fried Eggs	Retro	2	Programming stage 1	9RTU
Create-a-game	Action	1	Acceptance testing	11TF

- (i) Complete the Data Definition Language (DDL) statement to create the table `GAME_DEVELOPMENT`.

```
CREATE ..... (
    GameName VarChar,
    Genre VarChar,
    ..... ,
    DevelopmentStage VarChar,
    ManagerID VarChar,
    ..... (GameName)
);
```

[5]

- (ii) Another table, `PRODUCT_MANAGER`, is created.

```
PRODUCT_MANAGER(ManagerID, FirstName, LastName)
```

Complete the Data Manipulation Language (DML) statement to return the game name, genre and team number of all games managed by the product manager with the first name 'James' and the last name 'Fitz'.

```

..... GameName, Genre, TeamNumber

FROM GAME_DEVELOPMENT, PRODUCT_MANAGER

WHERE PRODUCT_MANAGER.FirstName = "James"

AND PRODUCT_MANAGER.LastName = "Fitz"

AND .....

= ..... ;

```

[3]

37. M/J 20/P13/Q6/a,b,c

6 Sheila creates a relational database for her hotel using a Database Management System (DBMS).

(a) Draw **one** line from each database term to its most appropriate description.

Database Term	Description
Primary key	A field in one table that links to a primary key in another table
Attribute	A collection of records and fields
Foreign key	The type of data that is being stored
Entity	A unique identifier for each tuple
	A data item, represented as a field within a table
	The concept or object in the system that we want to model and store information about

[4]

(b) Identify **three** tasks that Sheila can perform using the DBMS developer interface.

1

2

3

[3]

(c) Sheila creates the database `HOTEL` with the following table structure:

`ROOM (RoomNumber, RoomType)`

`BOOKING (BookingID, RoomNumber, CustomerID, StartDate)`

`CUSTOMER (CustomerID, FirstName, LastName, Address, Tel_Num)`

(i) The following table shows some sample data for the table `ROOM`.

RoomNumber	RoomType
1	Standard
2	Double
3	Executive
4	Standard

Complete the Data Definition Language (DDL) statement to create the table `ROOM`.

```
..... TABLE ROOM (
    RoomNumber Integer,
    RoomType .....,
    ..... (RoomNumber)
);
```

[3]

(ii) Room number **5** is a **Double** room.

Complete the Data Manipulation Language (DML) statement to add the details for room number 5 to the table `ROOM`.

```
INSERT ..... ROOM
VALUES (.....);
```

[2]

(iii) The table `BOOKING` needs an additional field to store the number of nights (for example, 3) a customer is staying.

Write a Data Definition Language (DDL) statement to add the new field to the table `BOOKING`.

[2]

38. O/N 19/P11/Q4/d

4 Customers of a bank can access their account information by logging in on the bank's website.

(d) The bank uses a relational database, `ACCOUNTS`, to store the information about customers and their accounts.

The database stores the customer's first name, last name and date of birth.

The bank has several different types of account. Each account type has a unique ID number, name (for example, regular or saving) and bonus (for example, \$5.00, \$10.00 or \$15.00).

A customer can have more than one account.

Each customer's account has its own ID number and stores the amount of money the customer has in that account.

The bank creates a normalised, relational database to store the required information. There are three tables:

- CUSTOMER
- ACCOUNT_TYPE
- CUSTOMER_ACCOUNT

(i) Write the attributes for each table to complete the database design for the bank.

CUSTOMER (.....
)
 ACCOUNT_TYPE (.....
)
 CUSTOMER_ACCOUNT (.....
)

[3]

(ii) Identify the primary key for each table that you designed in **part (d)(i)**.

CUSTOMER
 ACCOUNT_TYPE
 CUSTOMER_ACCOUNT

[2]

(iii) Identify **one** foreign key in one of the tables that you designed in **part (d)(i)**.

Table name

Foreign key

[1]

(iv) The following table has definitions of database terms.

Write the correct database term in the table for each definition.

Definition	Term
All the data about one entity	
The data in one row of a table	
A column or field in a table	

[3]

39. O/N 19/P12/Q4/a,b

4 Anushka needs to store information about bookings at a sports club.

(a) Anushka has a file-based storage system. She wants a relational database.

- (i) Describe the features of a relational database that address the limitations of Anushka's file-based system.

[4]

- (ii) The relational database design needs to be normalised. The following statements describe the three stages of database normalisation.

Complete the statements by filling in the missing words.

For a database to be in First Normal Form (1NF) there must be no groups of attributes.

For a database to be in Second Normal Form (2NF), it must be in 1NF, and contain no key dependencies.

For a database to be in Third Normal Form (3NF), it must be in 2NF, and all attributes must be fully dependent on the

[4]

(b) The **normalised** relational database, `SPORTS_CLUB`, has the following table design.

`MEMBER`(MemberID, FirstName, LastName, MembershipType)

`SESSION`(SessionID, Description, SessionDate, SessionTime, NumberMembers)

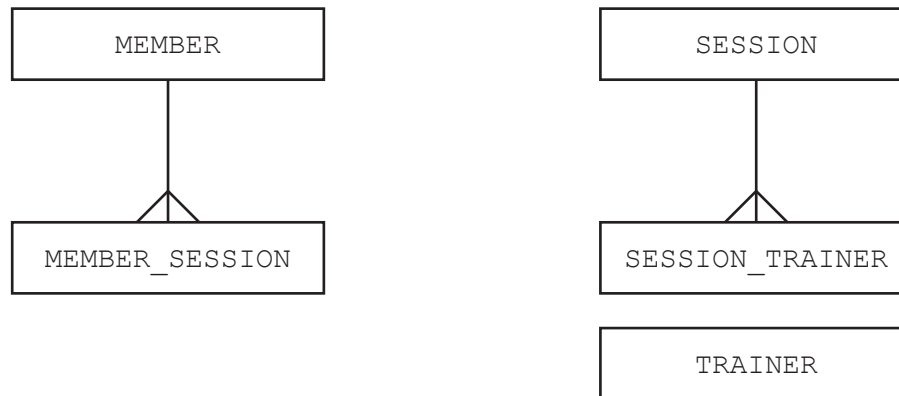
`TRAINER`(TrainerID, TrainerFirstName, TrainerLastName)

`MEMBER_SESSION`(MemberID, SessionID)

`SESSION_TRAINER`(SessionID, TrainerID)

- (i) Anushka has designed an entity-relationship (E-R) diagram for `SPORTS_CLUB`.

Complete the entity-relationship (E-R) diagram.



[2]

- (ii) Anushka first needs to create the database that she has designed.

Write a Data Definition Language (DDL) statement to create the `SPORTS_CLUB` database.

[1]

- (iii) The table shows some sample data for the table `SESSION`.

SessionID	Description	SessionDate	SessionTime	NumberMembers
21PL	Pilates junior	04/04/2020	18:00	15
13AE	Aerobics senior	04/04/2020	19:00	20
33WG	Weightlifting advanced	04/04/2020	10:00	10

Write a DDL script to create the table `SESSION`.

[5]

- (iv) Write a Data Manipulation Language (DML) script to return the first name and last name of all members who have `Peak` membership type.

[3]

40. O/N 19/P13/Q3/a,b,c

- 3 A hotel needs to record information about customers and their bookings.

- (a) The hotel has two types of room: double and family. Each room has a unique room number.

The hotel stores information about the customers including their name, address and contact details.

When a customer books a room, they give the start date and the number of nights they want to stay. If a customer wants more than one room, each room must have a separate booking. Each booking has an ID number.

The hotel creates a normalised, relational database to store the required information. There are three tables:

- CUSTOMER
- ROOM
- BOOKING

(i) Complete the database design for the hotel by writing the attributes for each table.

CUSTOMER (.....
)

ROOM (.....
)

BOOKING (.....
)

[3]

(ii) Identify the primary key for each table that you designed in **part (a)(i)**.

CUSTOMER

ROOM

BOOKING

[2]

(iii) Identify **one** foreign key in the tables that you designed in **part (a)(i)**.

Table name

Foreign key

[1]

(b) The hotel wants to use a Database Management System (DBMS) to set up and manage the database.

Describe, using examples, how the hotel can use the following DBMS tools:

Developer interface

Query processor

[5]

(c) The following table has four SQL scripts.

Tick (✓) **one** box in each row to identify whether the script is an example of a Data Definition Language (DDL) statement or a Data Manipulation Language (DML) statement.

Script	DDL	DML
CREATE TABLE FILMS		
SELECT FilmID FROM FILMS		
ALTER TABLE FILMS ADD PRIMARY KEY (FilmID)		
CREATE DATABASE MYDATA		

[2]

41. M/J 19/P11/Q2/b(i),c

2 A software company produces software and distributes it under different software licences.

(b) The software company stores information about customers and the software licences they have purchased. The company considers a file-based approach for the storage and retrieval of data.

(i) Give **three** limitations of a file-based approach to store the data.

1

2

3

[3]

(c) The database has the following tables:

CUSTOMER(CustomerID, CompanyName)

SOFTWARE(SoftwareID, SoftwareName, OperatingSystem, Description)

LICENCE(LicenceID, CustomerID, SoftwareID, DateOfPurchase,
LicenceType, Cost, ExpiryDate)

(i) Identify the type of relationship that exists between the tables CUSTOMER and LICENCE.

[1]

(ii) Describe how the relationship is created between the tables CUSTOMER and LICENCE.

[2]

(iii) The company needs a list of all software licences that have an expiry date on or before 31/12/2019.

Write an SQL query to return the fields CustomerID, SoftwareID, LicenceType, Cost and ExpiryDate for all licences that expire on, or before 31/12/2019. Group the output by CustomerID, and in ascending order of cost.

[5]

42. M/J 19/P12/Q5/a,b(ii,iii)

5 Moheem is creating a relational database to store data about his customers.

(a) Moheem has been told a relational database addresses some of the limitations of a file-based approach by reducing data redundancy.

(i) State what is meant by the term **data redundancy**.

[1]

(ii) Explain **how** a relational database can help to reduce data redundancy.

[3]

(b) Moheem uses a Database Management System (DBMS) to ensure the security and integrity of the data.

(ii) Name **and** describe **two** security features provided by a DBMS.

Feature 1

Feature 2

[4]

(iii) The DBMS provides software tools for the database developer.

Fill in the names of the missing software tools in the following statements.

A allows a developer to extract data from a database.

A enables a developer to create user-friendly forms and reports.

[2]

43. M/J 19/P13/Q3/a,b,c,d,e

3 A company uses a relational database, `EMPLOYEES`, to store data about its employees and departments.

(a) The company uses a Database Management System (DBMS).

(i) The DBMS has a data dictionary.

Describe what the data dictionary stores.

[2]

(ii) The DBMS has a query processor.

Describe the purpose of a query processor.

[2]

(b) Relationships are created between tables using primary and foreign keys.

Describe the role of a primary and a foreign key in database relationships.

[2]

(c) In the company:

- An employee can be a manager.
- A department can have several managers and several employees.
- An employee can only belong to one department.

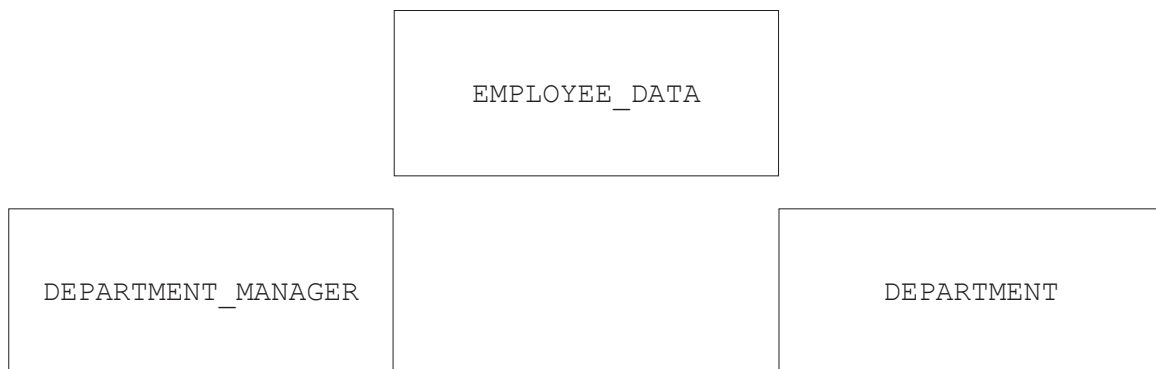
The EMPLOYEES database has three tables:

EMPLOYEE_DATA(EmployeeID, FirstName, LastName, DateOfBirth, Gender, DepartmentNumber)

DEPARTMENT(DepartmentNumber, DepartmentName)

DEPARTMENT_MANAGER(DepartmentNumber, EmployeeID, role)

Complete the entity-relationship (E-R) diagram for the EMPLOYEES database.



[3]

(d) Give **three** reasons why the EMPLOYEES database is fully normalised.

- 1
- 2
- 3

[3]

(e) Part of the EMPLOYEE_DATA table is shown.

EmployeeID	FirstName	LastName	DateOfBirth	Gender	DepartmentNumber
156FJEK	Harvey	Kim	12/05/1984	Male	S1
558RRKL	Catriona	Moore	03/03/1978	Female	F2
388LMDV	Oscar	Ciao	01/01/1987	Male	F2

(i) Write a Data Definition Language (DDL) statement to create the EMPLOYEES database.

[1]

- (ii) Write a DDL statement to define the table `EMPLOYEE_DATA`, and declare `EmployeeID` as the primary key.

[5]

- (iii) Write a Data Manipulation Language (DML) statement to return the first name and last name of all female employees in the department named Finance.

[5]

44. O/N 18/P11/Q7/a,b,c,d

- 7 A movie theatre has a relational database that stores the movie schedule, and information about the movies. The theatre has several screens that play movies at the same time.

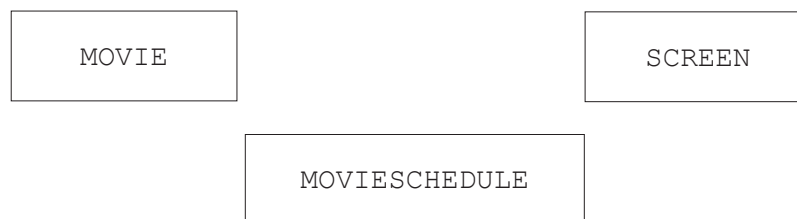
The database has three tables to store information about the movies, the screens and the movie schedule.

```
MOVIE(MovieID, Title, Length, Rating)
```

```
SCREEN(ScreenNumber, NumberSeats)
```

```
MOVIESCHEDULE(ScheduleID, MovieID, ScreenNumber, Time)
```

- (a) Complete the entity-relationship (E-R) diagram to show the relationships between these tables.



[2]

- (b) Explain how primary and foreign keys are used to link the tables in the movie theatre database.

[4]

- (c) The database needs to store the name of the company that produced each movie, for example, *Rocking Movies*.

Write an SQL script to add the attribute `ProductionCompany` to the `MOVIE` table.

[2]

- (d) Write an SQL script to display the title and rating of all movies scheduled to play on screen number 3.

[4]

45. M/J 18/P12/Q7/a,b,c,d

- 7** A social media website has a relational database, `WEBDATA`, that stores the site's information.

The database has three tables to store users' details, and details of the images and text that they post.

`USER`(`UserName`, `FirstName`, `SecondName`, `DateOfBirth`)

`PHOTO`(`PhotoID`, `UserName`, `Comment`, `UploadDate`)

`TEXTPOST`(`PostID`, `UserName`, `DateOfPost`, `TheText`)

- (a) (i)** Explain how the relationship between the tables `USER` and `PHOTO` has been implemented.

[2]

- (ii)** Draw the entity-relationship (E-R) diagram to show the relationships between the three tables.

[2]

- (b)** A database administrator decides to enforce referential integrity.

Use an example from the database `WEBDATA` to explain what is meant by **referential integrity**.

[3]

- (c)** The database has been normalised to Third Normal Form (3NF).

Define the three stages of database normalisation.

1NF

2NF

3NF

[3]

- (d)** The following shows sample data from the `USER` table.

<code>UserName</code>	<code>FirstName</code>	<code>SecondName</code>	<code>DateOfBirth</code>
gem123	John	Smith	01/01/1995
purpleSky	Muhammed	Ali	23/02/1956
OpenWindow	Sunny	Amir	03/03/1997
bluebird127	Raziya	Bello	04/03/1982

- (i)** Write an SQL script to create the `USER` table.

[5]

- (ii) The database administrator needs to alter the `USER` table. A new field, `Country`, needs to be added.

Write an SQL script to add the field `Country` to the `USER` table.

[2]

46. M/J 18/P13/Q2/a,b,c,d

- 2 A company writes applications (apps) for smartphones. The company has a relational database, `PURPLEGAME`, which stores the information for one of its online game apps.

The database has three tables to store player's details, dates when they have logged into the app and in-app purchase details.

`LOGIN(LoginID, PlayerID, Date)`

`PURCHASE(PurchaseID, PlayerID, PurchaseDate, Cost)`

`PLAYER(PlayerID, PlayerName, SkillLevel)`

- (a) Draw the entity-relationship (E-R) diagram to show the relationships between the three tables.

[2]

- (b) The database manager is concerned about data integrity.

State what is meant by **data integrity**. Give an example of how the manager can ensure data integrity in the `PURPLEGAME` database.

[2]

- (c) The database designer states that the `PURPLEGAME` database is in Third Normal Form (3NF).

Tick (✓) **one** box to indicate whether this statement is true or false.

True	False

Justify your choice.

[3]

- (d) (i) The following table shows some sample data for the `PLAYER` table.

PlayerID	PlayerName	SkillLevel
fly918	Kylie	3
elephant11	Mehrdad	9
candy22	Suzi	15
greenGrass	Jason	22

Write an SQL script to create the `PLAYER` table.

[5]

(ii) The table, `PLAYER`, needs to be altered. A new field, `DateOfBirth`, needs to be added.

Write an SQL script to add the `DateOfBirth` field to the `PLAYER` table.

[2]

47. O/N 17/P12/Q7/a,b,c,d,e

7 A company takes customer service for its clients very seriously.

The client

- The client names are unique.

A visit

- The company arranges a date for a visit to gather feedback from a client.
- A visit to a client never takes more than one day.
- Over time, the client receives many visits.

Staff (Interviewers)

- One or more staff attend the visit.
- If there is more than one staff member visiting, each performs a separate interview.

Interviews

- Each interview is classified as either 'general' or by some specialism, for example, marketing, customer service or sales.
- A report is produced for each interview, `InterviewText`.
- Each interview is conducted by a single staff member.

The client, visit, staff and interview data will be stored in a relational database.

(a) (i) Underline the primary key for each table in the following suggested table designs.

`STAFF(StaffID, StaffName, Department)`

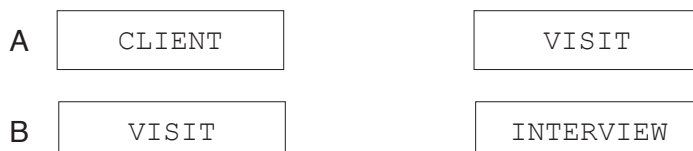
`CLIENT(ClientName, Address, Town)`

`VISIT(ClientName, VisitDate)`

`INTERVIEW(ClientName, VisitDate, StaffID, SpecialistFocus, InterviewText)`

[3]

(ii) For each of the pairs of entities, A, B and C, draw the relationship between the two entities.



C

INTERVIEW

STAFF

[3]

- (b) The company decides to produce a visit report, `VisitReportText`, for each visit made.

This text will be produced from the one or more interview texts obtained at the visit.

State how one or more of the given table designs can be changed to add this attribute.

[1]

- (c) Client ABC Holdings are now trading under the name of Albright Holdings.

- (i) Write an SQL script to update this client's record in the database.

UPDATE

SET

WHERE

[3]

- (ii) Describe why this update could cause problems with the existing data stored.

[2]

- (d) Write an SQL script to display the Staff ID of each member of staff who performed an interview when they visited New Age Toys on 13/10/2016.

[3]

- (e) At present, all interviews are performed in the UK. Many clients now operate in other countries in Europe. The company wants to perform interviews with the client's staff in other countries. Not all interview staff are willing to travel outside of the UK.

State how one or more of the table designs should be revised to store this information.

[1]

48. O/N 17/P13/Q7/a,b,c,d,e

- 7 A clinic is staffed by several doctors. The clinic serves thousands of patients. Each day and at any one time, there is only one doctor in the clinic available for appointments.

The clinic stores patient, doctor and appointment data in a relational database.

- (a) (i) Underline the primary key for each table in the following suggested table designs.

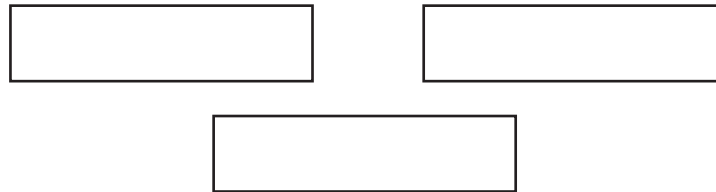
PATIENT(PatientID, PatientName, Address, Gender)

DOCTOR(DoctorID, Gender, Qualification)

APPOINTMENT(AppointmentDate, AppointmentTime, DoctorID, PatientID)

[2]

- (ii) Complete the following entity-relationship (E-R) diagram for this design.



[2]

- (b) The doctors are concerned that many patients make appointments but do not attend them.

Describe the changes to the table designs that could be made to store this information.

[2]

- (c) The doctors are about to set up a new clinic in the neighbouring village, SITE-B.

The original location is identified as SITE-A.

A new table is designed to store the ID of the doctor who is able to work at each site.

DOCTOR-AVAILABILITY (DoctorID, Site)

Five entries stored in the table are:

DoctorID	Site
098	SITE-A
074	SITE-A
117	SITE-B
098	SITE-B
033	SITE-B

- (i) State what this data shows about the availability of the doctor with the ID of 098.

[1]

- (ii) Opening a new clinic in the neighbouring village will not require any additional table for storing appointments. It will need a change to the existing appointment table design.

Show the revised APPOINTMENT table.

APPOINTMENT (.....

) [1]

- (d) The doctor with the ID of 117 has recently been allocated a new DoctorID of 017.

- (i) Write an SQL script to update this doctor's record in the database.

UPDATE

SET

WHERE

[3]

- (ii) Describe why this update could cause problems with the existing data stored.

[2]

- (e) Write an SQL script to display the date and time of all appointments made by the patient with the `PatientID` of 556.

[3]

49. M/J 17/P11/Q1/a,b,c,d

- 1 A hospital is divided into two areas, Area A and Area B. Each area has several wards. All the ward names are different.

A number of nurses are based in Area A. These nurses always work on the same ward. Each nurse has a unique Nurse ID of `STRING` data type.



- (a) Describe the relationship shown above.

[1]

- (b) A relational database is created to store the ward and nurse data. The two table designs for Area A are:

A-WARD (WardName, NumberOfBeds)

A-NURSE (NurseID, FirstName, FamilyName,

- (i) Complete the design for the A-NURSE table.

[1]

- (ii) Explain how the relationship in **part (a)** is implemented.

[2]

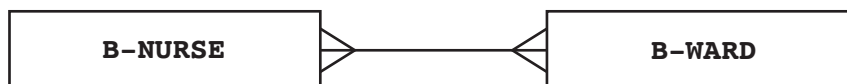
- (c) In Area B of the hospital, there are a number of wards and a number of nurses.

Each Area B ward has a specialism.

Each Area B nurse has a specialism.

A nurse can be asked to work in any of the Area B wards where their specialism matches with the ward specialism.

The relationship for Area B of the hospital is:



- (i) Explain what the degree of relationship is between the entities B-NURSE and B-WARD.

[1]

- (ii) The design for the Area B data is as follows:

B-NURSE (NurseID, FirstName, FamilyName, Specialism)

B-WARD (WardName, NumberOfBeds, Specialism)

B-WARD-NURSE (.....)

Complete the attributes for the third table. Underline its primary key.

[2]

- (iii) Draw the relationships on the entity-relationship (E-R) diagram.



[2]

- (d) Use the table designs in **part (c)(ii)**.

- (i) Write an SQL query to display the Nurse ID and family name for all Area B nurses with a specialism of 'THEATRE'.

[3]

- (ii) Fatima Woo is an Area B nurse with the nurse ID of 076. She has recently married, and her new family name is Chi.

Write an SQL command to update her record.

UPDATE

SET

WHERE

[3]

50. M/J 17/P12/Q1/a,b,c

- 1 Some shops belong to the Rainbow Retail buying group. They buy their goods from one or more suppliers.

Each shop has:

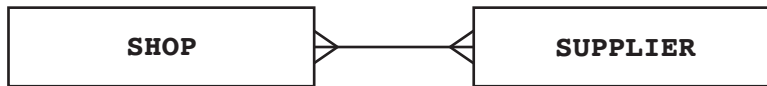
- a unique shop ID
- a single retail specialism (for example, food, electrical, garden).

Each supplier has:

- a unique supplier ID
- a similar single specialism recorded.

Rainbow Retail creates a relational database to record data about the shops and their suppliers.

The entity-relationship (E-R) diagram for the relationship between the `SHOP` and `SUPPLIER` tables is shown.



(a) Explain what the degree of relationship is between the entities `SHOP` and `SUPPLIER`.

[1]

The database design is as follows:

`SHOP(ShopID, ShopName, Location, RetailSpecialism)`

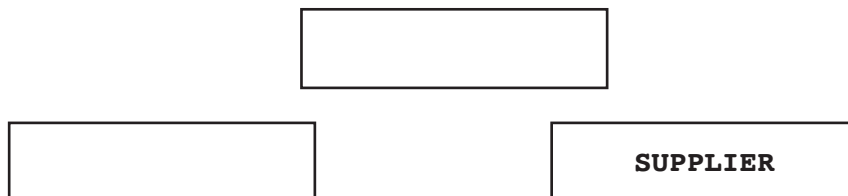
`SUPPLIER(SupplierID, SupplierName, ContactPerson, RetailSpecialism)`

`SHOP-SUPPLIER(ShopID, SupplierID)`

The `SHOP-SUPPLIER` table stores the suppliers that each shop has previously used.

Primary keys are not shown.

(b) (i) Label the entities and draw the relationships to complete the revised E-R diagram.



[3]

(ii) Complete the following table to show for each database table:

- the primary key
- the foreign key(s) (if any):
 - Each table may contain none, one or more foreign key(s).
 - For a table with no foreign key, write 'None'.
- an explanation for the use of any foreign key.

Table	Primary key	Foreign key(s) (if any)	Explanation
SHOP			
SUPPLIER			
SHOP-SUPPLIER			

[5]

- (iii) The database designer has implemented `SUPPLIER.ContactPerson` as a secondary key.

Describe the reason for this.

[2]

- (c) (i) Write an SQL query to display the shop ID and location of all shops with a 'GROCERY' specialism.

[3]

- (ii) The existing shop with ID 8765 has just used the existing supplier SUP89 for the first time.

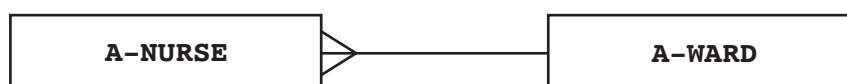
Write an SQL script to add this data to the database.

[3]

51. M/J 17/P13/Q1/a,b,c,d

- 1 A hospital is divided into two areas, Area A and Area B. Each area has several wards. All the ward names are different.

A number of nurses are based in Area A. These nurses always work on the same ward. Each nurse has a unique Nurse ID of `STRING` data type.



- (a) Describe the relationship shown above.

[1]

- (b) A relational database is created to store the ward and nurse data. The two table designs for Area A are:

A-WARD (WardName, NumberOfBeds)

A-NURSE (NurseID, FirstName, FamilyName,)

- (i) Complete the design for the A-NURSE table. [1]
- (ii) Explain how the relationship in **part (a)** is implemented. [2]

- (c) In Area B of the hospital, there are a number of wards and a number of nurses.

Each Area B ward has a specialism.

Each Area B nurse has a specialism.

A nurse can be asked to work in any of the Area B wards where their specialism matches with the ward specialism.

The relationship for Area B of the hospital is:



- (i) Explain what the degree of relationship is between the entities B-NURSE and B-WARD. [1]

- (ii) The design for the Area B data is as follows:

B-NURSE (NurseID, FirstName, FamilyName, Specialism)

B-WARD (WardName, NumberOfBeds, Specialism)

B-WARD-NURSE (.....)

Complete the attributes for the third table. Underline its primary key. [2]

- (iii) Draw the relationships on the entity-relationship (E-R) diagram.



- (d) Use the table designs in **part (c)(ii)**. [2]

- (i) Write an SQL query to display the Nurse ID and family name for all Area B nurses with a specialism of 'THEATRE'. [3]

- (ii) Fatima Woo is an Area B nurse with the nurse ID of 076. She has recently married, and her new family name is Chi.

Write an SQL command to update her record.

UPDATE

SET

WHERE

[3]

52. O/N 16/P11/Q1/a,b

- 1 (a) Five descriptions and seven relational database terms are shown below.

Draw a line to link each description to its correct database term.

Description	Database term
Any object, person or thing about which it is possible to store data	Secondary key
Dataset organised in rows and columns; the columns form the structure and the rows form the content	Candidate key
Any attribute or combination of attributes that can act as a unique key	Entity
Attribute(s) in a table that link to the primary key in another table to form a relationship	Foreign key
Attribute or combination of attributes that is used to uniquely identify a record	Primary key
	Table
	Tuple

[5]

- (b) Explain what is meant by referential integrity.

[3]

53. O/N 16/P12/Q9/a,b,c

9 A health club offers classes to its members. A member needs to book into each class in advance.

- (a)** The health club employs a programmer to update the class booking system. The programmer has to decide how to store the records. The choice is between using a relational database or a file-based approach.

Give **three** reasons why the programmer should use a relational database.

1

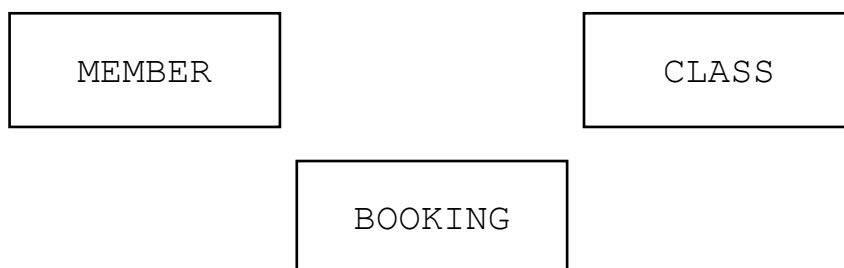
2

3

[6]

- (b)** The programmer decides to use three tables: `MEMBER`, `BOOKING` and `CLASS`.

Complete the Entity-Relationship (E-R) diagram to show the relationships between these tables.



[2]

- (c)** The `CLASS` table has primary key `ClassID` and stores the following data:

ClassID	Description	StartDate	ClassTime	NoOfSessions	AdultsOnly
DAY01	Yoga beginners	12/01/2016	11:00	5	TRUE
EVE02	Yoga beginners	12/01/2016	19:00	5	FALSE
DAY16	Circuits	30/06/2016	10:30	4	FALSE

Write an SQL script to create the `CLASS` table.

[6]

54. O/N 16/P13/Q1/a,b

- 1 (a)** Five descriptions and seven relational database terms are shown below.

Draw a line to link each description to its correct database term.

Description	Database term
Any object, person or thing about which it is possible to store data	Secondary key
Dataset organised in rows and columns; the columns form the structure and the rows form the content	Candidate key
Any attribute or combination of attributes that can act as a unique key	Entity
Attribute(s) in a table that link to the primary key in another table to form a relationship	Foreign key
Attribute or combination of attributes that is used to uniquely identify a record	Primary key
	Table
	Tuple

[5]

- (b)** Explain what is meant by referential integrity.

[3]

55. M/J 16/P11/Q8/a,b

- 8** A school stores a large amount of data. This includes student attendance, qualification, and contact details. The school's software uses a file-based approach to store this data.

(a) The school is considering changing to a DBMS.

(i) State what DBMS stands for.

[1]

(ii) Describe **two** ways in which the Database Administrator (DBA) could use the DBMS software to ensure the security of the student data.

1

2

[4]

(iii) A feature of the DBMS software is a query processor.

Describe how the school secretary could use this software.

[2]

(iv) The DBMS has replaced software that used a file-based approach with a relational database.

Describe how using a relational database has overcome the previous problems associated with a file-based approach.

[3]

(b) The database design has three tables to store the classes that students attend.

STUDENT (StudentID, FirstName, LastName, Year, TutorGroup)

CLASS (ClassID, Subject)

CLASS-GROUP (StudentID, ClassID)

Primary keys are not shown.

There is a one-to-many relationship between **CLASS** and **CLASS-GROUP**.

(i) Describe how this relationship is implemented.

[2]

(ii) Describe the relationship between **CLASS-GROUP** and **STUDENT**.

[1]

(iii) Write an SQL script to display the `StudentID` and `FirstName` of all students who are in the tutor group 10B. Display the list in alphabetical order of `LastName`.

[4]

(iv) Write an SQL script to display the `LastName` of all students who attend the class whose `ClassID` is CS1.

[4]

56. M/J 16/P12/Q8/a,b

- 8** A school stores a large amount of data. This includes student attendance, qualification, and contact details. The school's software uses a file-based approach to store this data.

(a) The school is considering changing to a DBMS.

(i) State what DBMS stands for.

[1]

(ii) Describe **two** ways in which the Database Administrator (DBA) could use the DBMS software to ensure the security of the student data.

1

2

[4]

(iii) A feature of the DBMS software is a query processor.

Describe how the school secretary could use this software.

[2]

(iv) The DBMS has replaced software that used a file-based approach with a relational database.

Describe how using a relational database has overcome the previous problems associated with a file-based approach.

[3]

(b) The database design has three tables to store the classes that students attend.

STUDENT (StudentID, FirstName, LastName, Year, TutorGroup)

CLASS (ClassID, Subject)

CLASS-GROUP (StudentID, ClassID)

Primary keys are not shown.

There is a one-to-many relationship between **CLASS** and **CLASS-GROUP**.

(i) Describe how this relationship is implemented.

[2]

(ii) Describe the relationship between **CLASS-GROUP** and **STUDENT**.

[1]

(iii) Write an SQL script to display the `StudentID` and `FirstName` of all students who are in the tutor group 10B. Display the list in alphabetical order of `LastName`.

[4]

(iv) Write an SQL script to display the `LastName` of all students who attend the class whose `ClassID` is CS1.

[4]

57. M/J 16/P13/Q5/a,b,c,d

5 (a) A Database Management System (DBMS) provides the following features.

Draw a line to match each feature with its description.

Feature**Description**

Data dictionary

A file or table containing all the details of the database design

Data security

Data design features to ensure the validity of data in the database

Data integrity

A model of what the database will look like, although it may not be stored in this way

Methods of protecting the data including the uses of passwords and different access rights for different users of the database

[3]

A school stores a large amount of data that includes student attendance, qualification and contact details. The school is setting up a relational database to store these data.

(b) The school needs to safeguard against any data loss.

Describe **three** factors to consider when planning a backup procedure for the data.

Justify your decisions.

1

2

3

[6]

(c) The database design has three tables to store the qualifications and grades each student has attained. The following is a sample of the data from each table.

STUDENT

StudentID	FirstName	LastName	Tutor
001AT	Ahmad	Tan	11A
003JL	Jane	Li	11B
011HJ	Heather	Jones	10A

QUALIFICATION

QualCode	Level	Subject
CS1	IGCSE	Computer Science
MT9	IGCSE	Maths
SC12	IGCSE	Science

STUDENT-QUALIFICATION

QualCode	StudentID	Grade	DateOfAward
SC12	011HJ	A	31/8/2014
SC12	003JL	C	31/8/2014
CS1	003JL	B	31/8/2014

- (i) Draw an Entity-Relationship (E-R) diagram to show the relationships between these three tables.

[2]

- (ii) State the type of relationship that exists between **STUDENT** and **STUDENT-QUALIFICATION**.

[1]

- (iii) Describe how the relationship between **QUALIFICATION** and **STUDENT-QUALIFICATION** is implemented.

[2]

- (d) (i) The database will store each student's date of birth.

Write an SQL script to add a date of birth attribute to the appropriate table.

[2]

- (ii) Write an SQL script to display the **StudentID**, **Grade** and **DateOfAward** for the **QualCode** value of SC12.

[3]

- (iii) Write an SQL script to display the `FirstName` and `LastName` and `QualCode` for all `STUDENT-QUALIFICATIONS` for which the `Grade` value is `A`.

[4]

58. M/J 15/P11/Q9/a,b,c

- 9 A database has been designed to store data about salespersons and the products they have sold.

The following facts help to define the structure of the database:

- each salesperson works in a particular shop
- each salesperson has a unique first name
- each shop has one or more salespersons
- each product which is sold is manufactured by one company only
- each salesperson can sell any of the products
- the number of products that each salesperson has sold is recorded

The table `ShopSales` was the first attempt at designing the database.

FirstName	Shop	ProductName	NoOfProducts	Manufacturer
Nick	TX	television set	3	SKC
		refrigerator	2	WP
		digital camera	6	HKC
Sean	BH	hair dryer	1	WG
		electric shaver	8	BG
John	TX	television set	2	SKC
		mobile phone	8	ARC
		digital camera	4	HKC
		toaster	3	GK

- (a) State why the table is **not** in First Normal Form (1NF).

[1]

- (b) The database design is changed to:

`SalesPerson` (`FirstName`, `Shop`)

`SalesProducts` (`FirstName`, `ProductName`, `NoOfProducts`, `Manufacturer`)

Using the data given in the first attempt table (*ShopSales*), show how these data are now stored in the revised table designs.

Table: *SalesPerson*

FirstName	Shop

Table: *SalesProducts*

FirstName	ProductName	NoOfProducts	Manufacturer

[3]

(c) (i) A relationship between the two tables has been implemented.

Explain how this has been done.

[2]

(ii) Explain why the *SalesProducts* table is **not** in Third Normal Form (3NF).

[2]

(iii) Write the table definitions to give the database in 3NF.

[2]

59. M/J 15/P12/Q9/a,b,c

9 A database has been designed to store data about salespersons and the products they have sold.

The following facts help to define the structure of the database:

- each salesperson works in a particular shop
- each salesperson has a unique first name
- each shop has one or more salespersons
- each product which is sold is manufactured by one company only
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The table `ShopSales` was the first attempt at designing the database.

FirstName	Shop	ProductName	NoOfProducts	Manufacturer
Nick	TX	television set	3	SKC
		refrigerator	2	WP
		digital camera	6	HKC
Sean	BH	hair dryer	1	WG
		electric shaver	8	BG
John	TX	television set	2	SKC
		mobile phone	8	ARC
		digital camera	4	HKC
		toaster	3	GK

(a) State why the table is **not** in First Normal Form (1NF).

[1]

(b) The database design is changed to:

`SalesPerson` (FirstName, Shop)

`SalesProducts` (FirstName, ProductName, NoOfProducts, Manufacturer)

Using the data given in the first attempt table (*ShopSales*), show how these data are now stored in the revised table designs.

Table: *SalesPerson*

FirstName	Shop

Table: *SalesProducts*

FirstName	ProductName	NoOfProducts	Manufacturer

[3]

- (c) (i) A relationship between the two tables has been implemented.

Explain how this has been done.

[2]

- (ii) Explain why the *SalesProducts* table is **not** in Third Normal Form (3NF).

[2]

(iii) Write the table definitions to give the database in 3NF.

[2]

Answers Section

1. M/J 26/P11/Q4/a,b,c

4(a)	1 mark for each correct point - CUSTOMER BORROW is a one-to-many relationship // 1:M - TOY_INSTANCE to TOY is a many-to-one relationship // M:1 - Identifying the correct Primary Key and matching Foreign Key in both relationships - Stating how the keys form the relationship	4	
4(b)	1 mark for each correct point - Selecting the contact name, customer ID and totalling the number of days - needs all 3 - Using an appropriate name - Linking or joining the correct tables - Grouping the results Example 1: <pre>SELECT CUSTOMER.ContactFirstName, CUSTOMER.CustomerID, SUM(NumberDays) AS TotalDays FROM BORROW, CUSTOMER WHERE CUSTOMER.CustomerID = BORROW.CustomerID GROUP BY CUSTOMER.CustomerID;</pre> MP1 MP2 MP3 MP4 Example 2: <pre>SELECT CUSTOMER.ContactFirstName, CUSTOMER.CustomerID, SUM(NumberDays) AS TotalDays FROM BORROW, INNER JOIN CUSTOMER ON CUSTOMER.CustomerID = BORROW.CustomerID GROUP BY CUSTOMER.CustomerID;</pre> MP1 MP2 MP3 MP4	4	CustomerID needs the table reference Accept any correct method of joining the tables, but no mix and match between methods
4(c)(i)	1 mark for each correct statement - Altering the correct table - Adding an appropriately named field with an appropriate data type <pre>ALTER TABLE BORROW ADD Returned BOOLEAN;</pre> accept VARCHAR/CHAR(3)/BOOL	2	Award just 1 mark if statements are in the wrong order or if there are any additional statements Accept any sensible field name
4(c)(ii)	1 mark for each correct statement - Updating the correct table - Setting the additional field to the correct data - Where clause data type must match 4(c)(i) Example: <pre>UPDATE BORROW SET Returned = "Yes" WHERE BorrowID = "A2226";</pre>	3	Award a maximum of 2 marks if statements are in the wrong order or if there are any additional statements

2. M/J 26/P12/Q3/a,b,c

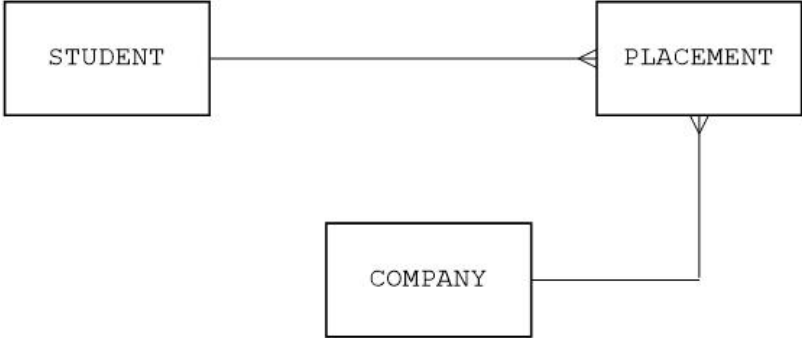
3(a)	1 mark for each correct relationship - The relationship between PLAYER and CHARACTER is one to many /1:M // one player can have many characters, but each character can only be created by one player - The relationship between GROUP and CHARACTER is many to many / M:M // one group can have many characters, and one character can be in many groups	2	The second option in each case needs both 'halves' Accept a m:m relationship between PLAYER and GROUP Accept user for player Mark the writing first, but accept accurate E-R diagram(s), if needed for clarification
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3(b)(i)	1 mark for each correct line number and the appropriate correction - Line 02 - CharacterID VARCHAR, // CharacterID VARCHAR (5), - Line 05 Level INT, - Line 08 - PRIMARY KEY (CharacterID) - Line 09 - FOREIGN KEY (PlayerID) REFERENCES PLAYER (PlayerID)	4	Ignore extra NOT NULL on line 02 but not on line 05 or line 08
3(b)(ii)	1 mark for each correct point - Selecting the player's name and counting a field - Using an appropriate name - Linking or joining the correct tables - Grouping the results - Ordering in descending order by player ID Example 1: <pre>SELECT PlayerName, COUNT(CharacterID) MP1 AS TotalCharacters MP2 FROM CHARACTER, PLAYER WHERE CHARACTER.PlayerID = PLAYER.PlayerID MP3 GROUP BY PlayerName MP4 ORDER BY PLAYER.PlayerID DESC; MP5</pre> Example 2: <pre>SELECT PlayerName, COUNT(CharacterID) MP1 AS TotalCharacters MP2 FROM CHARACTER INNER JOIN PLAYER ON CHARACTER.PlayerID = PLAYER.PlayerID MP3 GROUP BY PlayerName MP4 ORDER BY PLAYER.PlayerID DESC; MP5</pre>	5	Accept any correct method of joining the tables, but no mix and match between methods The only attribute that needs a table reference is the PlayerID. Ignore other table references as long as the correct table is used Take care with the DESC
3(c)	1 mark for two data dictionary examples For example: - data types of individual fields/attributes - type of validation - primary key - foreign key 1 mark for each remaining term (4 marks) - data dictionary - logical - access rights - developer interface The data dictionary stores data about the data in the database. The table names, field names and relationships will be stored. Two other examples that will be stored are example 1 and example 2 When designing a database, the logical schema is the design of the database tables, fields and relationships independent of any Database Management System (DBMS). A DBMS provides security features to protect the data. Access rights are used to give user accounts permission to read data, some to read and write data and others may be unable to read the data. When creating a database a developer interface provides tools to create visual components for example forms and reports.	5	Mark the first answer in each space Do not accept table names or field names or relationships, they are given in the question There are many answers for the examples, the list is suggestions only Do not accept metadata as an example or instead of data dictionary Just 'key' is NE for primary or foreign keys Just data type is TV for the second example

3. M/J 26/P13/Q4/a,b,c

4(a)	1 mark each 3 tables with suitable names e.g. Student, Book, Borrow - All tables given have a suitable PK - Book has fields: author, title, category, number of pages - Student has fields: FName LName, class - Borrow has fields: book PK, student PK, date borrowed, field to identify book is late - FKs in borrow are clearly identified as the Book PK and the Student PK e.g. Book(<u>BID</u>, Author, Title, Category, Pages) Student(<u>SID</u>, FName, LName, Class) Borrow(<u>BwlD</u>, <u>StudentID</u>, <u>BookID</u>, DateBorrowed, Late)	6	Allow composite primary key for Borrow table. Borrow table must contain sensible linking related fields Do not accept Title or Name as PKs. Allow alternatives to late e.g. DateBorrowed, DateDueBack, DateReturned.
4(b)	1 mark each to max 2 - Ensure consistency of data // up-to-date // not redundant data - Ensure relationships between data remain valid and prevent orphaned records - If a change is made in one place it is made everywhere else // cascading updates/deletes - Every foreign key has a corresponding primary key	2	
4(c)(i)	1 mark for each error line and correction - Line 01 CREATE DATABASE SCHOOLDATA; - Line 03 SubjectCode VARCHAR NOT NULL, - Line 04 SubjectName VARCHAR, - Line 07 FOREIGN KEY (DepartmentID) REFERENCES DEPARTMENT (DepartmentID)	4	Do not award mark for Line 04 if extra Not Null has been added
4(c)(ii)	1 mark for each correct point - Selecting department name and counting subject code ... - Using suitable field name - From statement where with join / other join method - Grouping by department name / department ID - Ordering by department name in ascending order Example 1: <pre> SELECT DepartmentName, COUNT(SubjectCode) (MP1) AS "NumberSubjectCodes" (MP2) FROM DEPARTMENT, SUBJECT WHERE DEPARTMENT.DepartmentID = SUBJECT.DepartmentID (MP3) GROUP BY (DepartmentID) (MP4) ORDER BY DepartmentName; (MP5) </pre> Example 2: <pre> SELECT DepartmentName, Count(Subject.SubjectCode) (MP1) AS CountOfSubjectCode (MP2) FROM DEPARTMENT (INNER) JOIN SUBJECT ON DEPARTMENT.DepartmentID = SUBJECT.DepartmentID (MP3) GROUP BY DEPARTMENT.DepartmentName (MP4) ORDER BY (DEPARTMENT.DepartmentName) ASC; (MP5) </pre> Please note that these two lines are equivalent to MP1 and MP2 <pre> SELECT DepartmentName, NumberSubjectCodes = Count(SUBJECT.subjectcode) </pre> MP1: SELECT DepartmentName, and a correct COUNT function MP2: CountOfSubject =	5	Accept == for = Accept any correct method of joining the tables, but no mix and match between methods In example 1 the join mark needs correct FROM and WHERE clauses In example 2 the join mark needs correct FROM... JOIN and ON clauses

4. O/N 25/P11/Q2/a,b,c,d

2(a)	1 mark for each correct relationship, max 2 marks  <pre> graph LR STUDENT[STUDENT] --- mandatory one PLACEMENT[PLACEMENT] PLACEMENT[PLACEMENT] --- mandatory one COMPANY[COMPANY] </pre>	2
2(b)	1 mark per bullet point, max 2 marks • There are no repeating groups of attributes • There are no many-to-many relationships • There are no partial key dependencies // no non-key dependencies // no transitive dependencies	2
2(c)(i)	1 mark per bullet point, max 2 marks • DELETE FROM and correct table • Correct condition Example answer: DELETE FROM PLACEMENT WHERE Complete = TRUE;	2
2(c)(ii)	1 mark per bullet point, max 4 marks • SELECT COUNT any appropriate field • AS and an appropriate name • FROM and correct table and WHERE and one correct condition • Two AND clauses with the other two correct conditions Example answer: SELECT COUNT (CompanyID) AS TotalPlacements FROM PLACEMENT WHERE CompanyID = "NEAM" AND StudentID = "LDEA01" AND Complete = TRUE;	4

2(d)	1 mark per bullet point, max 3 marks Generic mark points (max 2) • Referential integrity ensures that related data is consistent • Referential integrity ensures that every foreign key has a corresponding primary key • Referential integrity provides for cascading update / delete • Referential integrity ensures that if a primary key is deleted or modified all linked records in foreign table will be deleted or modified // Referential integrity stops "orphaned records"—records in a table that point to an entry in another table that no longer exists Specific mark points (max 2) for good examples e.g. • CompanyID is a foreign key in PLACEMENT table and is dependent on the primary key CompanyID in COMPANY table • If a record is deleted from the STUDENT table, then all records with the same StudentID will be deleted from the PLACEMENT table • If a CompanyID is modified in the company table, all records with that CompanyID in the placement table will also be modified	3
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5. O/N 25/P12/Q4/a,b,c,d

4(a)	1 mark per bullet point, max 2 marks • The relationship between SHIP and CONTAINER is one-to-many (1:M) • The primary key <u>ShipID</u> in the SHIP table is linked to the foreign key ShipID in the CONTAINER table	2
4(b)	1 mark per bullet point, max 2 marks • ALTER TABLE statement • Including the additional field with a suitable field name and suitable field type Example answer <pre>ALTER TABLE CONTAINER ADD InspectionDate DATE;</pre>	2

4(c)	1 mark per bullet point, max 4 marks • SELECT <u>COUNT</u> statement • Using the correct tables • Joining the tables • The condition for the name Example Answer 1: <pre>SELECT COUNT(ContainerID) FROM CONTAINER, SHIP WHERE CONTAINER.ShipID = SHIP.ShipID AND ShipName = "Caledonia";</pre> Example Answer 2: <pre>SELECT COUNT(ContainerID) FROM CONTAINER INNER JOIN SHIP ON CONTAINER.ShipID = SHIP.ShipID WHERE ShipName = "Caledonia";</pre>	4
4(d)	1 mark per bullet point, max 2 marks • To allow the user to create / modify / delete tables // maintain the database • To allow the user to set up / modify relationships • To allow the user to create a form for data input • To allow a user to add tools to a form // for example, drop-down boxes / buttons etc. • To allow a user to design a report (to show the output in an organised manner) • To allow a user to add a menu to enable a choice of different actions • To allow a user to inspect the database contents and metadata • ... by creating and/or running different SQL queries	2

6. O/N 25/P13/Q5/a,b,c,d

5(a)	1 mark for each correct relationship, max 3 marks <pre> graph TD CUSTOMER --> COMPLAINT PRODUCT --> COMPLAINT COMPLAINT --> STAFF </pre>	3
------	---	----------

5(b)	1 mark per bullet point, max 4 marks • Create table with opening and closing brackets and commas separating the attributes • Appropriate data types for StaffFirstName, StaffLastName and Department • Appropriate data types for StaffID and RemoteWorker • Primary key correctly defined Example Answer 1: <pre>CREATE TABLE STAFF(StaffID INTEGER, StaffFirstName VARCHAR, StaffLastName VARCHAR, Department CHAR, RemoteWorker BOOLEAN, PRIMARY KEY (StaffID));</pre> Example Answer 2: <pre>CREATE TABLE STAFF(StaffID INTEGER NOT NULL PRIMARY KEY, StaffFirstName VARCHAR, StaffLastName VARCHAR, Department CHAR, RemoteWorker BOOLEAN);</pre>	4
5(c)	1 mark per bullet point, max 5 marks • SELECT and the correct attributes • FROM and the correct tables • Tables joined correctly • Correct condition for the rating • Correct ORDER BY clause Example Answer 1 <pre>SELECT PRODUCT.ProductID, ProductName, ComplaintDetails FROM PRODUCT, COMPLAINT WHERE PRODUCT.ProductID = COMPLAINT.ProductID AND Rating <=5 ORDER BY Rating DESC;</pre> Example Answer 2 <pre>SELECT PRODUCT.ProductID, ProductName, ComplaintDetails FROM PRODUCT INNER JOIN COMPLAINT ON PRODUCT.ProductID = COMPLAINT.ProductID WHERE Rating <=5 ORDER BY Rating DESC;</pre>	5

5(d)	One mark for identification of method one mark for corresponding description max 4 marks • Authentication methods / passwords / biometrics / 2-factor authentication can be implemented • ... which prevents unauthorised access to the customer's data • Access rights / privileges can be set • ... so that only those with correct permissions can read / edit customer's data • Regular backups can be scheduled • ... so that a second copy of the customer's data is available in case of loss/damage • The data can be encrypted • ... so that the customer's data cannot be understood by anyone who gains unauthorized access • Different views can be created • ... so that not everyone can see the customer's data	4
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7. M/J 25/P11/Q5/a,b,c,d,e

5(a)	1 mark for each relationship. Max 2 if any extra <pre>graph TD; CUSTOMER --> CUSTOMER_CARD_DATA; CUSTOMER --> ORDER; ORDER --> ORDER_ITEM;</pre>	3								
5(b)	1 mark for: CardNumber	1								
5(c)	1 mark per row to max 2 <table><thead><tr><th>Table</th><th>Foreign key</th></tr></thead><tbody><tr><td>ORDER_ITEM</td><td>OrderID</td></tr><tr><td>ORDER</td><td>CustomerID</td></tr><tr><td>CUSTOMER_CARD_DATA</td><td>CustomerID</td></tr></tbody></table>	Table	Foreign key	ORDER_ITEM	OrderID	ORDER	CustomerID	CUSTOMER_CARD_DATA	CustomerID	2
Table	Foreign key									
ORDER_ITEM	OrderID									
ORDER	CustomerID									
CUSTOMER_CARD_DATA	CustomerID									

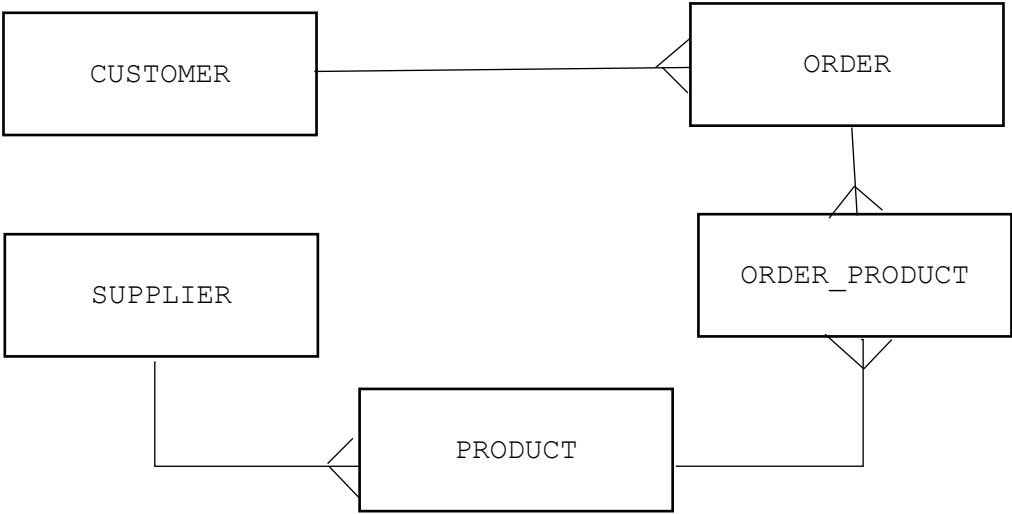
5(d)	1 mark each to max 3 e.g. • Each order would only be able to have one item • or the database would not be normalised • it would not be in 1NF • due to repeated groups of attributes • in the ORDER table	3
5(e)	1 mark each • Selecting the customer ID, customer name and sum of TotalCost with appropriate identifier • FROM clause with suitable join of tables (ON or WHERE) • ORDER.Paid = FALSE condition (with correct key word) • GROUP BY condition e.g. SELECT CUSTOMER.CustomerID, CUSTOMER.Name, Sum(ORDER.TotalCost) AS TotalOwed FROM CUSTOMER INNER JOIN ORDER ON CUSTOMER.CustomerID = ORDER.CustomerID WHERE ORDER.Paid = FALSE GROUP BY CUSTOMER.CustomerID; Alternative JOIN Statement SELECT CUSTOMER.CustomerID, CUSTOMER.Name, Sum(ORDER.TotalCost) AS TotalOwed FROM CUSTOMER,ORDER WHERE CUSTOMER.CustomerID = ORDER.CustomerID AND ORDER.Paid = FALSE GROUP BY CUSTOMER.CustomerID;	4

8. M/J 25/P12/Q5/a,b,c,d

5(a)	1 mark each to max 2 • To remove the many-to-many relationship • between CHARACTER and ITEM tables • To allow each character to have many items // to allow each item to be purchased for many characters • by creating a linking table • between the characters and the items purchased for each one	2
5(b)	1 mark for the three single Primary Keys 1 mark for the composite key USER(<u>Username</u> , Password, DateOfBirth) CHARACTER(<u>CharacterName</u> , <u>CharacterID</u> , Username, Level, Money) ITEM(<u>ItemName</u> , MinimumLevel, Cost) CHARACTER_ITEM(<u>CharacterID</u> , <u>ItemName</u>)	2

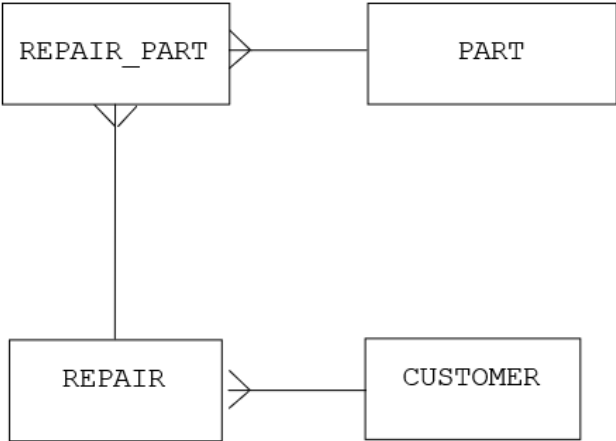
5(c)(i)	1 mark for method 1 mark for matching explanation e.g. • Access rights • appropriate permissions for the table USER are needed to read or edit the data • A password for the database or the USER table • prevents users without the password from accessing the data • Encrypting the database • stops users without the decryption key from decoding / understanding the data • Views • users can be given a view to the database that does not include the data in the table USER	4
5(c)(ii)	1 mark each to max 2 e.g. • Validation • Enforce referential integrity • Cascade update / delete • Ensuring the database is normalised	2
5(d)(i)	1 mark for each point • Select COUNT • FROM (using correct tables) and joining • WHERE (or AND) condition for correct username Example 1: SELECT COUNT(ItemName) FROM CHARACTER_ITEM INNER JOIN CHARACTER ON CHARACTER.CharacterID = CHARACTER_ITEM.CharacterID WHERE CHARACTER.Username = "KAT123"; Example 2: SELECT COUNT(ItemName) FROM CHARACTER_ITEM, CHARACTER WHERE CHARACTER.CharacterID = CHARACTER_ITEM.CharacterID AND CHARACTER.Username = "KAT123";	3
5(d)(ii)	1 mark for each point • UPDATE character • SET level and money • WHERE condition e.g. UPDATE CHARACTER SET Level = 3, Money = 10000.00 WHERE CharacterID = "0002";	3

9. M/J 25/P13/Q6/a,b,c,d

6(a)	1 mark for each correct relationship  <pre> graph LR CUSTOMER --> ORDER ORDER --> ORDER_PRODUCT SUPPLIER --> PRODUCT ORDER_PRODUCT --> PRODUCT </pre>	4
6(b)	1 mark each • INSERT INTO PRODUCT • VALUES with opening and closing brackets • Inserting ProductID, ProductName and SupplierID correctly as strings including quotation marks into correct fields • Inserting quantity and cost correctly into correct fields Example 1: INSERT INTO PRODUCT VALUES ("002323", "Blue ball point 2 mm", 50, 5.00, "SFX223"); Example 2: INSERT INTO PRODUCT (ProductID, ProductName, QuantityInBox, Cost, SupplierID) VALUES ("002323", "Blue ball point 2 mm", 50, 5.00, "SFX223");	4

6(c)	1 mark each • Selection of customer name and counting any field from ORDER as an appropriate identifier • Joining tables ORDER and CUSTOMER • AND (or WHERE) clause: Collected = FALSE • Grouping by customer ID or customer name Example 1: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER, CUSTOMER WHERE ORDER.CustomerID = CUSTOMER.CustomerID AND Collected = FALSE GROUP BY CUSTOMER.CustomerID Example 2: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER INNER JOIN CUSTOMER ON ORDER.CustomerID = CUSTOMER.CustomerID WHERE Collected = FALSE GROUP BY CUSTOMER.CustomerID	4
6(d)(i)	1 mark each to max 3 e.g. • Relationships • Views • Data types • Validation rules	3
6(d)(ii)	1 mark each to max 3 e.g. • To create / modify / delete database objects • To create a form for data input • To add tools to a form • for example, drop-down boxes / buttons etc. • To design a report to show the output in an organised manner • To add a menu to enable users to choose different actions / run different queries	3

10. O/N 24/P11/Q2/a,b,c

2(a)	1 mark for limitation and 2 marks for corresponding explanation e.g. Data redundancy // data duplication • Separate linked tables are used • Data items are stored once so reduces data duplication / data redundancy Data inconsistency / Poor data integrity • Data changed once in one place will automatically update elsewhere • Linked data cannot be entered differently in two tables • Referential integrity can be enforced The data structure used depends on the application • Changes to the data structure are managed by the DBMS • ... and queries are not dependent on the structure of the data • ... and changes to the data do not require programs to be re-written	3
2(b)(i)	1 mark for each correct relation 	3
2(b)(ii)	1 mark for each bullet point • Create table REPAIR_PART with opening and closing brackets, all statements inside the brackets • PartID and RepairNumber as varchar (or equivalent) • Quantity as integer • At least 1 appropriate constraint • Dual primary key For example: <pre>CREATE TABLE REPAIR_PART(PartID VARCHAR(20) NOT NULL, RepairNumber VARCHAR(4) NOT NULL, Quantity INT NOT NULL, PRIMARY KEY (PartID, RepairNumber));</pre>	5

2(b)(iii)	1 mark for each bullet point • Select SUM of AmountDue • From the correct table and one correct condition • Second correct condition <pre>SELECT SUM(AmountDue) FROM INVOICE WHERE SupplierID = "JK675" AND Paid = FALSE;</pre>	3								
2(c)	1 mark for each correct definition <table><tr><th>Database Term</th><th>Definition</th></tr><tr><td>Referential Integrity</td><td>All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key</td></tr><tr><td>Candidate Key</td><td>A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value</td></tr><tr><td>Tuple</td><td>A row / record in a table // one instance of an entity in a table</td></tr></table>	Database Term	Definition	Referential Integrity	All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key	Candidate Key	A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value	Tuple	A row / record in a table // one instance of an entity in a table	3
Database Term	Definition									
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Tuple	A row / record in a table // one instance of an entity in a table									

11. O/N 24/P12/Q6/a,b

6(a)	1 mark each to max 3 e.g. - There is reduced data redundancy // less repeated data ... because each item of data is only stored once - Data consistency is maintained // Data integrity is improved ... changes in one table will automatically update in another ... linked data cannot be entered differently in two tables - Program-data independence is ensured ... changes to the data do not require programs to be re-written // queries are not dependent on the structure of the data - Complex queries are easier to run - Different views can be provided so users can only see specific aspects of the database - Multiple concurrent access is possible ... through record locking	3
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6(b)(i)	1 mark each to max 6 - CUSTOMER to JOB is 1 to many ... implemented by Primary Key in CUSTOMER is Foreign Key in JOB - EMPLOYEE to LOGIN_DATA is 1 to 1 ... implemented by Primary Key in EMPLOYEE is Foreign Key in LOGIN_DATA - JOB to JOB_EMPLOYEE is 1 to many ... implemented by Primary Key in JOB is Foreign Key in JOB_EMPLOYEE - EMPLOYEE to JOB_EMPLOYEE is 1 to many ... implemented by Primary Key in EMPLOYEE is Foreign Key in JOB_EMPLOYEE	6
6(b)(ii)	1 mark each - Select SUM of Amount - From the correct table and one correct condition - Remaining correct condition Example: <pre>SELECT SUM(Amount) FROM INVOICE WHERE Paid = "Y" AND DateSent >= #01/01/2023# AND DateSent <= #31/12/2023#</pre>	3

12. O/N 24/P13/Q4/a,b,c,d,e

4(a)	1 mark each: - Foreign key: BatchID, table BATCH - Foreign key: CustomerID, table CUSTOMER	2
4(b)	1 mark each: - SELECT SUM(Quantity) - FROM SALE WHERE and one correct condition - AND with remainder correct conditions e.g. <pre>SELECT SUM(Quantity) FROM SALE WHERE CustomerID = "0034E" AND Date >= #01/01/2023# AND Date <= #31/12/2023#;</pre>	3

4(c)(i)	1 mark for each bullet point • Create table BATCH with opening and closing brackets, all statements within brackets • BatchID, Type and Flavour as varchar or equivalent • ...with suitable constraint(s) • Size as decimal, SellingPrice as currency, EndDate as date (or equivalent) • Primary key identified as BatchID e.g. <pre>CREATE TABLE BATCH (BatchID VARCHAR(6) NOT NULL, Type VARCHAR (20) NOT NULL, Flavour VARCHAR (20) NOT NULL, Size FLOAT, SellingPrice CURRENCY, EndDate DATE, PRIMARY KEY (BatchID));</pre>	5
4(c)(ii)	1 mark for each bullet point each (max 4) e.g. • Ice cream table with an appropriate name • ... containing type, flavour, size, selling price • ... with suitable primary key • ... foreign key identified in BATCH that links to the primary key in ICECREAM Example table definitions – not example answer: <pre>BATCH(BatchID, IceCreamID, EndDate) ICE_CREAM(IceCreamID, Type, Flavour, Size, SellingPrice)</pre>	4
4(d)	1 mark for each definition Entity: • A real-life object that is represented as a table Attribute: • An item of data about an entity	2
4(e)	1 mark for each bullet point each (max 3) • Referential integrity is enforced • ... such as cascade update/delete // if the data is changed in one place it is updated in every other place • ... and ensures each foreign key has a corresponding primary key	3

13. M/J 24/P11/Q6/a,b,c

6(a)	1 mark each: • User table with the username as the Primary Key • ... containing at least email address, date of birth / age and rating • Quiz table with Quiz ID or date or file name as the Primary Key. • ... containing at least the other field(s) not used as the PK • A joining table with an appropriate name including at least fields for user identification, quiz identification and score • ... with an appropriate Primary Key • ... and Foreign Keys matching the Primary Keys of the other two tables USER(<u>Username</u>, Email, DateOfBirth, Rating) QUIZ(<u>QuizID</u>, Date, Filename) USER QUIZ(<u>Username</u>, <u>QuizID</u>, Score)	6
6(b)	1 mark each to max 2 for data dictionary and max 2 for logical schema: Data dictionary: • Data about the data in the database // metadata • Identifies the characteristics of the data that will be stored • Appropriate example e.g. field names, table name, validation rules, data types, primary / foreign keys, relationships etc. Logical schema: • Conceptual design • Platform/database independent overview of the database • Is used to design the physical structure • Appropriate example e.g. Design of entities / E-R diagram / views	4
6(c)(i)	1 mark for each correct clause: • Alter table EVENT • Adding foreign key as PlayerID referencing correct table ALTER TABLE EVENT ADD FOREIGN KEY(PlayerID)REFERENCES PLAYER(PlayerID);	2
6(c)(ii)	1 mark each: • Selecting PlayerID from EVENT • Counting EventID • Grouping by the PlayerID Example: SELECT PlayerID, COUNT(EventID) FROM EVENT GROUP BY PlayerID;	3

14. M/J 24/P12/Q4/a,b,c,d

4(a)	1 mark for: 1-to-many	1
4(b)	1 mark each: • Creating table EXAM with opening and closing brackets • All fields with appropriate data types and commas at end of lines • ExamID as primary key Example: <pre>CREATE TABLE EXAM(ExamID varchar NOT NULL, Subject varchar, Level int, TotalMarks int, PRIMARY KEY (ExamID));</pre>	3
4(c)	1 mark each: • Altering table EXAM_QUESTION • Linking ExamID to ExamID in EXAM Example. <pre>ALTER TABLE EXAM_QUESTION ADD FOREIGN KEY (ExamID) REFERENCES EXAM (ExamID);</pre>	2
4(d)	1 mark each to max 5: • STUDENT table identified with suitable Primary Key • A linking table between STUDENT and EXAM with suitable Primary Key and appropriate name • ... that includes the Primary Key of the STUDENT table as a Foreign Key to join with STUDENT • ... and includes the Primary Key of the EXAM table as a Foreign Key to join with EXAM • A linking table between STUDENT and EXAM_QUESTION with suitable Primary Key and appropriate name • ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 • ... that stores the ExamQuestionID and the mark for that question	5

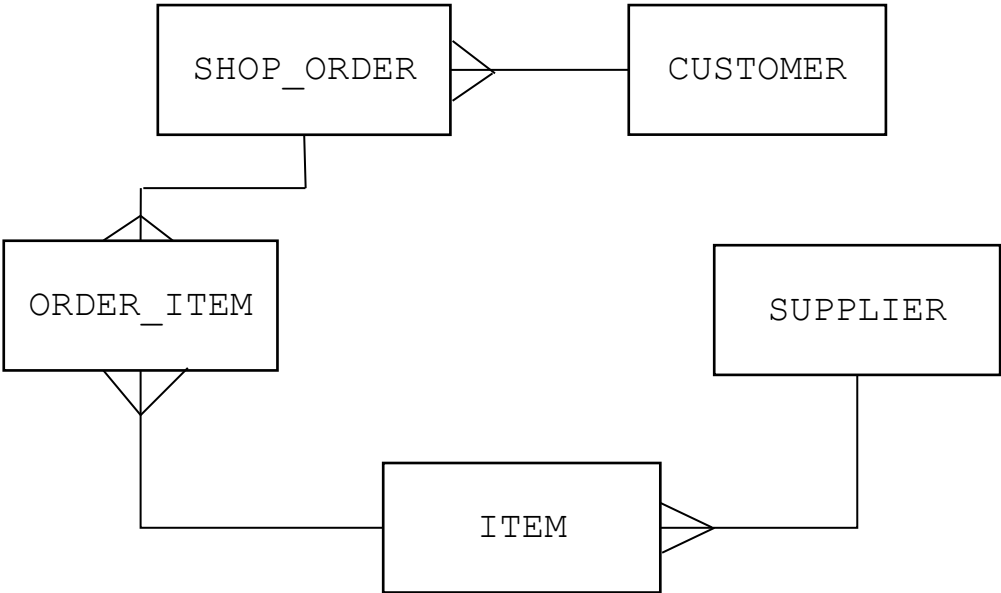
15. M/J 24/P13/Q4/a,b,c,d

4(a)	1 mark for: many to 1 // there are many performances of each show	1
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4(b)	1 mark each for: • Creating table <code>PERFORMANCE</code> with opening and closing brackets • Setting all four attributes with appropriate data types • Setting <code>PerformanceID</code> as primary key • Setting <code>ShowID</code> as foreign key referencing <code>SHOW</code> table Example: <pre>CREATE TABLE PERFORMANCE(PerformanceID varchar NOT NULL, ShowID varchar, ShowDate Date, StartTime Time, PRIMARY KEY (PerformanceID), FOREIGN KEY (ShowID) REFERENCES SHOW (ShowID));</pre>	4
4(c)	1 mark each for: • Selecting <code>COUNT</code> of an attribute in <code>PERFORMANCE</code> table with suitable name • <code>FROM</code> clause • Joining tables • Grouping by the title and selecting the title Example 1: <pre>SELECT SHOW.Title, Count (PERFORMANCE.PerformanceID) AS NumberOfShowings FROM PERFORMANCE, SHOW WHERE PERFORMANCE.ShowID = SHOW.ShowID GROUP BY SHOW.Title;</pre> Example 2: <pre>SELECT SHOW.Title, Count (PERFORMANCE.PerformanceID) AS NumberOfShowings FROM PERFORMANCE INNER JOIN SHOW ON PERFORMANCE.ShowID = SHOW.ShowID GROUP BY SHOW.Title;</pre>	4

4(d)	1 mark each to max 5: e.g. • CUSTOMER table identified with suitable Primary Key and appropriate name • ... and other suitable fields including name and email • BOOKING TABLE identified with suitable Primary Key and appropriate name • ... that stores the Primary Key of the CUSTOMER table as a Foreign Key to join with CUSTOMER table • ... and stores the Primary Key of the PERFORMANCE table as a Foreign Key to join with PERFORMANCE table • A linking table between Table 2 and SEAT with suitable Primary Key and appropriate name • ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 • ... that stores the SeatID.	5
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16. O/N 23/P11/Q3/a,b,c

3(a)	1 mark for each correct relationship or relationships • 1:M between CUSTOMER and SHOP_ORDER • 1:M between SUPPLIER and ITEM • 1:M between SHOP_ORDER and ORDER_ITEM and M:1 between ORDER_ITEM and ITEM  <pre> graph TD SHOP_ORDER[SHOP_ORDER] -- "1:M" --> CUSTOMER[CUSTOMER] SHOP_ORDER -- "1:M" --> ORDER_ITEM[ORDER_ITEM] ORDER_ITEM -- "M:1" --> ITEM[ITEM] SUPPLIER[SUPPLIER] -- "1:M" --> ITEM </pre>	3
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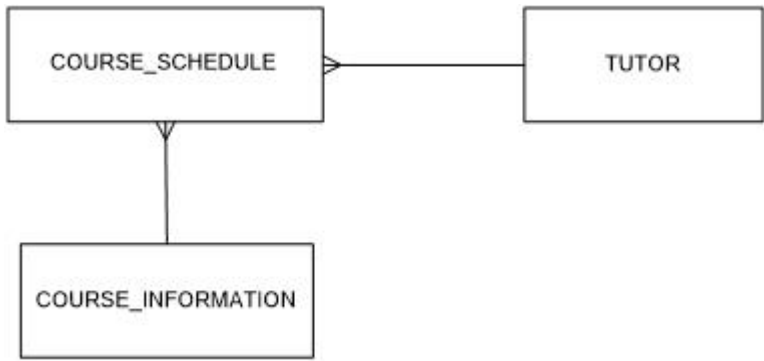
3(b)	1 mark for each bullet point (max 3) - Reduced data redundancy - Improved data integrity / consistency / referential integrity - Allows for views / improved privacy - Allows for program-data independence - Complex queries can be executed	3
3(c)(i)	1 mark for CREATE DATABASE SHOP;	1
3(c)(ii)	1 mark for each line (max 4) Either: SELECT SUM(Quantity) FROM ORDER_ITEM, SHOP_ORDER WHERE ORDER_ITEM.OrderNo = SHOP_ORDER.OrderNo AND SHOP_ORDER.CustomerID = 'HJ231'; OR SELECT SUM(Quantity) FROM ORDER_ITEM (INNER) JOIN SHOP_ORDER ON ORDER_ITEM.OrderNo = SHOP_ORDER.OrderNo WHERE SHOP_ORDER.CustomerID = 'HJ231';	4

17. O/N 23/P12/Q2/a,b,c

2(a)	1 mark for each term (max 3) Entity: - An object about which data can be stored Primary key: - The unique attribute / combination of attributes used to identify the record / tuple Referential integrity: - Makes sure that if data is changed in one place the change is reflected in all related records - cascading update/delete - Makes sure that data that does not exist cannot be referenced - Ensures that every foreign key has a corresponding primary key // A logical dependency of a foreign key on a primary key - Ensures that the data in the database is consistent / up to date - Prevents records from being added/deleted/modified incorrectly - Makes sure any queries return accurate and complete results	3
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2(b)	1 mark for each bullet point (max 4) Max 2 if no descriptions • Backup / recovery procedures • ... automatically takes copies of the database and store off site on a regular basis / weekly, etc. • ... so that the data can be recovered if lost • Use of access rights • ... some users are given different access permissions to different tables • ... read/write, read only, full access, etc. • Views • ... different users are able to see different parts of the database • ... only see what users need to see // by example • Record and table locking • ... prevents simultaneous access to data • ... so updates are not lost // data is not overwritten • Encryption • ... the data is turned into ciphertext • ... so it cannot be understood without a decryption key	4
2(c)	1 mark for each bullet point (max 4) • Identify repeating groups of attributes ... • ... Subject and SubjectCode • Ensure each field is atomic • ... StudentName should be split into e.g. FirstName and LastName • Identify the primary key for the table	4

18. O/N 23/P13/Q3/b

3(b)(i)	1 mark for both 1:M relationships as follows:  <pre> graph TD CS[COURSE_SCHEDULE] --> T[TUTOR] CS --> CI[COURSE_INFORMATION] </pre>	1
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3(b)(ii)	1 mark for each bullet point. • <code>SELECT Count (CourseID)</code> • <code>AS NumOfCourses</code> • <code>FROM COURSE_SCHEDULE</code> • <code>WHERE DateStarted > "09/09/23";</code> <code>SELECT Count (CourseID) AS NumOfCourses FROM COURSE_SCHEDULE WHERE DateStarted > "09/09/23";</code>	4
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19. M/J 23/P11/Q2/a,b

2(a)(i)	1 mark for definition: • Data about the data in the database // data about the structure of the database // metadata for a database 1 mark for a suitable example Examples: • table names • data types • field names	2						
2(a)(ii)	1 mark for definition • Methods of making sure the data is consistent 1 mark for example Examples: • Enforcing referential integrity • If data in one table is deleted/edited all tables are updated // cascading update/delete • Validation/verification rules	2						
2(b)(i)	1 mark for each field name and table <table><tr><th>Foreign key</th><th>Database table</th></tr><tr><td>BirdID</td><td>BIRD_SEEN</td></tr><tr><td>PersonID</td><td>BIRD_SEEN</td></tr></table>	Foreign key	Database table	BirdID	BIRD_SEEN	PersonID	BIRD_SEEN	2
Foreign key	Database table							
BirdID	BIRD_SEEN							
PersonID	BIRD_SEEN							

2(b)(ii)	1 mark for all 3 correct lines <table><thead><tr><th>Normal Form</th><th>Definition</th></tr></thead><tbody><tr><td>First Normal Form (1NF)</td><td>All fields are fully dependent on the primary key.</td></tr><tr><td>Second Normal Form (2NF)</td><td>There are no repeating groups of attributes.</td></tr><tr><td>Third Normal Form (3NF)</td><td>There are no partial dependencies.</td></tr></tbody></table>	Normal Form	Definition	First Normal Form (1NF)	All fields are fully dependent on the primary key.	Second Normal Form (2NF)	There are no repeating groups of attributes.	Third Normal Form (3NF)	There are no partial dependencies.	1
Normal Form	Definition									
First Normal Form (1NF)	All fields are fully dependent on the primary key.									
Second Normal Form (2NF)	There are no repeating groups of attributes.									
Third Normal Form (3NF)	There are no partial dependencies.									
2(b)(iii)	1 mark each • CREATE TABLE start and end bracket • Bird ID as CHAR/VARCHAR • Name and size as VARCHAR/CHAR • Bird ID as primary key Example answer: <pre>CREATE TABLE BIRD_TYPE(BirdID CHAR(4) NOT NULL, Name VARCHAR(9), Size VARCHAR(6), PRIMARY KEY (BirdID));</pre>	4								
2(b)(iv)	1 mark for each correctly completed space <pre>SELECT BIRD_TYPE.Size, COUNT(BIRD_TYPE.BirdID) AS NumberOfBirds FROM BIRD_TYPE, BIRD_SEEN WHERE BIRD_SEEN.PersonID = "J_123" AND BIRD_TYPE.BirdID = BIRD_SEEN.BirdID GROUP BY BIRD_TYPE.Size;</pre>	5								

20. M/J 23/P12/Q2/a,b,c(ii,iii)

2(a)	1 mark for each correct feature or description <table><tr><th>Feature</th><th>Description</th></tr><tr><td>Data dictionary</td><td>Data about the data in the database // data about the structure of the database // metadata for a database</td></tr><tr><td>Query processor</td><td>Software that allows the user to enter criteria, then finds and returns the appropriate result // software that processes and executes queries written in SQL</td></tr><tr><td><u>Logical schema</u></td><td>A model of a database that is not specific to one DBMS</td></tr><tr><td><u>Developer interface</u></td><td>A software tool that allows the user to create items such as tables, forms and reports</td></tr></table>	Feature	Description	Data dictionary	Data about the data in the database // data about the structure of the database // metadata for a database	Query processor	Software that allows the user to enter criteria , then finds and returns the appropriate result // software that processes and executes queries written in SQL	<u>Logical schema</u>	A model of a database that is not specific to one DBMS	<u>Developer interface</u>	A software tool that allows the user to create items such as tables, forms and reports	4
Feature	Description											
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<u>Developer interface</u>	A software tool that allows the user to create items such as tables, forms and reports											
2(b)	1 mark each to max 3 • Referential Integrity makes sure data is consistent • Referential Integrity makes sure all data is up-to-date • Referential integrity ensures that every foreign key has a corresponding primary key • Referential Integrity prevents records from being added / deleted / modified incorrectly • Referential Integrity makes sure that if data is changed in one place the change is reflected in all related records • Referential Integrity makes sure any queries return accurate and complete results	3										
2(c)(ii)	1 mark each • SELECT field Name • FROM table HORSE • WHERE with Intermediate / Beginner • OR with Beginner / Intermediate Example answer: SELECT Name FROM HORSE WHERE HorseLevel = "Intermediate" OR HorseLevel = "Beginner";	4										

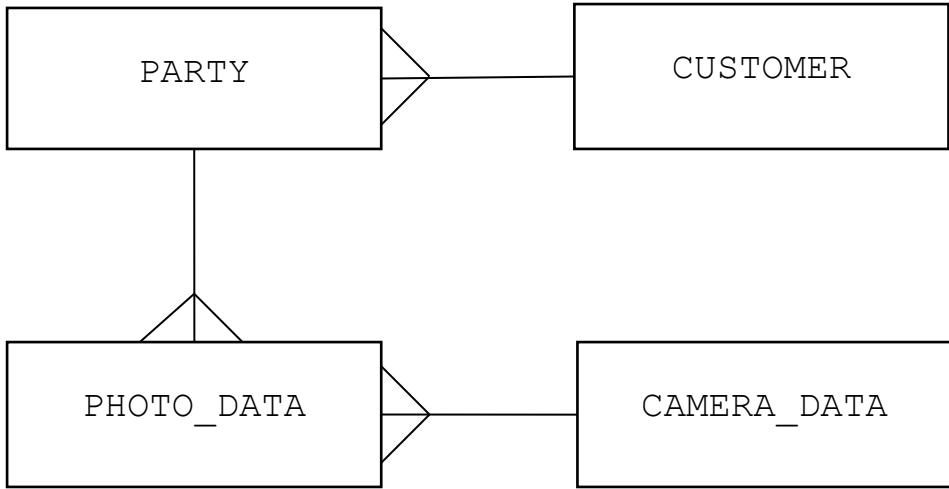
2(c)(iii)	1 mark each - SUM should be COUNT // SELECT COUNT(STUDENT.RiderLevel) - The WHERE statement needs the table names before each field name // WHERE STUDENT.StudentID = LESSON.StudentID - The OR should be AND // AND Date = #09/09/2023# - Beginner is missing the speech marks // STUDENT.RiderLevel = "Beginner";	4
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21. M/J 23/P13/Q4/a,b,c

4(a)	1 mark for each bullet point. Mark in pairs. Max 2 for each description. • Reduces data redundancy • ... because linked tables mean that each data item is stored only once • Reduces program-data dependency • ... because the data is separate from the software so changes to the data do not require programs to be re-written • Reduces data inconsistency // improves data integrity • ... because by only storing data once it only needs to be updated once // changes in one table will automatically update in another // linked data cannot be entered differently in two tables • Complex queries are easier to run • Can provide different views •so users can only see specific aspects of the database	4										
4(b)	1 mark each <table><tr><th>Term</th><th>Description</th></tr><tr><td>Entity</td><td>An object that data is stored about.</td></tr><tr><td>Tuple</td><td>A row of data in a table about one instance of an object.</td></tr><tr><td>Secondary key</td><td>An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.</td></tr><tr><td>Foreign key</td><td>A field in one table that is linked to a primary key in another table.</td></tr></table>	Term	Description	Entity	An object that data is stored about.	Tuple	A row of data in a table about one instance of an object.	Secondary key	An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.	Foreign key	A field in one table that is linked to a primary key in another table.	4
Term	Description											
Entity	An object that data is stored about.											
Tuple	A row of data in a table about one instance of an object.											
Secondary key	An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.											
Foreign key	A field in one table that is linked to a primary key in another table.											

4(c)	1 mark each - Only 3 tables with appropriate identifiers (i.e. one table for customer, one for booking and one for car) - Appropriate Primary key in each table underlined - Booking table includes Primary key from car and Primary key from customer as Foreign keys - All original fields are in correct tables Example answer: BOOKING(<u>BookingID</u>, CarRegistration, CustomerID, StartDate, EndDate) CAR(<u>CarRegistration</u>, CarModel, CarColour) CUSTOMER(<u>CustomerID</u>, CustomerFirstName, CustomerLastName, EmailAddress, TelephoneNumber)	4
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22. O/N 22/P11/Q4/a,b,c,d

4(a)	1 mark for each correct relationship: 1:M between CUSTOMER and PARTY 1:M between PARTY and PHOTO_DATA 1:M between CAMERA_DATA and PHOTO_DATA 	3
4(b)	1 mark for each bullet point (max 3): - no repeating groups of attributes // data is atomic - no partial key dependencies - no non-key dependencies // no transitive dependencies	3
4(c)(i)	1 mark for the definition, 1 mark for the example: - definition: a single row in a table - example: from the PHOTO_DATA table	2

4(c)(ii)	1 mark for each correctly completed empty space: • COUNT • PhotoID • PHOTO_DATA • 'CAN*' // 'CAN%' <pre>SELECT COUNT(PhotoID) FROM PHOTO_DATA WHERE CameraID LIKE 'CAN*'; // WHERE CameraID LIKE 'CAN%';</pre>	4
4(d)	1 mark for each bullet point: • ALTER TABLE CAMERA_DATA • ADD NumberStored INTEGER • , LastUsed DATE; <pre>ALTER TABLE CAMERA_DATA ADD NumberStored INTEGER, LastUsed DATE;</pre>	3

23. O/N 22/P12/Q5/a,b,c,d

5(a)	1 mark for each bullet point (max 3): Solution 1: • removing the many-to-many relationship between Owner and Tree • ... by removing TreeID and TreePosition from the Owner table • ... and creating a linking table between Owner and Tree • ... that contains OwnerID, TreeID and TreePosition • ... (composite) primary key of the linking table should be OwnerID and TreeID // insert a named new primary key in the linking table Solution 2: • removing the many-to-many relationship between Owner and Tree • move TreePosition into TREE table • ... put OwnerID into TREE table • create a new table with suitable name (for the species of tree) • ... containing ScientificName, MaxHeight and FastGrowing • ... with ScientificName as primary key // or another suitable primary key	3
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5(b)	1 mark for each bullet point: • INSERT INTO TREE • VALUES () and correct values • Values in correct order Option 1: INSERT INTO TREE(TreeID, ScientificName, MaxHeight, FastGrowing) VALUES('LOW_1276', 'Salix_Alba', 30.00, TRUE); Option 2: INSERT INTO TREE VALUES('LOW_1276', 'Salix_Alba', 30.00, TRUE);	3
5(c)	1 mark for: An attribute / field (or set of attributes / fields) that could be a primary key	1
5(d)(i)	1 mark for description • stores metadata about the database 1 mark for a correct example For example: • field / attribute names • table name • validation rules • data types • primary keys // foreign keys • relationships	2
5(d)(ii)	1 mark for each bullet point (max 2): • the overview of a database structure • models the problem / situation • ... by using methods such as an ER diagram • independent of any particular DBMS	2

24. O/N 22/P13/Q2/a,b,c,d,f

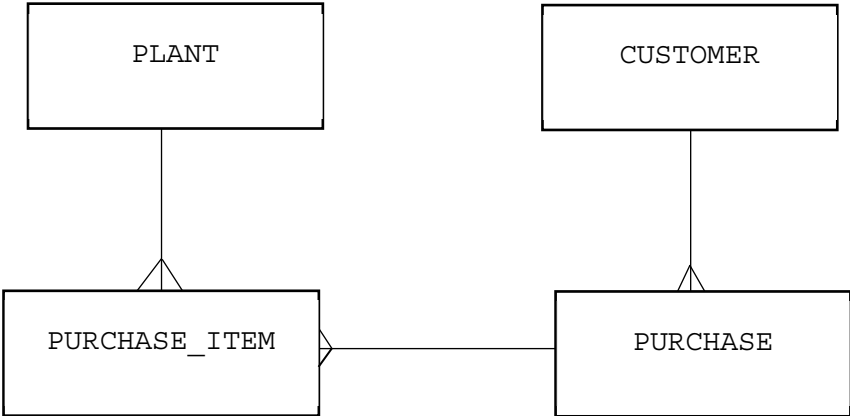
2(a)	1 mark for each correct answer. <table><tr><th>Item</th><th>Answer</th></tr><tr><td>a suitable field for the primary key in COMPANY</td><td>CompanyID</td></tr><tr><td>a candidate key in TELESCOPE</td><td>SerialNumber // TelescopeID</td></tr><tr><td>the degree of relationship between TELESCOPE and PHOTOGRAPH</td><td>1:M / 1 to many</td></tr></table>	Item	Answer	a suitable field for the primary key in COMPANY	CompanyID	a candidate key in TELESCOPE	SerialNumber // TelescopeID	the degree of relationship between TELESCOPE and PHOTOGRAPH	1:M / 1 to many	3
Item	Answer									
a suitable field for the primary key in COMPANY	CompanyID									
a candidate key in TELESCOPE	SerialNumber // TelescopeID									
the degree of relationship between TELESCOPE and PHOTOGRAPH	1:M / 1 to many									
2(b)	Logical schema	1								

2(c)	1 mark for each correctly completed missing part: SELECT COUNT (TelescopeID) FROM TELESCOPE WHERE CompanyID LIKE 'HW%' ;	4
2(d)	1 mark for each bullet point: - ALTER TABLE PHOTOGRAPH - ADD Resolution TEXT; ALTER TABLE PHOTOGRAPH ADD Resolution TEXT / VARCHAR(11);	2
2(f)	1 mark for each bullet point (max 2): - allows the user to enter criteria - searches for the data that meets the entered criteria - organises the results to be displayed to the user	2

25. O/N 21/P11/Q5/a,b,c

5(a)	1 mark for 2 or 3 correct ticks, 2 marks for 4 correct ticks	2																				
	<table><tr><th>Table</th><th>Field name</th><th>Primary Key (PK)</th><th>Foreign Key (FK)</th></tr><tr><td>MANAGER</td><td>ManagerID</td><td>✓</td><td></td></tr><tr><td>SHOP</td><td>ManagerID</td><td></td><td>✓</td></tr><tr><td>CAR</td><td>RegistrationNumber</td><td>✓</td><td></td></tr><tr><td>CAR</td><td>ShopID</td><td></td><td>✓</td></tr></table>	Table	Field name	Primary Key (PK)	Foreign Key (FK)	MANAGER	ManagerID	✓		SHOP	ManagerID		✓	CAR	RegistrationNumber	✓		CAR	ShopID		✓	
Table	Field name	Primary Key (PK)	Foreign Key (FK)																			
MANAGER	ManagerID	✓																				
SHOP	ManagerID		✓																			
CAR	RegistrationNumber	✓																				
CAR	ShopID		✓																			
5(b)	1 mark per bullet point - Access rights give managers / himself access to different elements ... by having different accounts / logins ... which have different access rights e.g. read only // no access / read / write - Specific <u>views</u> can be assigned to himself and to the managers ... e.g. managers can only see the data for their own shop(s)	3																				
5(c)(i)	1 mark per correctly completed statement SELECT COUNT(RegistrationNumber) FROM CAR GROUP BY ShopID	3																				
5(c)(ii)	1 mark for each correct statement INSERT INTO CAR VALUES ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")	2																				

26. O/N 21/P12/Q6/a,b,c

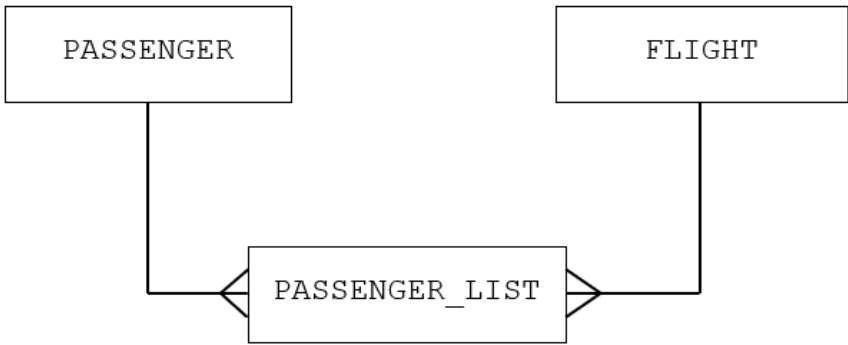
6(a)(i)	1 mark for 1 tick in the correct place 2 marks for all 3 ticks correct <table><thead><tr><th rowspan="2">Task</th><th colspan="3">Normalisation stage</th></tr><tr><th>0NF to 1NF</th><th>1NF to 2NF</th><th>2NF to 3NF</th></tr></thead><tbody><tr><td>Remove any partial key dependencies</td><td></td><td>✓</td><td></td></tr><tr><td>Remove any repeating groups of attributes</td><td>✓</td><td></td><td></td></tr><tr><td>Remove any non-key dependencies</td><td></td><td></td><td>✓</td></tr></tbody></table>	Task	Normalisation stage			0NF to 1NF	1NF to 2NF	2NF to 3NF	Remove any partial key dependencies		✓		Remove any repeating groups of attributes	✓			Remove any non-key dependencies			✓	2
Task	Normalisation stage																				
	0NF to 1NF	1NF to 2NF	2NF to 3NF																		
Remove any partial key dependencies		✓																			
Remove any repeating groups of attributes	✓																				
Remove any non-key dependencies			✓																		
6(a)(ii)	1 mark for each correct relationship 	3																			
6(b)	1 mark for description of purpose - Stores metadata about the database 1 mark for each example of contents to max 2 e.g. - field / attribute names - table name - validation rules - data types - primary keys // foreign keys - relationships	3																			
6(c)(i)	1 mark for each correctly completed space <pre>SELECT SUM(Quantity) FROM PURCHASE_ITEM WHERE PurchaseID = "3011A";</pre>	4																			

6(c)(ii)	1 mark per bullet point - ALTER TABLE PURCHASE - ADD OrderDate - Suitable data type, e.g. DATE <pre>ALTER TABLE PURCHASE ADD OrderDate DATE;</pre>	3
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27. O/N 21/P13/Q5/a,b,c

5(a)	1 mark for 2 or 3 correct ticks, 2 marks for 4 correct ticks	2																				
	<table><tr><th>Table</th><th>Field name</th><th>Primary Key (PK)</th><th>Foreign Key (FK)</th></tr><tr><td>MANAGER</td><td>ManagerID</td><td>✓</td><td></td></tr><tr><td>SHOP</td><td>ManagerID</td><td></td><td>✓</td></tr><tr><td>CAR</td><td>RegistrationNumber</td><td>✓</td><td></td></tr><tr><td>CAR</td><td>ShopID</td><td></td><td>✓</td></tr></table>	Table	Field name	Primary Key (PK)	Foreign Key (FK)	MANAGER	ManagerID	✓		SHOP	ManagerID		✓	CAR	RegistrationNumber	✓		CAR	ShopID		✓	
Table	Field name	Primary Key (PK)	Foreign Key (FK)																			
MANAGER	ManagerID	✓																				
SHOP	ManagerID		✓																			
CAR	RegistrationNumber	✓																				
CAR	ShopID		✓																			
5(b)	1 mark per bullet point - Access rights give managers / himself access to different elements ... by having different accounts / logins ... which have different access rights e.g. read only // no access / read / write - Specific <u>views</u> can be assigned to himself and to the managers ... e.g. managers can only see the data for their own shop(s)	3																				
5(c)(i)	1 mark per correctly completed statement SELECT COUNT(RegistrationNumber) FROM CAR GROUP BY ShopID	3																				
5(c)(ii)	1 mark for each correct statement INSERT INTO CAR VALUES ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")	2																				

28. 9608 O/N 21/P11/Q9/a,b

9(a)	One mark per correct relation 	2
9(b)	One mark per bullet point • Create table FLIGHT with open and close brackets (and ;) • FlightID as VARCHAR restricted to max 6 • FlightDate as DATE • FlightTime as TIME • Declaring FlightID as Primary Key Example <pre>CREATE TABLE FLIGHT (FlightID VARCHAR(6), FlightDate DATE, FlightTime TIME, PRIMARY KEY (FlightID));</pre>	5

29. M/J 21/P11/Q7/a,b,c

7(a)	1 mark per bullet point to max 3 • Flat-file has more data redundancy • ... because the same data is stored many times // data is stored in different tables which are linked • There is program-data dependence with flat-files • ... because any changes to the structure of the data means the programs that access that data have to be re-written • Flat-file has more data inconsistency // worse data integrity • ... because duplicated data might be stored differently //...because when data is updated in one place, it is not updated everywhere • It is not easy to perform complex searches /queries • ... because a new program has to be written each time • Flat files could have a lack of privacy • ... as user views cannot easily be implemented	3
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7(b)(i)	1 mark for each correct example one-to-one - e.g. customer to payment details // customer to login details one-to-many - e.g. customer to order many-to-many - e.g. order to product // customer to product	3								
7(b)(ii)	1 mark <table border="1"><thead><tr><th>Relationship</th><th>Tick (✓)</th></tr></thead><tbody><tr><td>one-to-one</td><td></td></tr><tr><td>one-to-many</td><td></td></tr><tr><td>many-to-many</td><td>✓</td></tr></tbody></table>	Relationship	Tick (✓)	one-to-one		one-to-many		many-to-many	✓	1
Relationship	Tick (✓)									
one-to-one										
one-to-many										
many-to-many	✓									
7(b)(iii)	1 mark CREATE DATABASE SHOPORDERS;	1								
7(c)	1 mark per item to max 3 - table name - field name // attribute - data type - type of validation - Primary Key - Foreign Key - relationships	3								

30. M/J 21/P12/Q1/a,b,c

1(a)	1 mark for definition, 1 mark for appropriate example in each	6								
	<table><tr><td>Term</td><td>Definition and example</td></tr><tr><td>Field</td><td>A column/attribute in a table e.g. CustomerID in the table CUSTOMER</td></tr><tr><td>Entity</td><td>Anything that data can be stored about e.g. A customer or a house</td></tr><tr><td>Foreign Key</td><td>A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u></td></tr></table>	Term	Definition and example	Field	A column/attribute in a table e.g. CustomerID in the table CUSTOMER	Entity	Anything that data can be stored about e.g. A customer or a house	Foreign Key	A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u>	
Term	Definition and example									
Field	A column/attribute in a table e.g. CustomerID in the table CUSTOMER									
Entity	Anything that data can be stored about e.g. A customer or a house									
Foreign Key	A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u>									
1(b)	1 mark per bullet point to max 2 - All fields in all tables are dependant fully on the PK and on no other fields - for example all fields in Customer table are fully dependent on CustomerID	2								

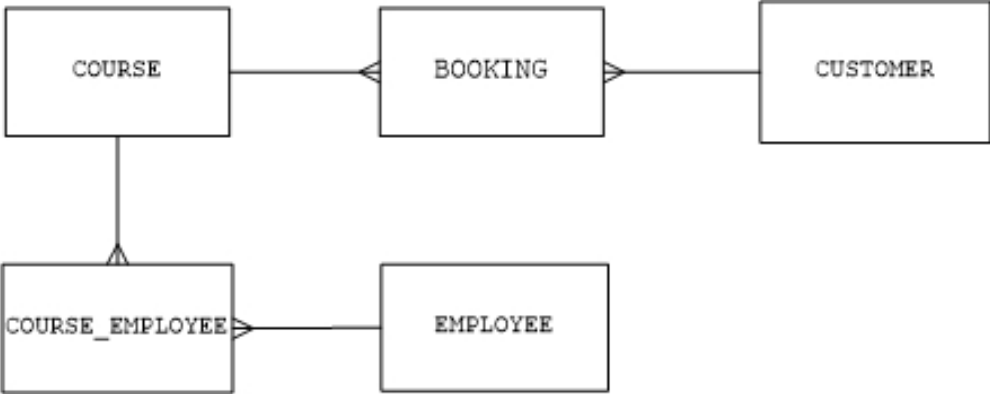
1(c)(i)	1 mark for each correctly completed line <pre>CREATE TABLE RENTAL (RentalID INTEGER NOT NULL, CustomerID INTEGER NOT NULL, HouseID VARCHAR (5) NOT NULL, MonthlyCost REAL/CURRENCY NOT NULL, DepositPaid BOOLEAN NOT NULL, PRIMARY KEY (RentalID));</pre>	4
1(c)(ii)	1 mark per bullet point • Select FirstName and LastName • From both tables • Where DepositPaid = No • Joining tables (either AND, or INNER JOIN) Example script: <pre>SELECT FirstName, LastName FROM CUSTOMER, RENTAL WHERE DepositPaid = No AND RENTAL.CustomerID = CUSTOMER.CustomerID;</pre>	4

31. M/J 21/P13/Q7/a,b,c

7(a)	1 mark per bullet point to max 3 • Flat-file has more data redundancy • ... because the same data is stored many times // data is stored in different tables which are linked • There is program-data dependence with flat-files • ... because any changes to the structure of the data means the programs that access that data have to be re-written • Flat-file has more data inconsistency // worse data integrity • ... because duplicated data might be stored differently //...because when data is updated in one place, it is not updated everywhere • It is not easy to perform complex searches /queries • ... because a new program has to be written each time • Flat files could have a lack of privacy • ... as user views cannot easily be implemented	3
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7(b)(i)	1 mark for each correct example one-to-one - e.g. customer to payment details // customer to login details one-to-many - e.g. customer to order many-to-many - e.g. order to product // customer to product	3								
7(b)(ii)	1 mark <table border="1"><thead><tr><th>Relationship</th><th>Tick (✓)</th></tr></thead><tbody><tr><td>one-to-one</td><td></td></tr><tr><td>one-to-many</td><td></td></tr><tr><td>many-to-many</td><td>✓</td></tr></tbody></table>	Relationship	Tick (✓)	one-to-one		one-to-many		many-to-many	✓	1
Relationship	Tick (✓)									
one-to-one										
one-to-many										
many-to-many	✓									
7(b)(iii)	1 mark CREATE DATABASE SHOPORDERS ;	1								
7(c)	1 mark per item to max 3 - table name - field name // attribute - data type - type of validation - Primary Key - Foreign Key - relationships	3								

32. 9608 M/J 21/P11/Q5/a,b,c,d

5(a)	1 mark for each link: • 1 CUSTOMER to many BOOKING • 1 COURSE to many BOOKING • 1 COURSE to many COURSE_EMPLOYEE • 1 EMPLOYEE to many COURSE_EMPLOYEE  <pre> graph LR COURSE --> BOOKING COURSE --> COURSE_EMPLOYEE BOOKING --> CUSTOMER COURSE_EMPLOYEE --> EMPLOYEE </pre>	4
5(b)	1 mark for description and 1 mark for application to the given tables e.g. (2 marks) • each value stored in the <code>CustomerID</code> (FK) field in the <code>Booking</code> table must have a corresponding value (1) in the <code>CustomerID</code> (PK) field in the <code>Customer</code> table (1) e.g. (1 mark) • Each foreign key value must have a matching value in the primary key of the linked table (1)	2
5(c)	1 mark each (max 3) e.g. • Tables • Fields/attributes • Indexes • Users • Primary Key • Foreign Key • Relationships • Views	3

5(d)	1 mark per syntactically correct bullet point: • Select correct fields • From correct table • Correct criteria for Role • Correct criteria for Language Example: SELECT FirstName, LastName FROM EMPLOYEE WHERE Role = "Leader" AND (Language = "French" OR Language = "English");	4
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33. 9608 M/J 21/P12/Q4/a,b,c

4(a)	1 mark per drawback identified, 1 mark for related description (max 6) • There is data duplication // redundant data • ...because the same data is stored multiple times // data changed in one file is not automatically changed in others // by example • There could be data inconsistency // reduced data integrity • ... because duplicated data might be stored as different values // by example • There is program-data dependency • ... if the data structure changes all the programs accessing that data must be changed too // by example • It is not easy to perform complex searches/queries • ... a new program has to be written each time • Lack of privacy • ...as access controls are usually to the system rather than the data // user views cannot easily be implemented	6
4(b)	1 mark per bullet point (max 3) • There is no unique <u>primary</u> key • Customer name is not atomic // Customer name needs to be split into first name and last name // by example • Customer name / Customer date of birth / Destination / Guide name / Trip have repeated groups of attributes	3

4(c)	1 mark per bullet point • Create table <code>GUIDE_TABLE</code> with brackets and semi-colon • <code>GuideID</code> of suitable data type • <code>Guide</code> and <code>Gender</code> of suitable data type(s) • <code>DateOfBirth</code> of correct data type • Declaring <code>GuideID</code> as Primary Key Example answer: <pre>CREATE TABLE GUIDE_TABLE(GuideID INT, Guide VARCHAR(25), DateOfBirth DATE, Gender CHAR(1), PRIMARY KEY (GuideID));</pre>	5
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34. 9608 M/J 21/P13/Q8/a,b,c,d

8(a)	1 mark for each correct join	2
8(b)	1 mark per bullet point • <code>EmployeeID</code> is the Primary Key in <code>EMPLOYEE</code> • ...links to <code>EmployeeID</code> which is the Foreign Key in <code>PROJECTTEAM</code> • <code>ProjectID</code> is the Primary Key in <code>PROJECT</code> • ...links to <code>ProjectID</code> which is the Foreign Key in <code>PROJECTTEAM</code>	4
8(c)	1 mark per bullet point • <code>ALTER TABLE EMPLOYEE</code> • <code>ADD Gender CHAR(1)</code> Example: <pre>ALTER TABLE EMPLOYEE ADD Gender CHAR(1);</pre>	2

8(d)	1 mark per bullet point • <code>SELECT FirstName, LastName, Salary</code> • <code>FROM EMPLOYEE</code> • <code>WHERE Salary > 17 500</code> Example: <code>SELECT FirstName, LastName, Salary</code> <code>FROM EMPLOYEE</code> <code>WHERE Salary > 17 500;</code>	3
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35. M/J 20/P11/Q7/a,b,c,d,e

7(a)	1 mark per bullet point to max 2 • Reduced data redundancy • Reduced data dependency • Improved data integrity • Improved data privacy • Program-data independence • Ability to create ad hoc queries	2
7(b)	1 mark for each correct link <pre> classDiagram INSTRUCTOR --> LESSON LESSON --> STUDENT INSTRUCTOR --> INSTRUCTOR_CAR INSTRUCTOR_CAR --> CAR </pre>	4

7(c)	1 mark for each correctly completed statement - CREATE (line 1) - INTEGER (line 6) - PRIMARY KEY (line 7) <pre>CREATE TABLE INSTRUCTOR (InstructorID VARCHAR(5), FirstName VARCHAR(15), LastName VARCHAR(15), DateOfBirth DATE, Level INTEGER, PRIMARY KEY (InstructorID));</pre>	3
7(d)	1 mark per bullet point - Alter table student - Add an appropriate identifier with suitable data type <pre>ALTER TABLE STUDENT ADD TelNum VARCHAR;</pre>	2
7(e)	1 mark per bullet point - Select lesson date and lesson time - From table LESSON - Where InstructorID = "Ins01" - And lesson date is greater than today's date <pre>SELECT LessonDate, LessonTime FROM LESSON WHERE InstructorID = "Ins01" AND LessonDate > #####;</pre>	4

36. M/J 20/P12/Q6/a,b,c

6(a)	1 mark per bullet point to max 2 - There are partial dependencies in the SOFTWARE_PURCHASED table // SoftwareDescription is dependent only on SoftwareName and not both SoftwareName and CustomerID - There is a non-key dependency in the SOFTWARE_PURCHASED table // LicenceCost is dependent on LicenceType	2
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6(b)	1 mark for a suitable example for each <table><tr><td>Term</td><td>Example</td></tr><tr><td>Entity</td><td>SOFTWARE_PURCHASED //CUSTOMER_DETAILS</td></tr><tr><td>Foreign Key</td><td>CustomerID (in SOFTWARE_PURCHASED table)</td></tr><tr><td>Attribute</td><td>Any valid example of an attribute from the tables</td></tr></table>	Term	Example	Entity	SOFTWARE_PURCHASED //CUSTOMER_DETAILS	Foreign Key	CustomerID (in SOFTWARE_PURCHASED table)	Attribute	Any valid example of an attribute from the tables	3
Term	Example									
Entity	SOFTWARE_PURCHASED //CUSTOMER_DETAILS									
Foreign Key	CustomerID (in SOFTWARE_PURCHASED table)									
Attribute	Any valid example of an attribute from the tables									
6(c)(i)	1 mark for each correct entry (in bold) CREATE TABLE GAME_DEVELOPMENT (GameName VarChar, Genre VarChar, TeamNumber Integer, DevelopmentStage VarChar, ManagerID VarChar, PRIMARY KEY (GameName));	5								
6(c)(ii)	1 mark for each correct entry (shown in brackets) SELECT (1) GameName, Genre, TeamNumber FROM GAME_DEVELOPMENT, PRODUCT_MANAGER WHERE PRODUCT_MANAGER.FirstName = "James" AND PRODUCT_MANAGER.SecondName = "Fitz" AND PRODUCT_MANAGER.ManagerID (1) = GAME_DEVELOPMENT.ManagerID (1) ;	3								

37. M/J 20/P13/Q6/a,b,c

6(a)	1 mark for each correct line <table><thead><tr><th>Database Term</th><th>Description</th></tr></thead><tbody><tr><td>Primary key</td><td>A field in one table that links to a primary key in another table</td></tr><tr><td>Attribute</td><td>A collection of records and tables</td></tr><tr><td>Foreign key</td><td>The type of data that is being stored</td></tr><tr><td>Entity</td><td>A unique identifier for each tuple</td></tr><tr><td></td><td>A data item, represented as a field within a table</td></tr><tr><td></td><td>The concept or object in the system that we want to model and store information about</td></tr></tbody></table>	Database Term	Description	Primary key	A field in one table that links to a primary key in another table	Attribute	A collection of records and tables	Foreign key	The type of data that is being stored	Entity	A unique identifier for each tuple		A data item, represented as a field within a table		The concept or object in the system that we want to model and store information about	4
Database Term	Description															
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Entity	A unique identifier for each tuple															
	A data item, represented as a field within a table															
	The concept or object in the system that we want to model and store information about															
6(b)	1 mark per task to max 3 • Create a table • Set up relationships between tables • Create / design a form • Create / design a report • Create / design a query (NOT run a query)	3														
6(c)(i)	1 mark for each completed line <pre>CREATE TABLE ROOM(RoomNumber Integer, RoomType Varchar, PRIMARY KEY (RoomNumber));</pre>	3														
6(c)(ii)	1 mark for each completed line <pre>INSERT INTO ROOM VALUES (5, "Double");</pre>	2														

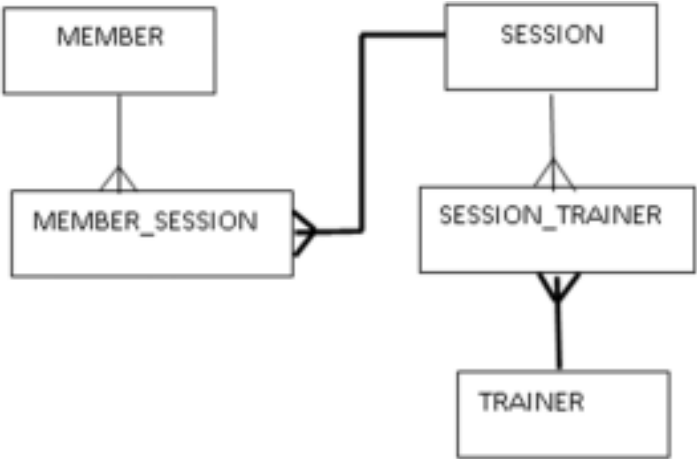
6(c)(iii)	1 mark per bullet point - Alter table booking - Add number of nights with appropriate field name and data type <pre>ALTER TABLE BOOKING ADD NumberNights Integer;</pre>	2
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38. O/N 19/P11/Q4/d

4(d)(i)	1 mark per table - Table CUSTOMER with fields FirstName, LastName, DateOfBirth, CustomerID - Table ACCOUNT_TYPE with fields AccountID, Name, Bonus - Table CUSTOMER_ACCOUNT with fields ID, CustomerID, AccountID, Amount CUSTOMER (<u>CustomerID</u>, FirstName, LastName, DateOfBirth) ACCOUNT_TYPE (<u>AccountID</u>, Name, Bonus) CUSTOMER_ACCOUNT (<u>ID</u>, CustomerID, AccountID, Amount)	3								
4(d)(ii)	1 mark for 1 or 2 correct Primary Keys, 2 marks for 3 correct Primary Keys CUSTOMER: CustomerID ACCOUNT_TYPE: AccountID CUSTOMER_ACCOUNT: ID	2								
4(d)(iii)	1 mark for both table name and Foreign Key Table: CUSTOMER_ACCOUNT Foreign Key: CustomerID / AccountID	1								
4(d)(iv)	1 mark for each correct term <table><tr><th>Definition</th><th>Term</th></tr><tr><td>All the data about one entity</td><td>Table / Relation</td></tr><tr><td>The data in one row of a table</td><td>Tuple / Record</td></tr><tr><td>A column or field in a table</td><td>Attribute</td></tr></table>	Definition	Term	All the data about one entity	Table / Relation	The data in one row of a table	Tuple / Record	A column or field in a table	Attribute	3
Definition	Term									
All the data about one entity	Table / Relation									
The data in one row of a table	Tuple / Record									
A column or field in a table	Attribute									

39. O/N 19/P12/Q4/a,b

4(a)(i)	1 mark per bullet point, max 3 marks from any group to max 4 • Multiple tables are linked together • ... which eliminates / reduces data redundancy / duplication • ... and increases <u>data</u> integrity / consistency • ... which reduces compatibility issues • ... so data need only be updated once • ... and associated data will be automatically updated // referential integrity can be enforced • ... which eliminates unproductive maintenance // which makes it easier to maintain the data • Program-data independence means that • ... the structure of data can change and does not affect program • ... the structure of programs can change and does not affect data • ... the data can be accessed by any appropriate program • Allows concurrent access to data • ... by the use of record locking • ... by restricting over-writing changes • Complex queries can be more easily written • ... to search / find specific data // specific example related to the sports club • Different users can be given different access rights • ... which improves security • Different users can be given different views of the data • ... so they do not see confidential information • ... and data privacy is maintained • ... accept a valid example related to the sports club	4
4(a)(ii)	1 mark for each word in the correct position For a database to be in First Normal Form (1NF) there must be no repeating groups of attributes. For a database to be in Second Normal Form (2NF), it must be in 1NF, and contain no partial key dependencies. For a database to be in Third Normal Form (3NF), it must be in 2NF, and all attributes must be fully dependent on the primary key.	4

4(b)(i)	1 mark for SESSION to MEMBER_SESSION (one-to-many) 1 mark for TRAINER to SESSION_TRAINER (one-to-many) 	2
4(b)(ii)	CREATE DATABASE SPORTS_CLUB;	1
4(b)(iii)	1 mark per bullet point • CREATE TABLE SESSION and (); • SessionID as char(4) / varchar / varchar(4) / text and Description as varchar / varchar(x) / char(x) / text • SessionDate as date and SessionTime as time • NumberMembers as integer / integer(2) / int / int(2) • SessionID set as a Primary Key Example 1: CREATE TABLE SESSION(SessionID Char(4), Description Varchar, SessionDate Date, SessionTime Time, NumberMembers Integer, PRIMARY KEY (SessionID)); Example 2: CREATE TABLE SESSION(SessionID Char(4) <u>NOT NULL</u> PRIMARY KEY, Description Varchar, SessionDate Date, SessionTime Time, NumberMembers Integer);	5

4(b)(iv)	1 mark per bullet point - Select FirstName and comma and LastName - From MEMBER - Where MembershipType = "Peak" <pre>SELECT FirstName, LastName FROM MEMBER WHERE MembershipType = "Peak";</pre>	3
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40. O/N 19/P13/Q3/a,b,c

3(a)(i)	1 mark per table - CUSTOMER table has at least customer ID, customer name, address and contact details - ROOM has at least room number, room type, - BOOKING has at least booking ID, room number, customer ID, start date, number of nights <pre>CUSTOMER (CustomerID, Name, Address, ContactDetails) ROOM (RoomNumber, RoomType) BOOKING (BookingID, RoomNumber, CustomerID, StartDate, NumberNights)</pre>	3
3(a)(ii)	1 mark for 1 or 2 correct Primary Keys, 2 marks for 3 correct Primary Keys <pre>CUSTOMER: CustomerID ROOM: RoomNumber BOOKING: BookingID</pre>	2
3(a)(iii)	1 mark for both table name and Foreign Key <pre>Table: BOOKING Foreign Key: CustomerID / RoomNumber</pre>	1
3(b)	1 mark per bullet point to max 2 plus 1 mark for suitable example for each DBMS tool Developer Interface - To create user friendly features e.g. forms to enter new bookings - To create outputs e.g. report of bookings on a given date - To create interactive features e.g. buttons and menus Query processor - To create SQL/QBE queries - To search for data that meets set criteria, e.g. all bookings for next week - To perform calculations on extracted data, e.g. number of empty rooms tomorrow	5

3c	1 mark for at least two correct rows, 2 marks for all four correct rows			2
	Script	DDL	DML	
	CREATE TABLE FILMS	✓		
	SELECT FilmID FROM FILMS		✓	
	ALTER TABLE FILMS ADD PRIMARY KEY (FilmID)	✓		
	CREATE DATABASE MYDATA	✓		

41. M/J 19/P11/Q2/b(i),c

2(b)(i)	1 mark per bullet point to max 3 • Data redundancy // data is repeated in more than one file • Data dependency // changes to data means changes to programs accessing that data • Lack of data integrity // entries that should be the same can be different in different places • Lack of data privacy // all users have access to all data if a single flat file	3
2(c)(i)	1-to-many // 1 customer to/has many licences	1
2(c)(ii)	1 mark per bullet point • <u>CustomerID</u> is the Primary key in <u>CUSTOMER</u> table • Links to <u>CustomerID</u> as a Foreign key in <u>LICENCE</u> table	2
2(c)(iii)	1 mark per bullet point • Select with correct 5 fields • From LICENCE • Where ExpiryDate <= '31/12/2019' (any appropriate date type) • Group by CustomerID • Order by Cost (with or without ASC, but not DESC) <pre>SELECT CustomerID, SoftwareID, LicenceType, Cost, ExpiryDate FROM LICENCE WHERE ExpiryDate <= '31/12/2019' GROUP BY CustomerID ORDER BY Cost;</pre>	5

42. M/J 19/P12/Q5/a,b(ii,iii)

5(a)(i)	1 mark for correct answer <u>Repeated</u> / <u>duplicated</u> data	1
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5(a)(ii)	1 mark per bullet point - Because each record/piece of data is stored once <u>and</u> is referenced by a (primary) key - Because data is stored in individual tables ...and the tables are linked by relationships - By the proper use of Primary and Foreign keys - By enforcing referential integrity - By going through the normalisation process	3
5(b)(ii)	1 mark for naming, 1 mark for description For example: - Access rights // User accounts - Restrict actions (e.g. read / read-write) of specific users // unauthorised users cannot access the database - Views - Restrict which parts of the database specific users can see - Password // Biometrics // PIN code - Prevents unauthorised access - Automatic Backup - Create regular copies of data in case of loss - Encryption - Data is incomprehensible to unauthorised users	4
5(b)(iii)	1 mark per bullet - Query Processor - Developer Interface	2

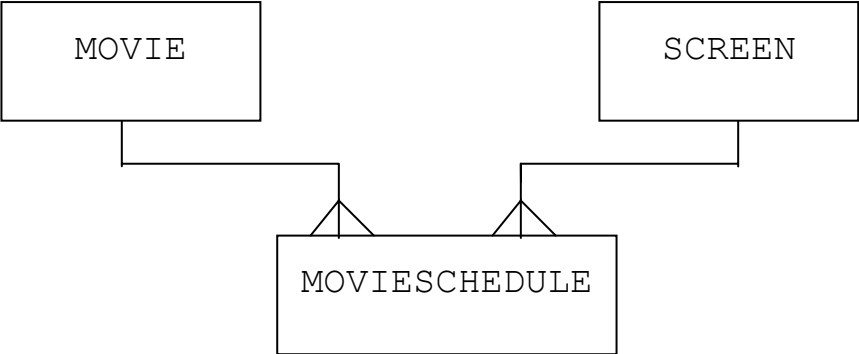
43. M/J 19/P13/Q3/a,b,c,d,e

3(a)(i)	1 mark per bullet point - Stores all the information about the database // data about the data // metadata about the data - For example, fields, data types, validation, keys	2
3(a)(ii)	1 mark per bullet point to max 2 - Allows the user to enter criteria - Searches for data which meets the entered criteria - Organises the results to be displayed to the user	2

3(b)	1 mark per bullet point to max 2 - Primary key uniquely identifies each tuple // Each tuple in the table is unique - Primary key can be used as a foreign key in another table to form a link/relationship between the tables By example: - Identification of a primary key in a table - Describing <u>that</u> primary key in another table as a foreign key	2
3(c)	1 mark for each correct link <pre> graph TD EMPLOYEE_DATA[EMPLOYEE_DATA] --- DEPARTMENT_MANAGER[DEPARTMENT_MANAGER] EMPLOYEE_DATA --- DEPARTMENT[DEPARTMENT] DEPARTMENT_MANAGER --- DEPARTMENT </pre>	3
3(d)	1 mark per bullet point to max 3 - There are no repeating groups (1NF) - There are no partial dependencies (2NF) - There are no non-key dependencies // There are no transitive dependencies (3NF)	3
3(e)(i)	1 mark for correct answer <pre>CREATE DATABASE EMPLOYEES;</pre>	1

3(e)(ii)	1 mark per bullet • Create table EMPLOYEE_DATA with open and close brackets • EmployeeID as VarChar restricted to max 7, Gender as a VarChar restricted to max 6, DepartmentNumber as a VarChar restricted to max 2 • FirstName and LastName as VarChar (any max lengths must be reasonable) • Date of birth as Date • Declaring EmployeeID as PK <pre>CREATE TABLE EMPLOYEE_DATA (EmployeeID VarChar(7), FirstName VarChar, LastName VarChar, DateOfBirth Date, Gender VarChar(6), DepartmentNumber VarChar(2), PRIMARY KEY (EmployeeID));</pre>	5
3(e)(iii)	1 mark per bullet • Select FirstName and LastName only • From both tables • Where DepartmentName = "Finance" • AND Gender = "Female" • Joining tables (either AND, or inner join) <pre>SELECT FirstName, LastName FROM EMPLOYEE_DATA, DEPARTMENT WHERE DepartmentName = "Finance" AND Gender = "Female" AND DEPARTMENT.DepartmentNumber = EMPLOYEE_DATA.DepartmentNumber;</pre>	5

44. O/N 18/P11/Q7/a,b,c,d

7(a)	1 mark for each correct join  <pre> graph TD MOVIE[MOVIE] --- MOVIESCHEDULE[MOVIESCHEDULE] SCREEN[SCREEN] --- MOVIESCHEDULE style MOVIESCHEDULE fill:none,stroke:none style MOVIE fill:none,stroke:none style SCREEN fill:none,stroke:none </pre>	2
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7(b)	1 mark per bullet point • <code>MovieID</code> is the Primary Key in <code>MOVIE</code> • ... <u>links to</u> <code>MovieID</code> which is the Foreign Key in <code>MOVIESCHEDULE</code> • <code>ScreenNumber</code> is the Primary Key in <code>SCREEN</code> • ... <u>links to</u> <code>ScreenNumber</code> which is the Foreign Key in <code>MOVIESCHEDULE</code>	4
7(c)	1 mark per bullet point • <code>ALTER TABLE MOVIE</code> • <code>ADD (COLUMN) ProductionCompany VARCHAR(25);</code>	2
7(d)	1 mark per bullet point Answer 1: • <code>SELECT Title, Rating</code> • <code>FROM MOVIE, MOVIESCHEDULE</code> • <code>WHERE MOVIE.MovieID = MOVIESCHEDULE.MovieID</code> • <code>AND MOVIESCHEDULE.ScreenNumber = 3;</code> Or Answer 2: • <code>SELECT Title, Rating</code> • <code>FROM MOVIE INNER JOIN MOVIESCHEDULE</code> • <code>ON MOVIE.MovieID = MOVIESCHEDULE.MovieID</code> • <code>WHERE MOVIESCHEDULE.ScreenNumber = 3;</code>	4

45. M/J 18/P12/Q7/a,b,c,d

7(a)(i)	1 mark per bullet • <u>UserName</u> is the <u>primary key</u> in <u>USER</u> • <u>UserName</u> is (included as) a <u>foreign key</u> in <u>PHOTO</u>	2
7(a)(ii)	1 mark for each correct relationship  <pre> graph LR PHOTO --> USER USER --> TEXTPOST </pre>	2

7(b)	1 mark per bullet to max 2 for explanation • Referential integrity is making sure tables do not try to reference data which does not exist // A value of one attribute of a table exists as a value of another attribute in a different table • A primary key cannot be deleted unless all dependent records are already deleted • Cascading delete • A primary key cannot be updated unless all dependent records are already updated • Cascading update / edit • Every foreign key value has a matching value in the corresponding primary key • The foreign keys must be the same data type as the corresponding primary key 1 mark for a suitable example e.g. • A <code>UserName</code> cannot be deleted from the <code>USER</code> table if they have a related <code>photo/textpost</code> • If <code>UserName</code> is updated in <code>USER</code> table, it must also be updated in <code>PHOTO</code> and <code>TEXTPOST</code> tables • Cannot create/edit a record in <code>TEXTPOST</code> / <code>PHOTO</code> without a matching entry in <code>USER</code> table	3
7(c)	Max 1 mark from each bulleted group 1NF • No repeated groups of attributes • All attributes should be atomic • No duplicate rows 2NF (in 1NF and) • No partial dependencies 3NF (in 2NF and) • No non-key dependencies • No transitive dependencies	3
7(d)(i)	1 mark per bullet • <code>CREATE TABLE USER and ();</code> • <code>UserName, FirstName and SecondName as VARCHAR and commas</code> • <code>DateOfBirth as DATE and comma</code> • <code>PRIMARY KEY(UserName)</code> • An appropriate <code>NOT NULL</code> <pre>CREATE TABLE USER(UserName: varchar(15) NOT NULL, FirstName: varchar(25), SecondName: varchar(25), DateOfBirth: Date, PRIMARY KEY(UserName));</pre>	5

7(d)(ii)	1 mark per bullet - ALTER TABLE USER - ADD COUNTRY varchar; ALTER TABLE USER ADD Country varchar;	2
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46. M/J 18/P13/Q2/a,b,c,d

2(a)	1 mark for each correct relationship 	2				
2(b)	1 mark for description - Ensure data is consistent / accurate // keep data consistent / accurate 1 mark for example from: e.g. - Validation rules - Referential integrity - Verification - Input masks - Setting data types - Removing redundant data - Backup data - Access controls - Audit trail	2				
2(c)	1 mark for the correct box ticked <table border="1" data-bbox="620 1337 991 1438"><tr><td>True</td><td>False</td></tr><tr><td>✓</td><td></td></tr></table> 1 mark per bullet for justification, max 2 - No repeated attributes // data is atomic // No partial dependencies (no dual keys) - No non-key / transitive dependencies	True	False	✓		3
True	False					
✓						

2(d)(i)	1 mark per bullet - CREATE TABLE PLAYER and (); - PlayerID and PlayerName as VARCHAR and commas - SkillLevel as INT and comma - PRIMARY KEY(PlayerID) - An appropriate NOT NULL <pre>CREATE TABLE PLAYER(PlayerID: varchar NOT NULL, PlayerName: varchar, SkillLevel: int, PRIMARY KEY(PlayerID),);</pre>	5
2(d)(ii)	1 mark per bullet - ALTER TABLE PLAYER - ADD DateOfBirth Date;	2

47. O/N 17/P12/Q7/a,b,c,d,e

7(a)(i)	1 Mark for correct primary key identified in both STAFF and CLIENT STAFF(StaffID, StaffName, Department) CLIENT(<u>ClientName</u> , Address, Town) 1 Mark for correct primary key identified in VISIT VISIT(<u>ClientName</u> , VisitDate) 1 Mark for correct primary key identified in INTERVIEW INTERVIEW(<u>ClientName</u> , <u>VisitDate</u> , StaffID, SpecialistFocus, InterviewText)	3
7(a)(ii)	1 Mark for each correct relationship <pre> graph LR CLIENT --- 1:M VISIT VISIT --- 1:M INTERVIEW INTERVIEW --- 1:M STAFF </pre>	3
7(b)	1 Mark for correct answer Add attribute VisitReportText to table <u>VISIT</u>	1
7(c)(i)	1 Mark for each correct line <pre>UPDATE CLIENT SET ClientName = 'Albright Holdings' WHERE ClientName = 'ABC Holdings';</pre>	3

7(c)(ii)	1 Mark per bullet, max 2 - Referential integrity should be maintained // Referential integrity could be violated - Data becomes inconsistent - There may be records in the <code>VISIT</code> and <code>INTERVIEW</code> tables / other tables with client name ABC Holdings - The <code>ClientName</code> in the <code>VISIT</code> and <code>INTERVIEW</code> tables / other tables might not be automatically updated - Records in the <code>VISIT</code> and <code>INTERVIEW</code> tables / other tables will become orphaned	2
7(d)	1 Mark for each correct line <pre>SELECT StaffID FROM INTERVIEW WHERE ClientName = 'New Age Toys' AND VisitDate = '13/10/2016'; (Accept clauses other way round)</pre>	3
7(e)	1 Mark for a correct answer Add a suitable attribute, for example, <code>EuropeTraveller</code> to the <code>STAFF</code> table // Add a suitable attribute, for example, <code>Country</code> to the <code>CLIENT</code> table	1

48. O/N 17/P13/Q7/a,b,c,d,e

7(a)(i)	<u>PatientID</u> } <u>DoctorID</u> (1) <u>AppointmentDate, AppointmentTime</u> (1)	2
7(a)(ii)	<pre> graph TD PATIENT((PATIENT)) DOCTOR((DOCTOR)) APPOINTMENT((APPOINTMENT)) PATIENT === DOCTOR PATIENT === APPOINTMENT DOCTOR === APPOINTMENT </pre> One <code>PATIENT</code> attends many <code>APPOINTMENTs</code> One <code>DOCTOR</code> takes many <code>APPOINTMENTs</code> Special case for 1 mark only (only if no one to many relationships shown) Many <code>PATIENTs</code> are seen by many <code>DOCTORs</code>	2
7(b)	Two marks from: Either: - Add an attribute (for example <code>Attended</code>) - To the appointment table // <code>APPOINTMENT</code> Or: - Add an attribute (for example <code>AppointmentsMissed</code>) - To the patient table // <code>PATIENT</code>	2

7(c)(i)	Available to work at both SITE-A and SITE-B	1
7(c)(ii)	APPOINTMENT(Site, AppointmentDate, AppointmentTime, DoctorID, PatientID)	1
7(d)(i)	One mark per line UPDATE DOCTOR SET DoctorID = '017' WHERE DoctorID = '117';	3
7(d)(ii)	1 Mark per bullet, max 2 - Referential integrity should be maintained // Referential integrity could be violated. - Data becomes inconsistent - There may be records in the APPOINTMENT table showing doctor ID 117 - The APPOINTMENT table might not be automatically updated - Records in the APPOINTMENT table will become orphaned	Max 2
7(e)	One mark per line SELECT AppointmentDate, AppointmentTime FROM APPOINTMENT WHERE PatientID = '556';	3

49. M/J 17/P11/Q1/a,b,c,d

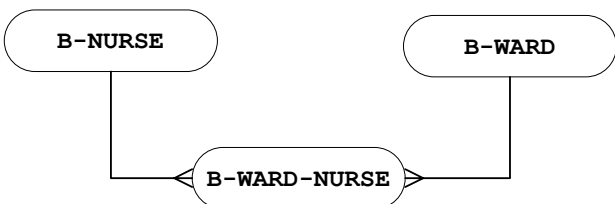
1(a)	Many-to-one	1
1(b)(i)	A-NURSE(<u>NurseID</u> , FirstName, FamilyName, WardName)	1
1(b)(ii)	- The primary key <u>WardName</u> in the A-WARD table links to the foreign key <u>WardName</u> in the A-NURSE table.	1 1 2
1(c)(i)	Many-to-many relationship	1
1(c)(ii)	B-WARD-NURSE(<u>WardName</u> , <u>NurseID</u>) Both attributes (with no additions) 1 Joint primary key correctly underlined 1	2
1(c)(iii)	Correct relationship between B-NURSE and B-WARD-NURSE 1 Correct relationship between B-WARD and B-WARD-NURSE 1	2
1(d)(i)	SELECT NurseID, FamilyName FROM B-NURSE WHERE Specialism = 'THEATRE';	1 1 1 3

1(d)(ii)	UPDATE B-NURSE SET FamilyName = 'Chi' WHERE NurseID = '076';	1 1 1	3
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50. M/J 17/P12/Q1/a,b,c

1(a)	Many-to-many relationship		1																
1(b)(i)	 Both entities correctly labelled 1 Correct relationship between SHOP and SHOP-SUPPLIER 1 Correct relationship between SUPPLIER and SHOP-SUPPLIER 1		3																
1(b)(ii)	<table border="1"> <thead> <tr> <th>Table</th><th>Primary key</th><th>Foreign keys(s) (if any)</th><th>Explanation</th></tr> </thead> <tbody> <tr> <td>SHOP</td><td>ShopID</td><td>None</td><td></td></tr> <tr> <td>SUPPLIER</td><td>SupplierID</td><td>None</td><td></td></tr> <tr> <td>SHOP-SUPPLIER</td><td>ShopID AND SupplierID</td><td>ShopID OR SupplierID (or both)</td><td>To create a link with the SHOP or SUPPLIER table.</td></tr> </tbody> </table> - SHOP has primary key ShopID and SUPPLIER has primary key SupplierID 1 - SHOP-SUPPLIER has primary key ShopID + SupplierID 1 - Both SHOP and SUPPLIER show foreign key as 'None' 1 - SHOP-SUPPLIER shows foreign key ShopID or SupplierID 1 - Explanation for SHOP-SUPPLIER foreign key describes ShopID or SupplierID creating a link 1	Table	Primary key	Foreign keys(s) (if any)	Explanation	SHOP	ShopID	None		SUPPLIER	SupplierID	None		SHOP-SUPPLIER	ShopID AND SupplierID	ShopID OR SupplierID (or both)	To create a link with the SHOP or SUPPLIER table.		5
Table	Primary key	Foreign keys(s) (if any)	Explanation																
SHOP	ShopID	None																	
SUPPLIER	SupplierID	None																	
SHOP-SUPPLIER	ShopID AND SupplierID	ShopID OR SupplierID (or both)	To create a link with the SHOP or SUPPLIER table.																
1(b)(iii)	Two from: - The database user will <u>frequently</u> want to search on contact name 1 - The contact name attribute has been indexed 1 - It allows for a <u>fast/faster</u> search using contact name 1		Max 2																
1(c)(i)	SELECT ShopID, Location FROM SHOP WHERE RetailSpecialism = 'GROCERY';	1 1 1	3																
1(c)(ii)	INSERT INTO SHOP-SUPPLIER (ShopID, SupplierID) VALUES (8765, 'SUP89');	1 1 1	3																

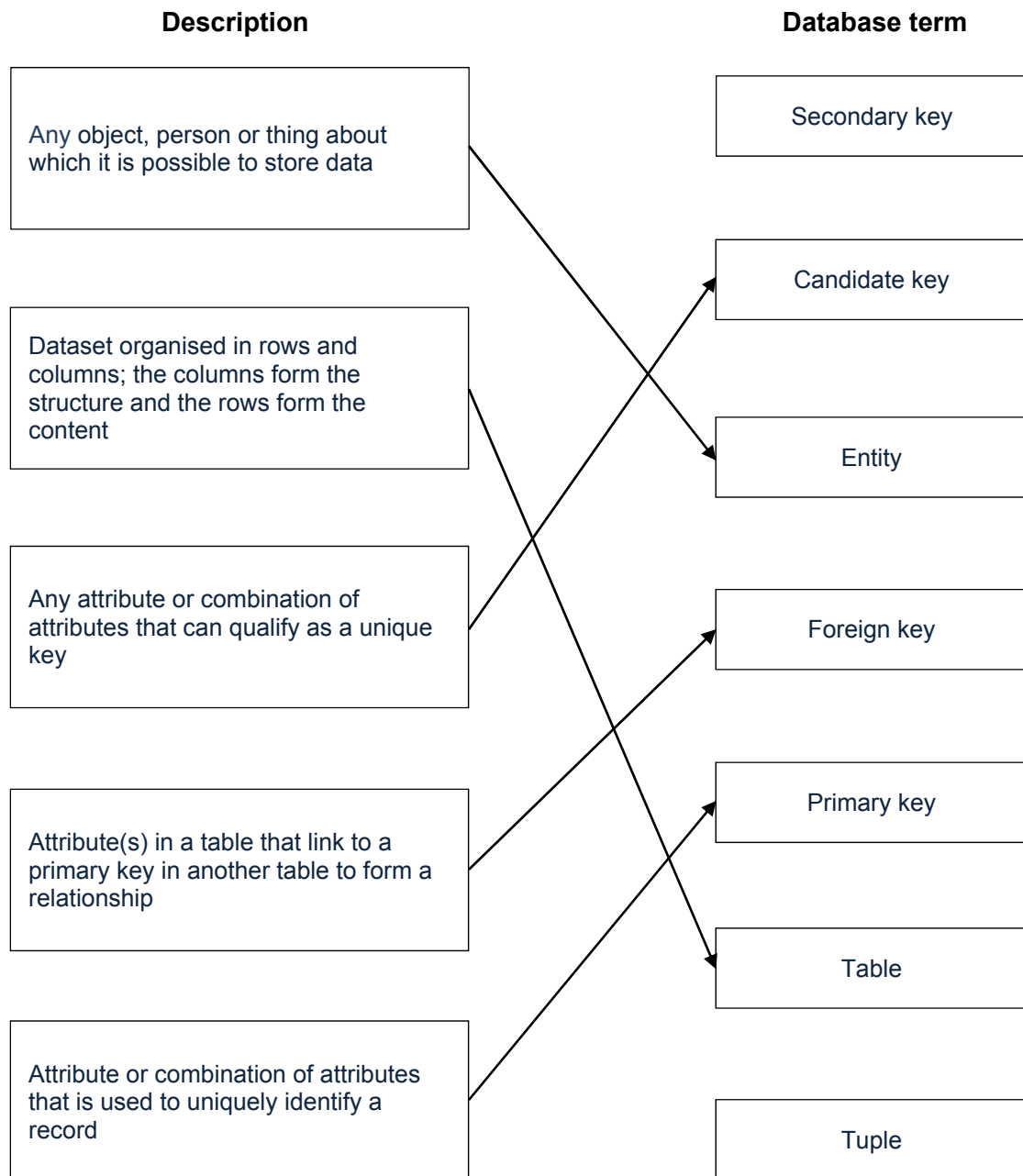
51. M/J 17/P13/Q1/a,b,c,d

1(a)	Many-to-one		1
1(b)(i)	A-NURSE (<u>NurseID</u> , FirstName, FamilyName, WardName)		1
1(b)(ii)	- The primary key <u>WardName</u> in the A-WARD table links to the foreign key <u>WardName</u> in the A-NURSE table.	1 1	2
1(c)(i)	Many-to-many relationship		1
1(c)(ii)	B-WARD-NURSE (<u>WardName</u> , <u>NurseID</u>) Both attributes (with no additions) Joint primary key correctly underlined	1 1	2
1(c)(iii)	 Correct relationship between B-NURSE and B-WARD-NURSE Correct relationship between B-WARD and B-WARD-NURSE	1 1	2
1(d)(i)	SELECT NurseID, FamilyName FROM B-NURSE WHERE Specialism = 'THEATRE';	1 1 1	3
1(d)(ii)	UPDATE B-NURSE SET FamilyName = 'Chi' WHERE NurseID = '076';	1 1 1	3

52. O/N 16/P11/Q1/a,b

1 (a) One mark for each correct line.

Two lines from any box on left means no mark for that description.



[5]

(b) Any **three** from:

- Ensures related data in tables are consistent
- If one table has a foreign key (the 'foreign' table)...
- ... then it is not possible to add a record to that table / the 'foreign' table
- ... unless there is a corresponding record in the linked table with a corresponding primary key (the 'primary' table)
- Cascading delete
- If a record is deleted in the 'primary' table...
- all corresponding linked records in 'foreign' tables must also be deleted
- Cascading update
- If a record in the 'primary' table is modified...
- ... all linked records in foreign tables will also be modified

[3]

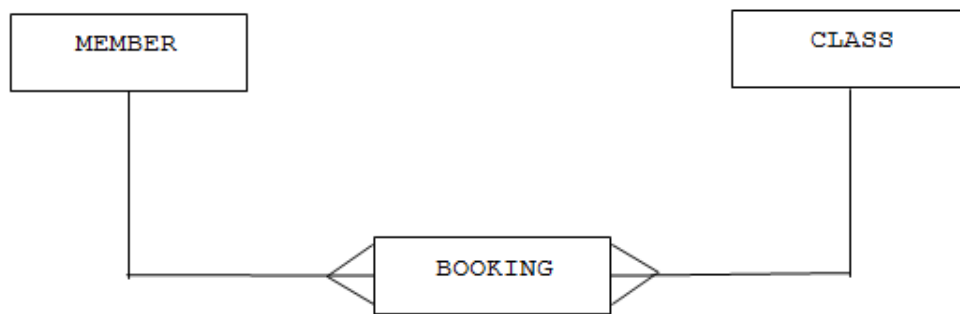
53. O/N 16/P12/Q9/a,b,c

9 (a) ONE mark for each reason and **ONE** mark for a further explanation. **MAX THREE** reasons.

- Reduced data redundancy / data duplication
- Data is stored in (separate) linked tables
- The database (generally) stores data only once / data need only be updated once
- Improved data consistency / integrity / associated data will be automatically updated / easier to maintain the data / elimination of unproductive maintenance
- Complex queries can be more easily written
- To search / find specific data // specific example related to the Health Club
- Fields can be more easily added to or removed from tables
- Without affecting existing applications (that do not use these fields)
- Program-data dependence is overcome
- Changes to the data (design) do not require changes to programs // changes to programs do not require changes to data // the data can be accessed by any appropriate program
- Security is improved
- Each application only has access to the fields it needs // different users can be given different access rights
- Different users can be given different views of the data / data privacy is maintained
- So they do not see confidential information
- Allows concurrent access
- Record locking prevents two users updating the same record at the same time // record locking assures data consistency

[6]

(b) **ONE** mark for each correct relationship as shown.



[2]

(c) An example of a script is shown, but different syntax may be used.

```
CREATE TABLE CLASS (  
    ClassID VARCHAR(5),  
    Description VARCHAR(30),  
    StartDate DATE,  
    ClassTime TIME,  
    NoOfSessions INT,  
    AdultsOnly BIT,  
    PRIMARY KEY(ClassID)  
);
```

Mark as follows:

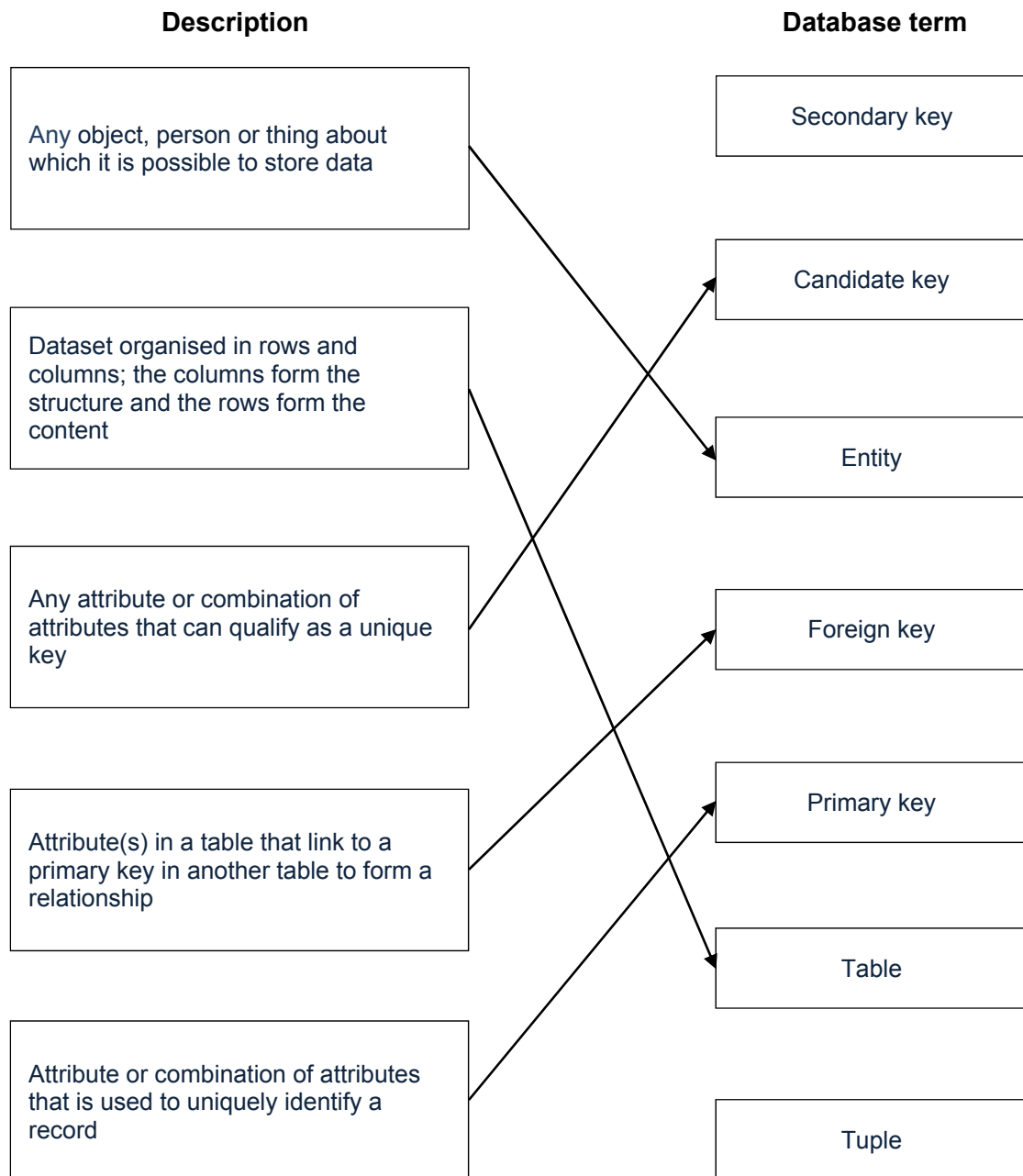
- 1 mark** for `CREATE TABLE CLASS` and `()`;
- 1 mark** for `PRIMARY KEY(ClassID)`
- 1 mark** for both `ClassID VARCHAR(5)`, and `Description VARCHAR(30)`,
- 1 mark** for both `StartDate DATE`, and `ClassTime TIME`,
- 1 mark** for `NoOfSessions INT`,
- 1 mark** for `AdultsOnly BIT`,

[6]

54. O/N 16/P13/Q1/a,b

1 (a) One mark for each correct line.

Two lines from any box on left means no mark for that description.



[5]

(b) Any **three** from:

- Ensures related data in tables are consistent
- If one table has a foreign key (the 'foreign' table)...
- ... then it is not possible to add a record to that table / the 'foreign' table
- ... unless there is a corresponding record in the linked table with a corresponding primary key (the 'primary' table)
- Cascading delete
- If a record is deleted in the 'primary' table...
- all corresponding linked records in 'foreign' tables must also be deleted
- Cascading update
- If a record in the 'primary' table is modified...
- ... all linked records in foreign tables will also be modified

[3]

55. M/J 16/P11/Q8/a,b

8 (a) (i) **Database Management System**

[1]

(ii) **One mark** for identifying the way in which the data security is ensured, and **one mark** for a further description.

Maximum of two marks per method. **Maximum of two** methods.

[4]

- Issue usernames and passwords...
 - stops unauthorised access to the data
 - any further expansion e.g. strong passwords / passwords should be changed regularly etc...
- Access rights / privileges...
 - so that only relevant staff / certain usernames can read/edit certain parts of the data
 - can be read only, or full access / read, write and delete
 - any relevant example e.g. only class tutors can edit details of pupils in their tutor group
- Create (regular / scheduled) backups...
 - in case of loss/damage to the live data a copy is available
 - any relevant example e.g. backing up the attendance registers at the end of each day and storing the data off-site/to a separate device
- Encryption of data...
 - if there is unauthorised access to the data it cannot be understood // needs a decryption key
 - any relevant example e.g. personal details of pupils are encrypted before being sent over the Internet to examination boards
- Definition of different views...
 - composed of one or more tables
 - controls the scope of the data accessible to authorised users
 - any relevant example e.g. teachers can only see their classes
- Usage monitoring / logging of activity...
 - creation of an audit /activity log
 - records the use of the data in the database / records operations performed by all users / all access to the data
 - any relevant example, e.g. Track who changed a student's grade

(iii) **Two points** from:

[2]

- Set up search criteria
- To find / retrieve / return the data that matches the criteria
- Any relevant example e.g. find pupils who were absent on a particular day

(iv) Three points from:

[3]

- By storing data in (separate) linked tables data redundancy is reduced / data duplication is controlled...
- Compatibility / data integrity issues are reduced as data only needs to be updated once / is only stored once.
- Unwanted or accidental deletion of linked data is prevented as the DBMS will flag an error.
- Program - data dependence is overcome.
- Changes made to the structure of the data have little effect on existing programs.
- Ad-hoc / complex queries can be more easily made as the DBMS will have a query language/ QBE form.
- Unproductive maintenance is eliminated as changes only need to be made once (rather than changing multiple programs).
- Fields can be added or removed without any effect on existing programs (that do not use these fields).
- Security / privacy of the data is improved as each application only has access to the fields it needs.
- There is better control of data integrity as the DBMS (uses its Data Dictionary) to perform validation checks on data entered.

(b) (i) Two points from:

[2]

- The Primary Key in **CLASS** is ClassID
- The Foreign Key of **CLASS-GROUP** is ClassID.
- The Primary Key of **CLASS** is also included in **CLASS-GROUP** as a Foreign Key, (which links to **CLASS** table)

(ii) Many-to-one

[1]

(iii) One mark per statement. Several statements may be on the same line.

[4]

```
SELECT StudentID, FirstName
FROM STUDENT
WHERE TutorGroup = "10B" // WHERE (TutorGroup = "10B")
ORDER BY LastName ASC;
```

(iv) One mark per statement. Several statements may be on the same line.

[4]

```
SELECT STUDENT.LastName
FROM STUDENT, CLASS-GROUP
WHERE ClassID = "CS1" // WHERE (ClassID = "CS1")
AND CLASS-GROUP.StudentID = STUDENT.StudentID;
```

One mark per statement. Several statements may be on the same line.

```
SELECT STUDENT.LastName
FROM STUDENT INNER JOIN CLASS-GROUP
ON CLASS-GROUP.StudentID = STUDENT.StudentID
WHERE ClassID = "CS1" // WHERE (ClassID = "CS1");
```

56. M/J 16/P12/Q8/a,b

8 (a) (i) Database Management System

[1]

- (ii) **One mark** for identifying the way in which the data security is ensured, and **one mark** for a further description.

Maximum of two marks per method. **Maximum of two** methods.

[4]

- Issue usernames and passwords...
 - stops unauthorised access to the data
 - any further expansion e.g. strong passwords / passwords should be changed regularly etc...
- Access rights / privileges...
 - so that only relevant staff / certain usernames can read/edit certain parts of the data
 - can be read only, or full access / read, write and delete
 - any relevant example e.g. only class tutors can edit details of pupils in their tutor group
- Create (regular / scheduled) backups...
 - in case of loss/damage to the live data a copy is available
 - any relevant example e.g. backing up the attendance registers at the end of each day and storing the data off-site/to a separate device
- Encryption of data...
 - if there is unauthorised access to the data it cannot be understood // needs a decryption key
 - any relevant example e.g. personal details of pupils are encrypted before being sent over the Internet to examination boards
- Definition of different views...
 - composed of one or more tables
 - controls the scope of the data accessible to authorised users
 - any relevant example e.g. teachers can only see their classes
- Usage monitoring / logging of activity...
 - creation of an audit /activity log
 - records the use of the data in the database / records operations performed by all users / all access to the data
 - any relevant example, e.g. Track who changed a student's grade

- (iii) **Two points** from:

[2]

- Set up search criteria
- To find / retrieve / return the data that matches the criteria
- Any relevant example e.g. find pupils who were absent on a particular day

(iv) Three points from:

[3]

- By storing data in (separate) linked tables data redundancy is reduced / data duplication is controlled...
- Compatibility / data integrity issues are reduced as data only needs to be updated once / is only stored once.
- Unwanted or accidental deletion of linked data is prevented as the DBMS will flag an error.
- Program - data dependence is overcome.
- Changes made to the structure of the data have little effect on existing programs.
- Ad-hoc / complex queries can be more easily made as the DBMS will have a query language/ QBE form.
- Unproductive maintenance is eliminated as changes only need to be made once (rather than changing multiple programs).
- Fields can be added or removed without any effect on existing programs (that do not use these fields).
- Security / privacy of the data is improved as each application only has access to the fields it needs.
- There is better control of data integrity as the DBMS (uses its Data Dictionary) to perform validation checks on data entered.

(b) (i) Two points from:

[2]

- The Primary Key in **CLASS** is ClassID
- The Foreign Key of **CLASS-GROUP** is ClassID.
- The Primary Key of **CLASS** is also included in **CLASS-GROUP** as a Foreign Key, (which links to **CLASS** table)

(ii) Many-to-one

[1]

(iii) One mark per statement. Several statements may be on the same line.

[4]

```
SELECT StudentID, FirstName
FROM STUDENT
WHERE TutorGroup = "10B" // WHERE (TutorGroup = "10B")
ORDER BY LastName ASC;
```

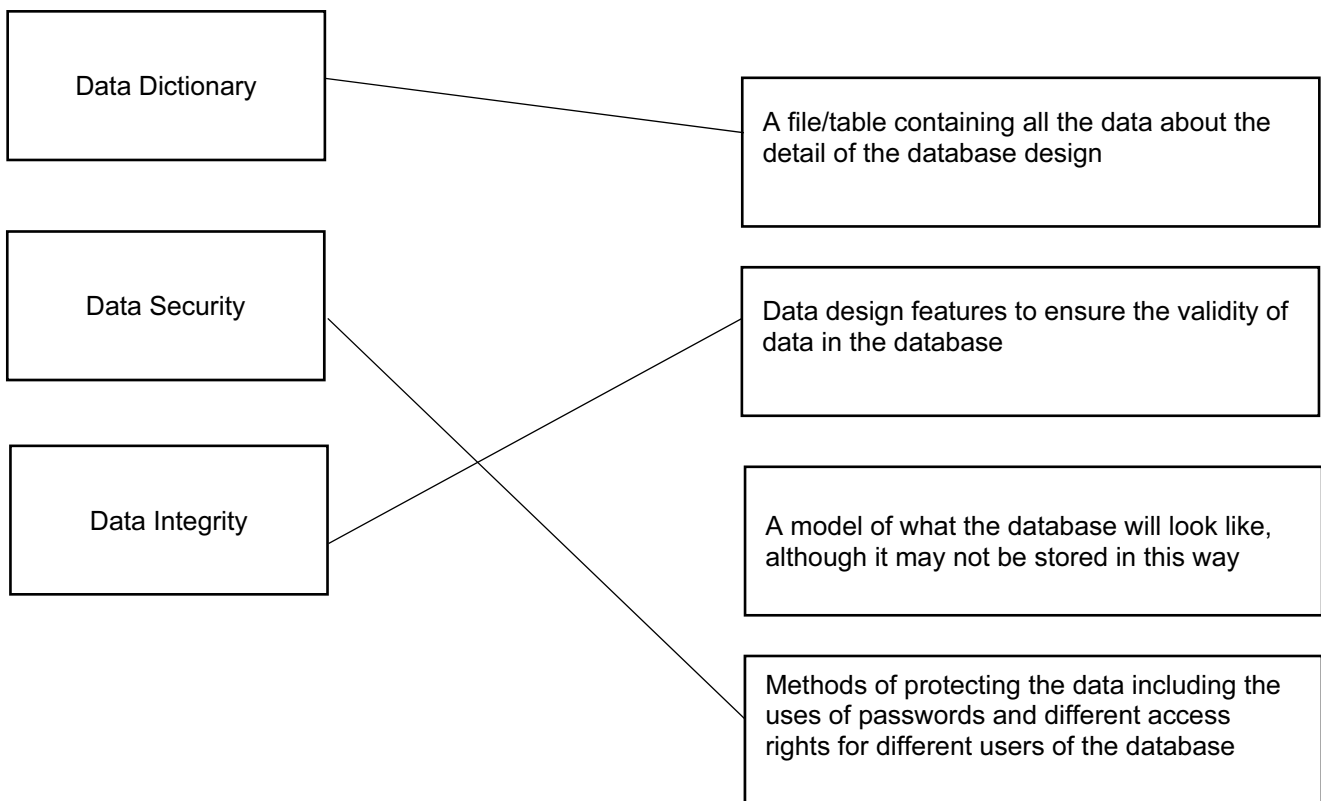
(iv) One mark per statement. Several statements may be on the same line.

[4]

```
SELECT STUDENT.LastName
FROM STUDENT, CLASS-GROUP
WHERE ClassID = "CS1" // WHERE (ClassID = "CS1")
AND CLASS-GROUP.StudentID = STUDENT.StudentID;
```

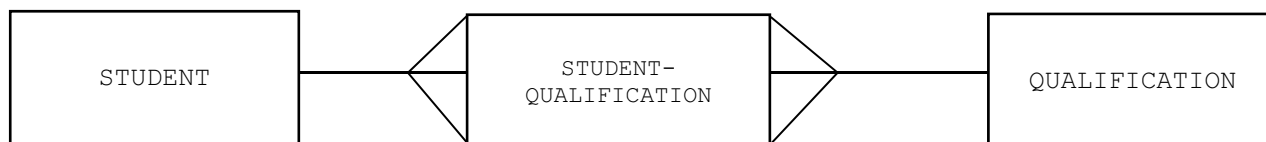
One mark per statement. Several statements may be on the same line.

```
SELECT STUDENT.LastName
FROM STUDENT INNER JOIN CLASS-GROUP
ON CLASS-GROUP.StudentID = STUDENT.StudentID
WHERE ClassID = "CS1" // WHERE (ClassID = "CS1");
```

57. M/J 16/P13/Q5/a,b,c,d**5 (a) One mark** for each correct line.**[3]****(b) One mark** for procedure point, **one mark** for justification.**[6]**Maximum **three** procedures.

- How often should the data be backed up? e.g. at the end of each day
- Justification e.g. student's progress may be edited each day and should not be lost
- What medium should the data be backed up to? e.g. external hard disk drive
- Justification e.g. it has large enough capacity
- Where should the backups be stored? e.g. off-site
- Justification e.g. so if the building is damaged only the original data are lost
- What is backed up? e.g. only updated files ...
- Justification e.g. There are a large number of files and they are not all updated each day
- When should the backup take place? e.g. overnight
- Justification e.g. the system is not likely to be used then
- Who is responsible for performing the backup?
- Justification e.g. otherwise it may not be done
- Make sure the procedure is written down and understood by staff
- Justification e.g. otherwise some data may not be backed up

(c) (i) **One mark** for each correct relationship. [2]



(ii) **One-to-many** [1]

(iii) **Two points** from: [2]

- The primary key in the QUALIFICATION table is QualCode.
- The foreign key in the STUDENT-QUALIFICATION table is QualCode.
- The primary key of QUALIFICATION is also included in QualCode.

(d) (i) **One mark** per statement. Several statements may be on one line. [2]

```
ALTER TABLE STUDENT
ADD DateOfBirth DATE;
```

(ii) **One mark** per statement. Several statements may be on one line. [3]

```
SELECT StudentID, Grade, DateOfAward
FROM STUDENT-QUALIFICATION
WHERE QualCode = 'SC12';
```

(iii) **One mark** per statement. Several statements may be on one line. [4]

```
SELECT STUDENT.FirstName, STUDENT.LastName, STUDENT-
    QUALIFICATION.QualCode
FROM STUDENT, STUDENT-QUALIFICATION
WHERE STUDENT-QUALIFICATION.Grade = 'A'
AND STUDENT.StudentID = STUDENT-QUALIFICATION.StudentID;
```

Alternative answer:

```
SELECT FirstName, LastName, STUDENT-QUALIFICATION.QualCode
FROM STUDENT, INNER JOIN STUDENT-QUALIFICATION
ON STUDENT.StudentID = STUDENT-QUALIFICATION.StudentID
WHERE Grade = 'A';
```

58. M/J 15/P11/Q9/a,b,c

9 (a) Any **one** from:

- (ShopSales) table has repeated group (of attributes)
- each sales person has a number of products
- FirstName, Shop would need to be repeated for each record

[1]

(b) One mark for SalesPerson table**table:** SalesPerson

FirstName	Shop
Nick	TX
Sean	BH
John	TX

table: SalesProducts

FirstName	ProductName	NoOfProducts	Manufacturer
Nick	television set	3	SKC
Nick	refrigerator	2	WP
Nick	digital camera	6	HKC
Sean	hair dryer	1	WG
Sean	electric shaver	8	BG
John	television set	2	SKC
John	mobile phone	8	ARC
John	digital camera	4	HKC
John	toaster	3	GK

(1 mark for FirstName column + 1 mark for remainder of table)

[3]

(c) (i) Any **two** from:

- primary key of SalesPerson table is FirstName
- links to FirstName in SalesProducts table
- FirstName in SalesProductsS table is foreign key

[2]

- (ii)
- There is a non-key dependency
 - Manufacturer is dependent on ProductName, (which is not the primary key of the SalesProducts table)

[2]

(iii) SalesPerson (FirstName, Shop)
-SalesProducts (FirstName, ProductName, NoOfProducts) **OR**
SalesProducts(SalesID, FirstName, ProductName, NoOfProducts)

-Product (ProductName, Manufacturer)

1 mark for correct attributes in SalesProducts and Product tables and **1 mark** for correct identification of both primary keys **[2]**

59. M/J 15/P12/Q9/a,b,c

9 (a) Any **one** from:

- (ShopSales) table has repeated group (of attributes)
- each sales person has a number of products
- FirstName, Shop would need to be repeated for each record

[1]

(b) **One mark** for SalesPerson table

table: SalesPerson

FirstName	Shop
Nick	TX
Sean	BH
John	TX

table: SalesProducts

FirstName	ProductName	NoOfProducts	Manufacturer
Nick	television set	3	SKC
Nick	refrigerator	2	WP
Nick	digital camera	6	HKC
Sean	hair dryer	1	WG
Sean	electric shaver	8	BG
John	television set	2	SKC
John	mobile phone	8	ARC
John	digital camera	4	HKC
John	toaster	3	GK

(1 mark for FirstName column + 1 mark for remainder of table)

[3]

(c) (i) Any **two** from:

- primary key of SalesPerson table is FirstName
- links to FirstName in SalesProducts table
- FirstName in SalesProductsS table is foreign key

[2]

- (ii)
- There is a non-key dependency
 - Manufacturer is dependent on ProductName, (which is not the primary key of the SalesProducts table)

[2]

- (iii) SalesPerson (FirstName, Shop)
-SalesProducts (FirstName, ProductName, NoOfProducts) **OR**
SalesProducts(SalesID, FirstName, ProductName, NoOfProducts)

-Product (ProductName, Manufacturer)

1 mark for correct attributes in SalesProducts and Product tables and **1 mark** for correct identification of both primary keys **[2]**